SECTION 3

Ethical issues in different engineering disciplines

Tom Børsen

Teaching ethics to engineering students often occurs in classes that enroll students from many engineering disciplines. The content in such interdisciplinary engineering classes is general and often not perceived as relevant for students specialized in particular engineering disciplines. There are, as we will see in the chapters in this section, issues in engineering ethics that are transversal to most, if not all, engineering disciplines and relevant for most, if not all, engineering students. This section of the handbook investigates how transversal ethical issues can be linked to specific engineering disciplines and, in this way, be perceived as relevant by specialized engineering students. We, the editors of this handbook, selected five engineering disciplines – civil engineering, aerospace and mechanical engineering, electrical and electronic engineering, chemical engineering, and software engineering – and invited prominent scholars of each discipline to author chapters where they identify, analyze, and discuss the ethical issues that they perceive relevant for students of their specific engineering area to become acquainted with during their education. Thus, this section contains five chapters coining ethical content of engineering ethics teaching in civil, mechanical, and aerospace, chemical, software, and electrical and electronic engineering.

Positionality

As a way to understand why this section appears as it does, it might be useful to make explicit the positionality of the section editor. Tom Børsen was originally educated as a chemist with a minor in philosophy, who after graduation engaged in research in science and engineering education with a focus on how to promote ethical and socio-ecological responsibility among technical experts. This led to several curriculum development projects bridging social science and humanities (SSH) and science, technology, engineering, and mathematics (STEM) disciplines. Most predominantly, Tom led the establishment of the BSc and MSc degrees in Techno-Anthropology at Aalborg University. He has also taught ethics at different engineering programs. Thus, after a process of self-reflection, Tom has identified two ways in which his positionality can have influenced the content of this theme: He has experienced that the proximity between engineering ethics content and the engineering discipline in which it is situated is imperative for engineering students' acceptance and sense-making of EEE. The section shows that Tom is not the only one who has had this experience. The second way the section editor's positionality may have had implications

DOI: 10.4324/9781003464259-17
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for the section is reflected in his engagement in promoting engineers and other technical experts to take responsibility for the broader implications of their work. This concern aligns well with how the chapters in the section are conceptualized.

Chapter topics

Here we provide a synopsis of each chapter in this section, highlighting the unique contributions of each.

Chapter 14, 'Ethical Considerations in Civil Engineering' by Irene Josa, Ester Gimenez-Carbo, and Christina Nick, overviews ethical issues inherent in civil engineering, a field vital for providing essential infrastructure such as water access, shelter, transportation, and communication. While distinct from military engineering, this branch of engineering often intersects with it, particularly in areas like transport infrastructures. The chapter underscores the global trend towards a more holistic and socially conscious approach to civil engineering, embraced by professional associations worldwide. Furthermore, it emphasizes the need for civil engineers to excel in technical skills and *also* demonstrate leadership and ethical decision-making abilities, navigating complex dilemmas by considering social, cultural, and philosophical dimensions. The authors stress the importance of a comprehensive perspective that encompasses the entire life cycle of infrastructure projects, from design and planning to decommissioning, in determining the ethical responsibilities of individual professionals.

The authors present a framework and set of tools (e.g., applicable case studies) to identify content for ethics education suggested to be delivered to civil engineering students (Table 14.1). They present ethical issues in civil engineering – professionalism, social responsibility, sustainability, health and safety, EDI (equality, diversity, and inclusion), and decolonization – and relate them to each stage of the life cycle of civil engineering projects. The chapter showcases a range of existing case studies and discussion topics for teachers wanting to integrate ethics teaching into their civil engineering courses.

The authors of Chapter 15, 'Ethical Issues in Mechanical and Aerospace Engineering,' Aaron W. Johnson, Corin L. Bowen, Cristiano Cordeiro Cruz, and Renato Bezerra Rodrigues, examine ethical issues within mechanical and aerospace engineering, encompassing the application of scientific and mathematical principles to develop mechanical systems, including aircraft and spacecraft. The chapter adopts a 'social justice' lens, viewing mechanical and aerospace engineering as potential avenues for empowering marginalized communities.

A vital aspect of this chapter is its deliberate utilization of a critical, emancipatory framework inspired by activist scholars (Freire, 1970; hooks, 1994). Departing from the traditional portrayal of mechanical and aerospace engineering as politically neutral domains, the authors challenge the prevalent notion by advocating for a shift towards macro-ethical considerations alongside micro-ethics. They argue for cultivating students' critical consciousness, emphasizing the importance of addressing systemic oppression perpetuated by engineering practices. While confronting mainstream perceptions of engineering ethics, this approach encourages readers to engage with discomfort, explore relevant literature, and get involved in discussions to broaden their understanding of these perspectives.

In Chapter 16, 'Ethical Issues in Electronic and Electrical Engineering,' Susan M. Lord and John E. Mitchell delve into the ethical dimensions within electronic and electrical engineering (E&EE), emphasizing the crucial alignment of ethical education with technical content. They address the scarcity of case studies specific to E&EE contexts, attributing it to the nature of E&EE components being often integrated into larger engineered systems and less visibly public-facing

compared to other engineering disciplines. The investigation by Lord and Mitchell highlights a recurring trend whereby electrical technology remains peripheral in traditional engineering ethics case studies, which lack focus on E&EE contexts. These case studies often drift outside the discipline or address irrelevant professional issues. The authors advocate for explicitly integrating ethics relevant to E&EE into technical studies and emphasize the importance of engineers teaching these topics. They explore various approaches in ethics education for E&EE students, ranging from discussions on professional responsibility to standalone ethics courses and integration across modules in a course.

Additionally, Lord and Mitchell examine the challenges of teaching ethics in E&EE, identifying opportunities to address ethical issues within the culture of electronic and electrical engineering. This includes scrutinizing offensive technical jargon and controversial imagery like the Lena photograph used in image processing. By identifying and confronting these persistent ethical challenges, the authors advocate for proactive measures to assist students in navigating the intricate ethical landscape of electronic and electrical engineering.

Chemical engineering encompasses the vast domain of large-scale chemical synthesis, production, transportation, storage, and diverse industrial and consumer product applications. In Chapter 17, 'Ethics in Chemical Engineering,' Jan Mehlich, Tom Børsen, and Dayoung Kim draw upon historical and contemporary ethical case studies from research journals to outline ethical issues. They emphasize using varied case studies in ethics education to foster different dimensions of responsibility among chemical engineering students.

Mehlich, Børsen, and Kim probe ethical issues pertinent to chemical engineering education, exploring concerns about the unpredictable nature of chemicals and the potential misuse of chemical engineering processes. The chapter advocates for reflections on the broader implications of chemical engineering projects as a fundamental aspect of ethical practice in chemical engineering. By examining moral, legal, and institutional responsibilities, as well as individual and collective responsibilities, the authors provide valuable insights for educators designing ethics education curricula in chemical engineering. They propose linking case studies with various forms of responsibility to enhance the effectiveness of teaching materials and encourage critical discourse among students.

In Chapter 18, 'Ethical Issues in Software Engineering,' Stephanie J. Lunn, Isis Hazewindus, Prajish Prasad, and Vivek Ramachandran aim to cultivate ethical awareness among students, teachers, and software engineers. They stress the importance of recognizing responsibilities in the creation, dissemination, and maintenance of software – emphasizing the need for educational environments that foster ethical learning alongside technical skills – and for continually reassessing ethics and ensuring accountability and transparency as technologies like artificial intelligence and machine learning dynamically evolve. Embracing 'Ethics by Design,' they propose tools, discussions, and assignments to promote ethical reflection and aid in problem-solving.

In exploring ethical issues in software engineering, the authors advocate for cultivating ethical mindsets throughout the software development and data life cycles. Linking these two cycles into a brand new 'ETHOS' model for facilitating ethical reflection, the authors provide a tool that can help scaffold dialogue and help people analyze ethical issues in software engineering practice as well as in engineering ethics classes.

Trends and implications

Transversal ethical issues are exemplified within the concrete discipline, as the observant reader of all five chapters in this section will notice, and there are overlaps among the different ethical

issues perceived as relevant by the authors of the chapters. This is not to say that all ethical issues are present in all chapters, because they are not; however, most issues are identified as central in three or four chapters. Few, if any, ethical issues are limited to only one engineering discipline.

Professional codes of ethics are one common ethical topic described in several chapters. Yet, authors across this section argue that codes cannot be the only content of engineering ethics education – additional topics must be included. Chapter 16 on electrical and electronic engineering underscores that students need to know the ethical codes of their engineering discipline *and* also be able to relate them to their own projects and technical work. Many engineering companies have business ethical schemes that students can apply in parallel to the professional ethical codes of conduct, for example, during an internship.

The broader implications of engineering work within different non-technical or socio-ecological spheres (e.g., the environment, human health, users' privacy, employees' well-being, vulnerable groups, and society as a whole) constitute another transversal ethical issue that appears in several chapters and is considered relevant in many engineering disciplines. Chapter 14 on ethical issues in civil engineering names this ethical topic 'social responsibility.' Socio-ecological responsibility is an extended predicate one can put on this transversal ethical issue. Chapter 17 on chemical engineering bridges this ethical issue with the previous one by suggesting that engineering ethical codes must include macro-ethical concerns regarding, for example, environmental and human health implications. Chapter 18 on ethical issues in software engineering encourages engineering solutions to actively strive for improvements in these non-technical spheres, bridging the following ethical issues.

Sometimes, harmful effects on workers, technology users, or local communities are or should have been known and could have been prevented. Sometimes, the implications for humans are unforeseen and emerge due to complex systems that are difficult to control. Chapter 17 on chemical engineering asserts this distinction as central for taking responsibility for the implications. Engineers and company managers must be held responsible for their neglect if harmful implications were or could have been known (e.g., were reasonably foreseeable). When engineering design, solutions, and innovation are new and emerging, and implications uncertain and difficult to predict, engineers and other actors can and should exercise precaution. Chapter 18 on software engineering contains scenarios of both intentional misuse and unforeseen implications. Chapter 15 on mechanical and aerospace engineering argues that the harmful implications often follow from hegemonic power structures. One way to gain knowledge about the human implications of technology is to involve those affected by engineering solutions and design. Engineering that engages employees and users is denoted 'human-centric engineering.'

Sustainable design is a transversal ethical issue related to the issues identified above, focused on the dual influences of engineering activities on the environment: engineering can contribute to sustainable development, but it can also pollute the environment. Chapter 14 on civil engineering argues that engineering projects must actively strive to minimize the use of resources and emissions of greenhouse gasses. Chapter 16 on electrical and electronic engineering suggests incorporating environmental concerns in technical design. Chapter 15 on mechanical and aerospace engineering warns that developing 'green' products (in a narrow sense) is not sustainable if the products enable or encourage unsustainable practices. The difference between this ethical issue and the two previous ones is that socio-ecological responsibility focuses on identifying the implications, human-centric engineering focuses on co-creation, and sustainable design creates socio-ecological beneficial consequences. Several chapters point toward cases where engineers and engineering solutions have had very positive implications for human communities; they assert that engineers in technical design should actively seek beneficial implications.

Some chapters mention military applications as an ethical issue to be discussed and reflected upon in engineering ethics education. Chapter 15 on mechanical and aerospace engineering draws attention to the military-industrial complex and notes that military equipment, which has been used in armed conflicts around the world, kills innocent civilians in high numbers. Chapter 17 on chemical engineering includes cases from World War I and the Vietnam War.

Social justice is the final transversal ethical issue highlighted in this section's introduction. It covers an array of topics: Chapter 14 on civil engineering discusses both 'equality, diversity and inclusion' and 'decolonization' as topics to be included in engineering ethics education. Chapter 16 on electrical and electronic engineering provides examples from the academic and professional engineering culture (such as terminology and jargon) that can adversely reduce the sense of belonging among women and students of color in engineering. Engineers' use of an image of Lena Forsén in image processing, an image originating from the sexist context of *Playboy* magazine, is one of several examples the authors critique and pose for in-class and professional dialogue.

None of the chapters in this section suggest that students must be able to provide clear-cut answers to ethical challenges by being acquainted with some of the ethical issues presented in this section's chapters. It is essential to dialogue, reflect, and act on these issues when they appear in students' disciplinary and cross-disciplinary contexts. Chapter 18 on software engineering includes a sophisticated reservoir of dialogical methods and reflection tools that can be applied in other engineering disciplines. We find connectivity among the different ethical issues presented by the authors of this section: discussions, reflections, and analyses should be part of engineering ethics education as they can probe the various issues and identify the relationships among the different ethical topics.

The section highlights some general aspects of EEE, some of which are discussed in this section of the handbook, and others in previous sections. The general ethical issues must be translated into discipline-specific cases, questions, exercises, and design projects. Thus, we find relationships between this section and the previous two ones in the way that the first section highlights the importance of evoking normative frameworks in disciplinary-specific EEE content, and the second section emphasizes that thorough descriptions of the emotional, social, organizational, cultural, and legal context must be included as parts of disciplinary-specific EEE content. Such an affinity to the concrete sociotechnical context is a central way to root the more general normative frameworks and transversal ethical issues in specific engineering disciplines. After this section has reflected on the content of disciplinary EEE, one next step can be to identify appropriate teaching and assessment methods, which are the topics addressed in the following sections.

Conclusions from the section editor

As the editor responsible for leading this section, Tom Børsen expects that the selection of chapters will also be relevant to read for engineering ethics teachers, students, practitioners, and researchers from other disciplines than those directly addressed in the chapters. They are not intended to only inspire those directly involved in the discipline discussed. A common denominator in all the chapters is a suggested balance between – on the one hand – disciplinary exercises, examples, case studies, and scenarios deeply rooted within the specific discipline and – on the other hand – more general or transversal ethical issues such as professional and socio-ecological responsibilities, human-centric engineering, sustainable design, military applications, and social justice. Our rationale is that the transversal ethical issues identified in these chapters can be translated into engineering disciplines other than those from which they originate. Moreover, the chapters outline the importance of translating transversal ethical issues into specific engineering disciplines, rather

than presenting ethical issues to students in a generic or general engineering context. The disciplinary proximity of engineering ethics education comes through disciplinary exercises, examples, cases, scenarios, and designs that illustrate more general engineering ethical issues but in ways that clearly link to the student's chosen discipline of focus.

Tom hopes readers will enjoy assessing how the above briefly presented transversal ethical issues – including professional codes of ethics, socio-ecological responsibility, human-centered engineering, sustainable design, military applications, and social justice – unfold in specific disciplinary contexts. That said, Tom and the larger editorial team recognize that the section does not provide a final and complete overview of ethical issues in different engineering disciplines; therefore, they welcome readers' comments on the following chapters and encourage additional research in the realm of EEE for specific engineering disciplines.

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14

ETHICAL CONSIDERATIONS IN CIVIL ENGINEERING

Irene Josa, Ester Gimenez-Carbo, and Christina Nick

Introduction

Civil engineering is the oldest branch of engineering, with civil engineering practices dating back to ancient times, with the establishment of the earliest urban settlements. The roads, canals, viaducts, and other public works developed by the Roman Empire are still visible today. Other notable examples of ancient civil engineering were constructed in Egypt, Greece, Mesopotamia, China, the Inca Empire, and Persia – thus illustrating the global impact and diversity of ancient civil engineering achievements. This predates the formal establishment of professional societies, which emerged later to organize and advance various engineering disciplines. Civil engineering was founded to serve and benefit society. Since its inception, the civil engineering profession has supported the development of societies and improved people's quality of life by providing access to basic needs such as water, shelter, transport, and communications. Over time, civil engineers began to formalize the scope, definition, and responsibilities under their remit, and today, professional bodies specific to civil and structural engineering practices proliferate worldwide. They include, for example, the US-based American Society for Civil Engineering (ASCE), the UK-based Institution of Civil Engineers (ICE), the Japan Society of Civil Engineers (JSCE), the International Association of Structural Engineers (IABSE), and the UK-based Institution of Structural Engineers (IStructE). Many subfields of civil engineering host professional organizations as well, such as the International Society for Soil Mechanics and Geotechnical Engineering (ISSMGE), the American Concrete Institute (ACI), and the Society of Fire Protection Engineers (SFPE).

In recent years, increasing attention has been paid to ethical issues within this profession – see, for instance, the 'vision of the 2025 engineer' by one of current society's most active professional engineering bodies, the ASCE (ASCE Steering Committee, 2006). Civil engineers' knowledge, skills, and work have great potential for enhancing humanity's quality of life and solving some of today's most pressing global challenges (Koehn, 1991). Nonetheless, undesired outcomes can arise if citizens' health, safety, and welfare are not placed at the center of civil engineers' work. Several controversial issues related to infrastructure have filled the news over the past decades, including corruption cases and infrastructure collapses.

DOI: 10.4324/9781003464259-18

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To avoid adverse societal and environmental effects resulting from civil engineers' work, future professionals must be trained to understand, identify, and solve ethical issues related to their work (Banik & Gouranga Banik, 2011). Engineering ethics education is vital in equipping engineers with the necessary knowledge and skills to navigate complex ethical dilemmas that may arise throughout their careers. In addition to the technical education essential for civil engineers, an effective education program should include a comprehensive exploration of ethical questions, case studies highlighting real-world ethical challenges civil engineers face, and opportunities for ethical decision-making and reflection.

Considering the above, this chapter aims to provide an overview of the ethical issues, dilemmas, and challenges that are fundamental in the education of civil engineering students. The chapter also addresses how the discipline's students and practitioners might address these issues, dilemmas, and challenges.

Positionality

How we approach our discipline – the questions we ask, the methodologies and case studies we select – are shaped by our positionality, so it is vital to reflect on this briefly.

Two of this chapter's authors are civil engineers, while the third is an applied ethicist with several years of experience teaching engineering ethics. The chapter is the result of an interdisciplinary exchange in which the authors have tried to bring together ethical theories and concepts with concrete applications relevant to the day-to-day practices of civil engineers.

All three authors identify as white and come from and currently work in Western European countries. As such, we have no lived experience of the racial oppression and global inequalities faced by many people. All three authors identify as cis heterosexual women. Although this makes us particularly attuned to the gendered dimensions of civil engineering, it also means that we have no first-hand experience of the marginalization faced by queer people. Finally, one of the authors identifies as disabled. This has certainly heightened our awareness of the impact that our choices have on disabled people. Still, the first-hand knowledge of a single person will necessarily provide only a limited guide to the lived experiences of a large and heterogeneous group.

The danger coming from our positionality is that, through our writing, we could re-center the voices of those in positions of social advantage. Attempting to mitigate this, we have consciously tried to problematize areas where civil engineering currently works to reinforce oppression and marginalization. We have placed emphasis on equality, diversity, and inclusion (EDI) throughout the chapter.

Overview of the chapter

The chapter starts by contextualizing the civil engineering discipline. The historical framework of the profession is described, as are the subdisciplines that make up civil engineering.

Two crucial aspects must be factored in when discussing ethics in civil engineering. One involves the different areas for ethical considerations, which we have grouped under the themes of (1) professionalism; (2) social responsibility; (3) sustainability; (4) health and safety; (5) equality, diversity, and inclusion (EDI); and (6) decolonization. Secondly, civil engineering projects have different life-cycle stages: (1) design and planning, (2) development and construction, (3) operation and maintenance, and (4) decommissioning. This chapter introduces the most important aspects of the different life-cycle stages that require consideration for each thematic area. Nevertheless, readers of this chapter might want to explore the contents according to either thematic groups or according to the life-cycle stages. To help the reader go through the contents in

the order they find most convenient, the different contents of this chapter have been summarized in Table 14.1. This table also includes additional content that readers can explore independently.

- Ethical issues explored
  - Case studies presented in this chapter
    - Other potential case studies

This chapter offers conceptual insights into ethical concerns and provides practical examples through case studies and discussion exercises for enriching students' understanding of ethical issues (Harris et al., 2014; Whitbeck, 2011). The selected case studies portray real-world situations, covering a diverse spectrum of civil engineering contexts and subdisciplines, aligning with the chapter's themes.[1]

The case studies serve as catalysts for student discussion. They encourage students to explore various scenarios, anchoring their arguments in established professional ethics codes while striving for solutions that align with these ethical guidelines. Teachers unfamiliar with this teaching approach can examine the discussions' structure, alignment with ethical codes, personal experiences in similar situations, and the resulting conclusions. (For more detailed advice on teaching via case studies, please see Chapter 20, and via discussion and reflection, Chapter 25.)

We suggest discussion questions that can be used in engineering classrooms to prompt discussion of core issues – but we encourage readers to consider these suggestions *alongside advice on applying case study teaching methods* as presented in Chapter 20 and *recommended methods for teaching engineering ethics using reflective and dialogical approaches* as presented in Chapter 25. The case study examples we draw from below are listed in Table 14.1, where key concepts, location within the construction process, the ethical issues explored, and other cases to be explored are also highlighted.

Lastly, civil engineering encompasses six primary subdisciplines: construction and buildings, energy infrastructure, environmental technology, water, transportation, and urbanism. This chapter addresses ethical concerns in each area, emphasizing specific issues where relevant. Each subdiscipline is represented with at least one case study, providing a comprehensive view of ethical challenges in civil engineering.

Contextualization

In 2006, the ASCE convened a summit on the 'Future of Civil Engineering' to establish a global vision for the profession in the twenty-first century. The resulting document, 'The Vision for Civil Engineering in 2025' (ASCE, 2006), outlines the role of civil engineers as stewards of the natural environment and resources, innovators and integrators of ideas and technology across sectors, and leaders in shaping public policy.

This vision has been embraced by civil engineering associations worldwide, recognizing the profession's responsibility in transforming the built and natural environments to achieve the '2030 Agenda' regarding the United Nations Sustainable Development Goals (SDGs). ASCE adopted a policy statement in support of the United Nations Sustainable Development Goals (ASCE, 2023), and ICE (2020) published the 'UN75 Sustainable Engineering in Action' to highlight civil engineering's role in this agenda.

The call for a more holistic and socially conscious approach to civil engineering is a global phenomenon that professional associations in many countries have embraced. Various professional engineering organizations worldwide have developed codes of conduct that establish ethical standards for civil engineers. These codes, such as the Code of Conduct of the European Federation

Table 14.1 Overview of possible contents for civil engineering ethics education (non-exhaustive list)

	Key Concepts	*Design and Planning*	*Development and Construction*	*Operation and Maintenance*	*Decommissioning*
Professionalism	• Codes of conduct • Registration, accreditations, charters	• Bribery and corruption • Whistle-blowing		• Conflicts of professional duties	
		- Maharashtra Irrigation Scam		○ NSPE Board of Ethical Review Case 93-13	
Social Responsibility	• Corporate social responsibility (CSR)	• Unforeseen consequences of innovation • Distribution of responsibility	• Working in foreign settings • Global supply chains	• Impact on users, local communities, and the environment • Maintenance of infrastructure to avoid disasters • Project failure responsibility	
		- Walkie Talkie Skyscraper London	○ Workers' Rights on Saadiyat Island Abu Dhabi	- Morandi bridge - L'Ambiance Plaza	- Energy site redevelopment in Italy
Sustainability	• Circular economy • Obsolescence	• Design with sustainability criteria • Sustainable use of materials • Ethically sourced materials • Minimisation of waste • Energy efficiency			
		- Dutch-style cycling lanes in Manchester - Sanergy, Kenya	- Green concrete and BioMason	○ Powerhouse Kjørbo - Shanghai Tower	- Dam Removal Europe

	Key Concepts	*Design and Planning*	*Development and Construction*	*Operation and Maintenance*	*Decommissioning*
Health and Safety	• Risk–Benefit Analysis	• Safety issues consideration in early stages • Calculating risk - I-35W Bridge Collapse - Grenfell Tower	• Workers' safety and protection - Panama Canal and contract/foreign workers	• Users' safety • Preventative and corrective maintenance - Secret repairs to Citicorp Building	• Hazardous materials ◦ La Scala Opera House Asbestos deaths
Equality, diversity and inclusion (EDI)	• Inclusive design • An inclusive profession	• User-centered design ◦ Trans-inclusive sanitation in Nepal		• Considering socio-historic contexts • Energy poverty • Universal access to water - Leeds City Council TIBB Project	• Impact on communities - Demolition as urban policy in the American Rust Belt
Decolonization	• Colonialism and neo-colonialism	• Working in foreign settings • Global power imbalance - Water access for nomadic pastoral communities	• Resource extraction • Modern slavery and global exploitation	• Considering socio-historic contexts ◦ The Central Corridor Tanzania	• Waste and environmental racism - Construction waste sent to landfills abroad

of National Engineering Associations (FEANI), the ethical code of the Federation of African Engineering Organizations (FAEO), the guidelines for ethical conduct of JSCE (Japan's society), and the Code of Ethics of Engineers Australia, share common themes of promoting responsibility to society and the environment. These examples illustrate that professional associations around the world are recognizing the need for civil engineers to be responsible stewards of the built and natural environment and to act with integrity and social awareness in their professional practice.

Therefore, the training of civil engineering graduates must include not only technical knowledge but also the development of ethics skills essential for their professional practice. In addition to being excellent technicians, civil engineers need the ability to lead, influence, integrate, and weigh up the various social issues that underpin optimal approaches regarding planning, design, and construction. Engineers can more effectively navigate complex ethical dilemmas by cultivating ethics skills and assessing and balancing social, cultural, and philosophical aspects while making decisions. If students do not learn to appreciate elements of science such as its history, its relationship to culture, religion, differing worldviews, commerce – and its philosophical, epistemological, ontological, and methodological assumptions – then the vast opportunity for science and engineering to enrich culture and human lives cannot be fully realized (Matthews, 1994).

It is therefore essential to ask ourselves what role universities play in training twenty-first-century civil engineers and what civil engineering model to use. Universities are responsible for cultivating a comprehensive educational experience that extends beyond technical knowledge. They should aim to equip future civil engineers with a broad skill set, including critical thinking, ethical reasoning, interdisciplinary collaboration, and a holistic understanding of their work's societal and environmental impacts. By incorporating these elements into their curriculum, universities can foster a new generation of civil engineers who are technically proficient, ethically conscious, socially responsible, and capable of addressing the complex challenges of our time. In doing so, universities can play a pivotal role in shaping the civil engineering profession and ensuring its alignment with the needs and aspirations of society.

Key ethical considerations for civil engineers

Professionalism

Civil engineers are part of a profession; their first and foremost responsibility is to the public good. The profession is regulated through several relevant professional bodies. They have a key role in supporting and guiding engineers to help them abide by professional and ethical standards. This kind of association serves several purposes, such as providing career support for members (Grigg, 1998), helping steer the direction of education (Hildreth & Gehrig, 2010), and offering a platform for exchanging ideas. These associations exist for most engineering disciplines, as in civil engineering.

Crucially, professional bodies lay out the professional responsibilities of their members in relevant codes of ethics (for more on professional organizations and codes of ethics, see Chapter 5). Professional ethics is essential for appropriately governing behavior among professionals of different disciplines. It differs from personal morality, which refers to those standards of conduct that apply to everyone in society rather than to members of a particular group. Instead, professional ethics are those standards of conduct that all members of a profession should follow (Harris et al., 2014). As such, civil engineering ethics applies primarily to civil engineers. Although the separation between professional and personal responsibility is commonly accepted, other authors believe that professionals must acknowledge personal responsibility for the consequences of professional conduct (Christians & Nordenstreng, 2004). This perspective highlights the importance of considering the ethical implications of professional actions on a personal level.

The construction of civil works usually involves large projects in which substantial amounts of public money are invested, and social agents, public administrations, and private companies are involved. For this reason, cases where the engineers in charge lack professionalism tend to receive much media attention. Such lack of professionalism is widespread during the work's planning, design, development, and construction phases. It ranges from bribes to influence the location of a particular infrastructure to the construction of unnecessary engineering works that allows construction companies to enrich themselves at the expense of public money (see the Maharashtra irrigation scam for a real-life example of this) to the choice of suppliers who are in some way linked to people who are to be favored financially.

Codes of conduct in civil engineering identify and describe various professional responsibilities, such as declaring conflicts of interest, maintaining client confidentiality, refusing bribery, and reporting corruption (ICE, 2022). These codes primarily prioritize the safety, health, and welfare of the public, along with values like honesty and objectivity (Zhang & Wang, 2018).

The responsibilities that are laid out by professional associations, however, cannot be adopted unthinkingly. Codes of conduct can only provide general advice as a starting point for engineers. For example, they do not guide what should be done when two or more professional responsibilities clash. Instead, they require engineers to interpret them (Davis, 1991). Common conflicts of duties that may arise are, for example, having to weigh up the duty of confidentiality to one's client with the impact on the public if any safety issues are found during maintenance or with the potential adverse environmental impacts of decisions taken during the decommissioning stage of civil engineering projects.

Case study: National Society of Professional Engineers Board of Ethical Review Case 97-13 (National Society of Professional Engineers Board of Ethical Review, 2008)

A civil engineer is subcontracted to inspect the current state of a bridge as part of a more comprehensive maintenance project. The scope of the engineer's work is restricted to identifying any existing pavement damage and reporting this to the client.

While conducting the work, the engineer notices a defect in the bridge's wall. Not long ago, a car driver had lost control of their vehicle, crashed through a part of the wall not far away from where the engineer is now spotting the defect, fell into the river, and died. The engineer believes that this defect could have contributed to the fatal crash.

Although it is outside of the scope of the work the engineer had been hired for, the engineer puts this detail into their notes and verbally informs the client of their suspicion, who then forwards this information to the public agency overseeing the maintenance project. The public agency contacts the engineer and asks them not to mention this supposition in their final report as it did not fall within the scope of their work. The engineer agrees to keep their notes but not to include the information in the final report. They do not raise their suspicion with anyone else.

Discussion activity: This case presents a clash between client confidentiality and public safety. *Do you think that the engineer was correct in retaining the information in their notes but not to include them in the final report? Do you think they were correct in not raising this information with anyone else?*

Social responsibility

Civil engineers bear a primary duty to serve the public good. One way this social responsibility plays out during the design and planning process is that they must scrutinize the potential conse-

quences of their innovations for society and the environment. *But what happens when unintended negative consequences occur despite their best efforts because of their innovation?* It is helpful here to distinguish between unforeseen and unforeseeable harms. An unforeseen harm is a negative consequence that the engineer neither anticipated nor intended. Yet, in principle, an unforeseen harm could have been envisioned if, for example, more detailed work and consideration had gone into the design and planning process. In contrast, unforeseeable harms are those in which the engineer would not reasonably be expected to have known that a particular harm could occur as a result of their design, for example, because the necessary scientific knowledge was not available at the time (Sucklin et al., 2021). We tend to hold civil engineers responsible for unintended but foreseeable consequences. In contrast, we tend to resist doing the same in cases where the consequences were unforeseeable during the planning process.

This is further complicated when we consider that civil engineering projects tend to require input from professionals from various disciplines, such as architecture or project management. This brings benefits as teams are more interdisciplinary, but also raises questions regarding the distribution of tasks and accountabilities in areas where responsibilities overlap. If a project fails, civil engineers may be held responsible for any design or construction errors that contributed to the failure. This responsibility may be legal or ethical, and civil engineers may face consequences such as lawsuits, loss of professional licenses, or damage to their professional reputation. However, in cases where failures result from a range of factors, including inadequate project management, lack of resources, changes in project requirements, or unforeseen environmental or economic conditions, responsibility may be shared among multiple stakeholders, and a thorough investigation may be required to determine the causes of the failure and the parties responsible.

During the development and construction phase, a particularly important aspect to consider is the relationship with the local inhabitants of the areas where the construction is carried out. This includes the contracts made with and treatment of local suppliers and the workers who carry out the construction, especially in countries that do not have strong protection for workers' rights. It is common for large engineering projects to be carried out by multinational companies that are based in the Global North. It is not permissible for these companies to exploit a lack of legal regulations and protections when contracting and negotiating the working conditions of local workers or when purchasing anything from local producers. A vital framework in this context is that of corporate social responsibility (CSR), according to which companies should integrate environmental and social considerations into their business operations and interactions with their stakeholders. According to stakeholder theory, businesses have duties not only to their shareholders but also to anyone who may be affected by their business, such as employees, users, local communities, and the environment. As such, CSR activities are seen as ethically justifiable, and indeed required, even when they do not directly maximize profits (Goodpaster, 1991).

Operation and maintenance also represent crucial stages in the life cycle of infrastructure in terms of social responsibility. During the operation phase, civil engineers are responsible for ensuring the efficiency and safety of the functioning of the infrastructure. For this, regular inspections, maintenance activities, and addressing any operational issues that may arise are fundamental.

In the last stages of the life cycle of infrastructure, civil engineers are responsible for ensuring that these structures are decommissioned or dismantled in a socially and environmentally responsible manner when they reach their serviceability limit. This includes minimizing the risk to human health and the environment, ensuring that materials and components of the infrastructure are disposed of or recycled in an environmentally responsible manner, and ensuring that the decommissioning or dismantling process does not cause unnecessary disruption to the local community.

Case study: Saadiyat Island Abu Dhabi and worker rights

Saadiyat Island, Abu Dhabi, hosts prominent cultural institutions like the Louvre Abu Dhabi and the Guggenheim Abu Dhabi. However, concerns arose regarding working conditions and workers' rights during its development (Human Rights Watch, 2009). Reports revealed exploitative conditions for migrant workers: extended work hours, low wages, and poor living conditions. Many workers paid fees to recruiters, leaving them vulnerable and in debt. Their passports were confiscated and their movement was restricted, effectively trapping them.

This situation prompted protests and calls for action, eventually leading cultural institution developers to partner with the International Labour Organization for improved worker conditions and compliance with labor standards. This case underscores the social responsibility of companies, urging them to prevent exploitation and human rights abuses. Civil engineers must promote ethical practices and safeguard worker rights during projects.

Discussion activity: *What are the social responsibilities of companies and organizations involved in large-scale development projects? How can civil engineers ensure that their projects are ethical and socially responsible? What measures can be taken to prevent exploitation and protect worker rights in large-scale development projects?*

Sustainability

Civil engineering is crucial for society's development, but its activities have significant environmental impacts, like resource depletion and greenhouse gas emissions (see also Chapters 6 and 11). As a result, civil engineers bear great responsibility for enhancing infrastructure while safeguarding the environment (Ramírez & Seco, 2012). Most codes of conduct and ethics established by professional associations place high importance on protecting the environment (Byrne, 2012). Zhang and Wang (2018) note that these codes acknowledge environmental protection as the top responsibility of engineers. Ethics and sustainability are closely linked, as sustainability challenges involve various stakeholders with conflicting needs (Curren & Metzger, 2017). Balancing societal demands and ethical considerations in finding solutions can be challenging. Hence, an understanding of ethics can be critical when deciding what techniques or strategies need to be employed (Biedenweg et al., 2013).

Designing and planning projects with sustainability in mind is of the utmost importance for civil engineers. Priorities include minimizing the use of energy, materials, and resources, and embracing circular design principles to reduce waste and regenerate resources. A thorough analysis of the supply chain is also essential to gauge the project's true environmental impact (Engineering Council, 2021). Relevant considerations for civil engineers can include, for example, the suitability of a particular location (e.g., *Would the project interfere with a wildlife corridor?*), the materials to be used (e.g., *Will the building supplies be sourced locally?*), and energy efficiency (e.g., *Does the design use natural light effectively?*). More fundamentally, however, sustainability ought to be a crucial guiding factor when we decide which projects to work on in the first place. Sustainability should guide project selection, favoring initiatives like expanding safe bike lanes over new motorways, even if the latter were designed sustainably.

The construction stage of a project offers opportunities to increase the quality and sustainability of the project; excessive ground disturbance can be avoided, the use of recycled materials and the re-use of resources can be increased, and waste can be appropriately managed while meeting the needs of all project stakeholders from designers to suppliers, workers, and users (Vanegas, 2003). It is important to remember that manufacturing and construction processes typically use non-

renewable resources. To address this, civil engineers must raise the use of recycled or regenerative materials and view demolition waste as a resource for recycling and reintegration into production (here again, Chapter 6 holds relevance).

Sustainability extends beyond these initial stages, remaining vital in the maintenance and operation of engineering projects. The operation and use of buildings is a key source of global greenhouse gas emissions (UN Environment Programme, 2022). Thus, prioritizing energy efficiency in new designs and retrofitting older buildings with energy-saving measures is critical. Notably, there is a distinction between nearly zero-energy buildings (NZEB) with high energy efficiency and a significant share of renewable energy, zero-emissions buildings (ZEB) achieving high energy efficiency entirely from local or on-site renewables, and net-positive buildings that generate more renewable energy than they consume throughout their life cycle (Cole & Fedoruk, 2015).

To enhance building sustainability, typical upgrades include improved insulation, shifting from gas central heating to ground-source heat pumps, and replacing incandescent bulbs with LEDs. When designing energy-efficient new buildings, civil engineers should prioritize natural light through well-placed windows and effective use of solar energy. At the same time, they need to consider the ability to effectively use solar energy to heat the building during colder months while avoiding overheating during warmer ones. Ways of using natural ventilation or the creation of green roofs and facades to aid in this process can be incorporated. Designers (including teams of civil engineers and architects) should also consider to what extent the building can be designed to create the energy it uses, for example, via solar panels.

Sustainability's role isn't confined to the project's initiation, operations, and maintenance but also includes decommissioning. Civil engineers often need to manage substantial volumes of construction and demolition waste; these rank among the most significant waste sources.

In the context of sustainability and ethics, circular economy practices aim to minimize waste and optimize resource utilization. The circular economy is based on two cycles: the technological cycle and the biological cycle (McDonough & Braungart, 2009). In the technological cycle, materials are repaired, re-used, re-purposed, and recycled; in the biological cycle, any nutrients are returned to nature. With the introduction of circular economy practices, decommissioning is no longer restricted to demolition but to any other process aimed at recirculating the material of the decommissioned infrastructure (Ellen McArthur Foundation, n.d.). Yet it is worth noting that critics of the circular economy argue that, no matter how circular our economy is, unless we stop aiming for constant economic growth, we will always exceed planetary boundaries (Hickel, 2020).

Infrastructure obsolescence is a critical factor to consider in sustainability efforts. Obsolescence occurs when an infrastructure component or system is no longer effective or efficient in meeting current demands. Civil engineers can take several approaches to confront obsolescence, including planning for obsolescence by using materials that can be easily dismantled, re-used, recycled, or returned to the natural environment and planning to delay obsolescence by carefully selecting materials – encouraging a shift from a use-and-throw-away mindset to one that prioritizes re-use and repair (Lawlor, 2015). Civil engineers can limit the effects of infrastructure obsolescence by incorporating long-term planning and life-cycle cost analysis into their work.

Case study: Powerhouse Kjørbo

Between 2013 and 2014, two 1980s-era office buildings on Oslo's outskirts underwent complete renovation, resulting in Powerhouse Kjørbo (UNFCCC, n.d.). This project became one of the

world's first renovations to generate more renewable energy than it consumes throughout its lifetime (encompassing construction, materials, maintenance, operation, and eventual demolition).

Preserving the old structure while enhancing insulation, cladding, and windows maximized energy efficiency. Solar energy generation was enabled via photovoltaic systems, and old building materials were repurposed, transforming external glass panels into interior office walls. Ground-source heat pumps are used to supply energy. Furthermore, the renovation encourages sustainable behaviors by providing bike storage and showers to promote cycling to work, and parking spaces equipped for electric vehicles.

Class activity: In groups, select a campus building and plan upgrades to achieve a desired energy efficiency level.

Health and safety

Health and safety are paramount in civil engineering projects throughout their life cycle. During the planning phase, engineers must carefully assess risks and benefits. This includes considering the nature of the potential harm, the affected parties' awareness and consent, the project's overall value, the availability of alternative and less risky approaches, and opportunities to take reasonable precautions to mitigate the potential harm. Engineers often face trade-offs between cost-effectiveness and safety, but they must prioritize ethical concerns alongside economic considerations (Toole, 2007).

During the construction phase, risks exist for both workers and local communities. Although zero risk is unattainable, most accidents are preventable. Occupational risk prevention should be integrated into projects. Strict adherence to health and safety measures and the provision of personal protective equipment are essential. Neglecting safety regulations or failing to provide necessary measures is unethical and may be illegal. Neglecting workers' well-being, including working hours and rest periods, can lead to exhaustion-related accidents. For instance, during Qatar's 2020 World Cup construction, over 38,000 workers suffered accidents, resulting in 37 deaths (Re & Tunon, 2021).

Ensuring occupational health and safety during operation and use must be a top priority (Lukhele et al., 2023). Given the longevity of these projects, continuous maintenance plays a vital role in sustaining user safety. Maintenance encompasses preventive and corrective aspects.

Preventive maintenance aims to anticipate and address issues before they arise, typically on a scheduled basis. It involves creating a maintenance plan that factors in the product's expected lifespan, material degradation rates, downtime costs, resource availability, and user safety. The goal is to maintain reliability and safety without wasting time or resources. Corrective maintenance addresses specific problems that arise during regular maintenance or in response to incidents (Stenström et al., 2016). Insufficient or infrequent preventive maintenance often leads to such issues. However, routine maintenance may not always revisit initial design choices, limiting maintenance personnel's ability to detect health and safety issues that are rooted in design flaws.

At the end of an infrastructure's life cycle, whether dismantling for re-use or demolition, potential associated risks must be considered. Decommissioning can involve handling hazardous materials like asbestos, lead, or radioactive substances, posing health risks to workers and the public. Civil engineers must ensure safe decommissioning in compliance with health and safety regulations. Environmental impacts, such as proper waste disposal and site remediation, also require attention.

Worker safety during decommissioning is critical, involving proper training, equipment, and safety protocols for handling hazardous materials. Long-term community effects should not be

overlooked, hazards should be effectively communicated to the public, and measures should be put in place to mitigate potential health risks.

Case study: La Scala Opera House asbestos deaths

In 2001, six retired workers from La Scala Opera House in Milan, Italy, died due to asbestos exposure (ACTS FACTS, 2016). Their exposure occurred while handling insulation materials containing asbestos fibers in the theatre's workshops, highlighting the grave risks of working in neglected older structures.

Asbestos, once valued for its insulation and fire-resistance properties, later proved detrimental to human health – causing lung cancer, asbestosis, and mesothelioma. Despite Italy's 1992 ban on asbestos, many older buildings still harbor asbestos-containing materials (ACMs). La Scala Opera House failed to safeguard its workers, failing to provide protective gear, professional removal, and education regarding health risks.

Discussion activity: *What are the potential health and safety risks associated with working in older buildings? What measures can civil engineers take to protect workers and occupants from hazardous materials like asbestos? How can communication and training be improved to raise awareness of health and safety risks in the workplace?* Class discussion of this topic can benefit from consideration of individual and collective dimensions of ethical decision-making in engineering as presented in Chapter 3.

Equality, diversity, and inclusion (EDI)

Although infrastructure is considered essential to satisfy the most basic human needs, civil engineering has inherently incorporated biases in its processes and methodologies, which have contributed to injustices. Civil engineers, therefore, must acknowledge how their projects impact people with different characteristics such as gender, age, socio-economic status, health status (Lowe et al., 2015), or location (Maswime, 2021). Not considering these characteristics can further perpetuate existing vulnerabilities (Field et al., 2022; Hao et al., 2023).

A civil engineer's responsibility to "promote equality, diversity and inclusion" (Royal Academy of Engineering and Engineering Council, 2017, p. 1) clearly applies to how they design and plan projects. Structures and products should be accessible to as many people as possible and ought not to discriminate unjustifiably based on characteristics such as the ones mentioned above. The primary principle that civil engineers should follow to achieve this, is to put users' needs at the center of their design (for more on this please see Chapter 12 on the foundations of engineering design). The planning process should start with considering who the users will be, identifying their different characteristics and needs, and embracing this diversity. Sometimes, this means that design solutions must offer users different options to engage with a product or space. The more flexibility and the ability to adapt to people's needs is built into the design, the more inclusive the final product will be. It is important that inclusion for civil engineers is not just about physical access to spaces but also includes emotional and intellectual factors such as, for example, whether a space is signposted and enjoyable to use. Furthermore, design should not only be inclusive but also avoid hostile features such as, for example, sectioned benches intended to deter rough sleeping.

The need to make inclusive design choices comes up in various civil engineering projects. Examples that readily spring to mind often relate to designs that increase accessibility for disabled users, such as, for example, integrating ramps into building plans to make them wheelchair accessible or incorporating tactile paving in city planning to help blind and visually impaired pedestrians

navigate streets safely. Yet inclusive engineering can also include issues such as, for example, the provision of public toilets in urban planning to address existing gender inequalities (Greed, 2019).

Another factor that impacts the inclusiveness of our design choices regards the makeup of the profession itself. In particular, the number of men practicing civil engineering is still significantly higher than the number of women; for example, in 2022 in the United States, only 17.1% of civil engineers were women (U.S. Bureau of Labor Statistics, 2022). This trend will not be corrected in the coming years since the number of women who access relevant university courses is still lower than that of men; for example, in 2017–2018 in the United States, only 25.9% of civil engineering undergraduate degrees were awarded to women (Roy, 2018).

To better understand just how widespread the idea is that civil engineering is a profession primarily practiced by men, let us briefly consider the well-known case of the Citicorp Center building in Manhattan. It was about to collapse due to a miscalculation during the project phase, which made it weak against forces caused by winds. Luckily, a Princeton student who was analyzing the structure of the building for their studies noticed the miscalculation and alerted the engineers' office in New York. The building was secretly repaired, and the truth only became publicly known, by chance, 20 years later. While this case is often used in ethics classes, it is interesting to note that for a long time, the student's name was not known and that everyone simply assumed he was a man. In fact, the Princeton student who discovered the calculation error and presented it in her undergraduate thesis was a woman: Diane Hartley.

When engineering projects are created, they are inevitably affected by the broader socio-political context at the time. This means that the end product may reflect existing inequalities and biases within the profession. This can happen both implicitly and explicitly. Regarding the former, we tend to design projects with people like us in mind. Even if we do not set out to favor one group at the expense of another, our biases mean that we can inadvertently create projects that do so. In a profession like engineering that historically has been dominated by men, for instance, many projects have not been created with women in mind. We can see this, for example, in urban transport systems, which usually do not account for the gendered dimensions of travel. For example, women have been found to make more non-commuting trips relating to domestic responsibilities and travel more during off-peak hours (Ng & Acker, 2018). Sometimes, however, existing inequalities and biases are used explicitly to justify creating discriminatory projects. There is, for example, a well-documented history of urban planning and infrastructure projects that were influenced by racist attitudes to explicitly serve white communities at the expense of minoritized ethnic groups (Reft et al., 2022). When thinking about the operation and maintenance of existing projects, it is therefore essential for civil engineers to bear the relevant socio-political context in mind and aim to counter unjustified discrimination through appropriate changes and upgrades.

While the end-of-life stage of infrastructure is sometimes seen as an opportunity to develop new infrastructure which might enhance society's quality of life, it can come at the cost of forcing dwellers to move or depriving users of a service that they were previously utilizing (e.g., a community center, a school). Demolition can particularly affect disadvantaged communities. Civil engineers can promote inclusive practice through transparency and stakeholder engagement in decommissioning. A successful example of EDI being effectively included at the project's end-of-life is the Kibera public space project (in Nairobi, Kenya), which involved rehabilitating a former dumping site to turn it into a public space.

Case study: trans-inclusive sanitation

Since 2011, Nepal has recognized a third gender category for those who don't identify as strictly male or female. In 2012, the first gender-inclusive public toilet opened, allowing use by any

gender. A different approach emerged later, featuring gender-segregated toilets with spaces for women, men, and third-gender individuals. Feedback from the third-gender community was positive, as they gained access to appropriate sanitation facilities, reducing the need for open defecation. Having a designated third-gender toilet also raised awareness and acceptance among the public. However, this approach raised concerns among trans individuals. People often assume all trans individuals identify as being third-gender, potentially leading to abuse and violence when trans men and women use gender-specific toilets. Balancing the needs of third-gender and trans individuals remains a challenge in ensuring equitable access to sanitation (Boyce et al., 2018).

Class activity: Consider inclusive design principles in a diversity impact assessment for both gender-inclusive and gender-segregated public toilets in Nepal. Discuss the best way to accommodate the diverse needs of various groups.

Decolonization

The global history of colonialism has shaped many existing engineering projects. During European colonialism, European powers occupied foreign territories to violently extract labor and natural resources for the sole benefit of the colonizers. They portrayed indigenous populations as backward and in need of civilization through European-based science and technology. This has left long-lasting legacies and created power imbalances between the Global North and the Global South, which have been reflected in development policies and practices. The aim of decolonization is to undo these harmful legacies by fundamentally questioning and undoing the unfair privileges held by the Global North and the systems it has created to uphold them (Maldonado-Torres, 2016). In this way, the aim of decolonization goes beyond equality, diversity, and inclusion initiatives; whereas the latter wants the inclusion of marginalized groups in the current system, the former wants to question and undo that very system. For more on this topic, please see Chapter 9 titled 'Sociological, Postcolonial, and Critical Theory Foundations of Engineering Ethics Education.'

Civil engineering is deeply intertwined with the project of colonialism – after all, it was civil engineers who built the ports, roads, and railways that enabled the slave trade and theft of natural resources from colonized countries (Muller, 2018). When engineers are designing and planning their projects, they need to be aware of how they are connected to this wider global context and how they may, even if only inadvertently, further entrench global inequalities.

When planning to work on projects in foreign settings, especially in countries in the Global South, civil engineers must be mindful of their positionality in the broader socio-political context of (neo-)colonialism. For example: *Does the project have local involvement or is it administered through a top-down approach driven by the interests of a multi-national organization? Are local collaborators meaningfully involved in the project's design, or are they relegated to physical or unskilled work? Are local collaborators sufficiently remunerated and given acknowledgment for the work that they are contributing?* More broadly, engineers need to ask themselves to which extent the projects they are designing are buying into unhelpful 'white savior' and 'deficit' narratives in which skilled and educated professionals from the Global North are conceptualized as coming to fix the problems experienced by the Global South (Eichhorn, 2020; Noxolo, 2017).

The influence of colonialism on engineering projects also extends to the construction phase. During colonial times, civil engineers were involved in building infrastructure that facilitated the exploitation of resources and the control of colonized territories. Today, even when working within a domestic context, the reliance on global supply chains means that civil engineering projects will take on global dimensions in which engineers must be cognizant of this historical context of

resource extraction and the potential perpetuation of unfair power dynamics. For example: *Does our design rely on materials created abroad by workers in unsafe conditions who are not paid a fair wage? Or are we planning on using materials from abroad that are causing environmental degradation and endangering local ecosystems and livelihoods?*

Initiating a process of decolonization also relates to the operation and maintenance of already existing structures. To see this, consider how colonizers created a false narrative that presented the building of infrastructure projects such as railways, roads, and ports as a benefit to local populations – a way of integrating them into the European world of enlightenment and progress (Carneiro et al., 2000). While many occupiers left once the former colonies gained independence, the infrastructure projects that they had built remained – and to this day crucially shape aspects of transport and the economy in formerly colonized countries. When working on maintaining and upgrading such structures, it is therefore essential for engineers to consider whether their continued use is sustaining colonial policies and ideas, albeit in a slightly different guise, today.

Many life-cycle stages in civil engineering, especially the decommissioning phase of projects, involve producing large amounts of waste. Here, civil engineers should be particularly aware of how the disposal of industrial waste can contribute to environmental racism by placing disproportionate environmental burdens on indigenous communities and communities of color through the positioning of landfills and toxic waste disposal facilities on their land or in their neighborhoods (Bullard, 1993), or by sending waste produced in the Global North to countries in the Global South (Okafor-Yarwood & Adewumi, 2020). What happens to the waste that civil engineers create can contribute to reinforcing racial disparities and global inequalities.

Case study: The Central Corridor, Tanzania

In the early twentieth century, German colonial rulers constructed a central railway line spanning Tanzania to facilitate the exploitation and export of the nation's natural resources. Today, the Tanzanian government is investing in the Central Corridor, which aligns with this historic railway. The Corridor comprises a transportation network featuring motorways, railways, ports, and pipelines, coupled with an economic focus on industries such as oil, gas, fishing, agriculture, and tourism. The government contends that industrialization along the Corridor will generate jobs and alleviate local poverty.

However, the new Corridor follows the footprint of colonial-era infrastructure, initially built to support German interests in an export-oriented economy. Similarly, the Central Corridor aims to boost Tanzanian exports, integrating the nation into the global market dominated by former colonial powers. While it enhances the movement of goods and capital, it does not necessarily increase the mobility of the rural population as urbanization is actively discouraged in order to maintain a sufficient workforce along the corridor. Thus, the colonial railway line serves as a conduit for carrying colonial policies, albeit with different manifestations, into the present day (Enns & Bersaglio, 2020).

Class activity: Choose a structure in your area with colonial history, describe its past and current use, and analyze whether its present function perpetuates colonial agendas.

Conclusions

This chapter has provided an overview of some ethical issues related to different areas and life-cycle stages that can be included in civil engineering education. However, it is essential to note

that the topic of ethics in civil engineering is vast and complex, and it is impossible to cover everything in a single chapter. Nonetheless, we hope that this chapter has served as a starting point for readers to reflect on the ethical responsibilities that come with the practice of civil engineering and that engineering education should be prepared to address.

We have highlighted the importance of considering the full life cycle of infrastructure projects, from design and planning to development and construction, to operation and maintenance, to decommissioning. We have also discussed some key aspects that should be considered to facilitate determining the ethical responsibilities of individual professionals. It is well worth also considering and actively engaging in discussion regarding the individual and collective dimensions of ethical decision-making in engineering (Chapter 3), because civil engineers need to learn to work together to shape their collective approach to ethics in our constantly changing world, in addition to monitoring their own individual behavior and the behavior of the firms and projects where they operate (see also Chance et al., 2021).

However, it is important to acknowledge that the considerations outlined in this chapter only scratch the surface of a complex and multifaceted topic. Ultimately, the responsible practice of civil engineering requires a deep understanding of the ethical challenges and considerations that arise throughout the project life cycle. We hope that this chapter will inspire readers to continue exploring these issues in greater depth and to approach their work with a commitment to ethical and sustainable engineering practices.

Note

1 More case studies from a variety of engineering disciplines can be found at the following website https://epc.ac.uk/resources/toolkit/ethics-toolkit/ethics-toolkit-case-studies/

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15

ETHICAL ISSUES IN MECHANICAL AND AEROSPACE ENGINEERING

Aaron W. Johnson, Corin L. Bowen, Cristiano Cordeiro Cruz, and Renato Bezerra Rodrigues

Introduction

This chapter discusses ethical issues in the related fields of mechanical and aerospace engineering from a critical perspective. These disciplines use mathematical and scientific principles to design, analyze, and construct mechanical systems. Mechanical engineering, which focuses on a wide variety of mechanical systems, is one of the oldest, most common, and broadest degree programs to be offered by colleges of engineering (American Society for Engineering Education, 2022; Dixit et al., 2017; Grayson, 1980). Mechanical engineers design commercial power-producing and power-using machines such as medical devices, kitchen appliances, industrial robots, and automobiles (US Bureau of Labor Statistics, 2023a). Aerospace engineering covers many of the same concepts but applies them to designing aircraft and spacecraft systems for commercial and military purposes (US Bureau of Labor Statistics, 2023b). As aerospace engineering can be considered a specific focus area within mechanical engineering, we address both fields in this chapter.

Undergraduate mechanical and aerospace engineering curricula cover a wide range of technical concepts such as statics and dynamics, strength of materials, thermodynamics, combustion, fluid mechanics, controls, machine design, and manufacturing. In addition, undergraduate mechanical and aerospace programs also commonly emphasize design processes and professional skills, such as communication, leadership, and ethics. Across the world, the structure of degree programs varies, as does the approach to the integration of professional skills into the curriculum. However, the incorporation of 'ethics,' by some definition, is common globally, particularly in the 23 countries that are signatories of the Washington Accord (International Engineering Alliance, 2021). The various conceptualizations of ethics education generally agree on the necessity of considering interactions between engineering technology and the people who both affect it and are affected by it. However, there are vastly differing approaches to *what* exactly 'ethics' encompasses and *how* ethics is taught – from presentations of case studies of engineering disasters to emphasis on the importance of communication with stakeholders to analyses of the perpetuation of colonization through technology. Accreditation documents, like the Washington Accord, do not address these questions. We argue that ethics education must engage mechanical and aerospace engineering students in deeper conversation to critically analyze the sociotechnical impacts of their fields and

DOI: 10.4324/9781003464259-19

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develop the critical consciousness (Freire, 1970) necessary to build toward social justice. (For more on reflective and dialogical approaches in engineering ethics education, see Chapter 25.)

In this chapter, we deliberately employ a critical, emancipatory framework as framed by activist scholars such as Paulo Freire (Freire, 1970) and bell hooks (hooks, 1994). In doing so, we aim to push against the Western[1] conceptualization of mechanical and aerospace engineering as apolitical, neutral fields (Cech, 2013). This typical framing of engineering ethics focuses on *micro-ethics*, the individual responsibilities of engineers within the established guidelines of their profession, rather than *macro-ethics*, the collective social responsibility of the profession (Herkert, 2005). (For more on individual and collective dimensions of ethical decision-making in engineering, see Chapter 3.) Instead, we argue that developing students' critical consciousness (as per Freire, 1970) should be a primary objective of engineering ethics education. To this end, we address how mechanical and aerospace engineering have historically facilitated oppression and inequity through their embrace of apolitical framings, their avoidance of macro-ethical topics, and their lack of development of critical consciousness in education. Our analyses focus very explicitly on systems of power as we attempt to address root causes of systemic oppression (thus complementing Chapters 8, 9, 35, and 36, which also adopt critical perspectives and provide helpful context beyond the fields of mechanical and aerospace engineering). This approach may be challenging to some readers because it confronts mainstream realities in engineering practice and critically questions the development of engineering identity (Cech, 2013). We encourage the reader to lean into this discomfort, explore some of the literature cited herein, and discuss it with colleagues, peers, and the authors of this chapter to understand this 'unfamiliar' perspective.

To structure this chapter, we present and dissect four topics central to mechanical and aerospace engineering: the military-industrial complex, automation, sustainability efforts, and humanitarian engineering. While these topics involve engineers from many disciplines, they directly relate to positions of employment commonly held by aerospace and mechanical engineers. These four topics involve various socio-political issues that render them useful for students' reflection and discussion within a critical ethics education. They serve as concrete examples that can be taught in mechanical and aerospace engineering courses to help students understand the macro-ethical implications of their work. Furthermore, this type of critical systemic analysis can be applied to other topics in mechanical and aerospace engineering. Critical education on these and other issues is imperative to build toward emancipatory sociotechnical orders[2] – not to undermine existing approaches and perspectives on ethics education, but to add a perspective that is commonly ignored and overlooked in engineering. To close the chapter, we include a discussion of a critical question for the future of the fields: *What does a conceptualization of mechanical and aerospace engineering education committed to social justice look like?* To do this, we highlight examples of mechanical or aerospace engineering practice and education committed to social justice and the empowerment of marginalized communities.

Positionality, frameworks, and approach

We first outline the positionality of the author team, which consists of our social backgrounds and an acknowledgment of their impact on our perspectives on and approach to engineering ethics education. Corin Bowen is a white cisgender woman from a rural community in the United States with a background in structural engineering and mechanics. She works as a tenure-track professor of engineering education research. Cristiano Cordeiro Cruz is a white Brazilian cisgender male with a background in philosophy and engineering who works as a visiting researcher at two Brazilian universities. He is a member of the Brazilian Network of Popular Engineering. Aaron Johnson is

a white, cisgender male from the United States with a background in aerospace engineering who conducts engineering education research as a tenure-track professor in an aerospace department. Renato Bezerra Rodrigues is a white Brazilian cisgender male with a background in engineering and philosophy who is currently pursuing a Ph.D. in engineering education. We all recognize the ways in which we each come from a privileged background, and we are all committed to exploring emancipatory approaches to our work as engineering researchers and teachers. Due to our personal experience with engineering education in the United States, Canada, and Brazil, we primarily analyze engineering and engineering education systems within the context of these settings and our own social positions within them.

In this chapter, we take an emancipatory approach to engineering ethics education. We recognize that the field of engineering and the educational processes within it enable, support, and perpetuate colonial,[3] cis-heteropatriarchal, racist, capitalistic, and militaristic systems (Cech, 2013). We assert that engineering education should prepare students to apply their engineering skills and knowledge not only in the service of mainstream capitalist beneficiaries within the domain of engineering industry, but also in the construction of new lived realities at both individual and community scales (Freire, 1970). In this way, engineering agency can be redirected toward social justice and decolonizing efforts toward equitable societal change (Alvear et al., 2021; J. A. Leydens & Lucena, 2018).

Technology and engineering are not politically or ontologically neutral (Cruz et al., 2024). They support, emulate, and/or create social orders and ways of living, such as the individualistic, consumerist, and Christian ones that are dominant in the Global North. Thus, any change in how we (want to) live and make sense of our lives – both individually and collectively – requires widening or decolonizing engineering practice, supporting knowledge, and research. Alternative framings exist that describe alternative ways of living and being, such as the Amerindian *Buen Vivir* (Acosta & Abarca, 2018; Estermann, 2006) and African *Ubuntu* (Ewuoso & Hall, 2019; Le Grange, 2019), which are centered upon values such as collective wellness, humanity, and care. Alternative framings for engineering and engineering education are capable of leading to the development of different technologies that bring about the construction of such different social arrangements and ways of living. Within engineering fields, if ethics education is to facilitate such emancipatory sociotechnical arrangements, then mechanical and aerospace engineering students in our programs must:

1. Realize that the hegemonic sociotechnical order we live in is constructed and moldable.
2. Be aware of the possibility of alternative sociotechnical arrangements.
3. Be able (to learn how) to construct the technological infrastructure of other possible worlds.

These are not trivial endeavors; they require a complete reframing of both engineering more generally and engineering labor and practice more specifically. As such, considerable dialogue will be needed to initiate these efforts. This chapter aims to contribute to and encourage such dialogue, especially within educational spaces. For this reason, we lean heavily into the domain of macro-ethics and attempt to challenge existing Western norms of acceptable macro-ethical behavior. We also advocate for sociotechnical education in which macro-ethics is integrated into technical engineering coursework rather than compartmentalized as 'separate' from the role of an engineer (e.g., Benham et al., 2021). This helps combat the prevailing engineering culture, which purports engineering to be purely technical, objective, and apolitical (Cech, 2013; Nieusma, 2015).

Sociotechnical issues in mechanical and aerospace engineering

The military-industrial complex

The military-industrial complex refers to the capitalistic enterprise of weapons research, development, and manufacturing that supplies Western imperial governments and military forces, a lucrative endeavor that frequently occurs at the expense of people and communities in the Global South (Christiansen, 2020). The fields of mechanical and aerospace engineering in the Global North are closely tied to the military-industrial complex in terms of financial support, research directions, and career prospects. The US aerospace industry is often referenced monolithically as the 'aerospace and defense' industry (*2023* Aerospace and Defense Industry Outlook, n.d.), conflating commercial and military aerospace technology. The design and optimization of the manufacture of weaponry are also undertaken mainly by mechanical engineers (*Aerospace Engineering*, n.d.; *How to Become a Firearm Engineer*, 2018). Given the career prospects of aerospace and mechanical engineers in Western countries, the relationship between engineering practice and colonial militaries is a common macro-ethical dilemma for students in these fields (Strehl et al., 2023). However, this relationship is not frequently acknowledged nor discussed within educational spaces. Instead, the connection is generally accepted (particularly within aerospace engineering) without critical questioning or discussion by engineers or engineering educators. Thus, we argue the need for aerospace and mechanical engineering programs in Western countries to directly confront the macro-ethical issue of the military-industrial complex in their programs and classrooms through critical questioning, intentional reframing, and dialogue.

Throughout modern history, imperial forces have utilized engineering and technological skills and development to advance the efficiency of weapons in producing mass death and destruction. The success of this agenda has been illustrated by the escalating efficiency of weaponry in terrorizing marginalized populations, typically perpetuated by powerful entities in the Global North at the expense of people and communities in the Global South. The growth and expansion of global capitalism, dominated and controlled by a wealthy elite in Western countries, is inextricably intertwined with ongoing processes of colonialism. Put simply, war is a profitable business. Arms sales from the five biggest 'defense' companies in the world (all of which are located in the United States) totaled over 191 billion US dollars in 2021 (SIPRI, 2021). An economic report in 2012 demonstrated that, within the for-profit corporate model, the defense industry is actually *more* profitable than other areas of industry (Wang & San Miguel, 2012), generating income that is directed to a wealthy elite consisting of corporate owners and shareholders while simultaneously producing weapons that are used to decimate populations in colonized regions. These private 'defense' corporations employ mechanical and aerospace engineers to use their technical training to produce weapons systems. An estimated 30%–60% of engineers employed in the United States work on projects sponsored by military funding sources (Meiksins & Smith, 1996; Papadopoulos & Hable, 2008).

In Western countries, mechanical and aerospace engineering educational spaces and programs fail to challenge this reality. Both *what* we teach and *how* we teach it are tied to military goals, practices, and structures (Lucena, 2011). Military entities and the for-profit corporations that supply them are lucrative sources of income for universities (McLaren & Farahmandpur, 2001). As such, the systemic structures of militarism and capitalism effectively feed on one another to drive forward engineering efforts to enact violence against oppressed peoples, including in spaces explicitly designated for educational purposes. Introducing macro-ethical content into mechanical – and, particularly, aerospace – engineering curricula should directly confront this reality through open acknowledgment and critical discussion.

One recent example that serves as a valuable case study of the macro-ethics of aerospace engineering is Raytheon's production of missiles and other weaponry that have been used by the Saudi Arabian government to commit genocide[4] in Yemen (Bachman, 2019; Wilken & Kane, 2020; Drysdale, 2022). Since 2014, the United States has been supportive of the Saudi-led campaign in Yemen, stemming from conflict between the political interests of the Saudi and Iranian governments. Yemen, the poorest nation in the Arab region, is experiencing ongoing genocide as a result of the conflict, with 233,000 people believed to have died since the start of the conflict (*UN humanitarian office*, 2020). From a macro-ethical perspective, the lived experience of the Yemeni people necessitates critical reflection on the role of the US military-industrial complex in the mass murder of Yemenis. Raytheon has sold billions of US dollars' worth of bombs and missiles to the Saudi Arabian government since the onset of the conflict (LaForgia & Bogdanich, 2020). The company has vehemently denied any wrongdoing, echoing arguments to evade responsibility or blame supported by the prevailing 'apolitical' framing of Western engineering (Cech, 2013). However, from a critical, macro-ethical perspective, engineers at Raytheon would certainly bear part of the responsibility for work that directly results in the murders of tens – if not hundreds – of thousands of innocent people.

Resistance to the military-industrial complex already exists within the body of Western engineering workers, for example, through organized groups such as Science for the People (Rullán, 2021). This movement was formed in the 1960s by students and academics in direct response to research activity that supported the Vietnam War (Wisnioski, 2003). If, throughout their education as mechanical and aerospace engineers, more students become exposed to the critical issue of militarism in engineering and have the opportunity to see resistance work as a possible *part* of being an engineer, this would open up new corridors for organized masses of engineers to radically transform the worlds they exist within, rather than solely using their engineering skills and knowledge to supply the violence of a continued militaristic and capitalistic world order.

Automation

Since the Industrial Revolution of the late eighteenth century, mechanical engineers have designed and built mechanical devices – and later computers – to fully or partially complete tasks without human assistance. Whatever the form, mechanical and aerospace engineers have and will continue to play a large role in developing automated machines like industrial robots or flight management systems. Thus, our educational programs are tasked with presenting to students which considerations and whose voices are worthy of influence within the design process.

A common argument supporting automation is that it has a net benefit for society by increasing safety, efficiency, and the amount of time humans have for leisure. However, history has shown that automation changes the nature of work rather than removing work (Autor et al., 2020; Bainbridge, 1983) and leads to deskilling (the loss of manual skill) in the long term (Feenberg, 2009; Wiener & Curry, 1980), while the benefits of automation are not equitably distributed across society, contributing to and widening the impacts of existing power relations.

As a case study of the societal-level impacts of automation, we look at the introduction of the mechanical tomato harvester in California, the United States, in the 1960s (Winner, 1980). Between 1961 and 1967, the number of harvesters in use grew from 25 to 1,000 (Schmitz & Seckler, 1970), and today nearly all tomatoes grown in the United States for processed food are harvested mechanically. While the tomato harvester was a technical success in the mechanical engineering and agricultural science fields, it significantly impacted agricultural labor in California. The machine displaced nearly a half-million person-hours of work in 1965 alone

(Schmitz & Seckler, 1970), and an estimated 32,000 jobs were lost by the late 1970s (de la Peña, 2013). Most of these jobs were previously filled by Mexican-American workers, who reasonably viewed the harvester as yet another manifestation of racial capitalism. The harvester also led to a consolidation of the tomato-growing industry, in which the number of tomato growers decreased by approximately 85% after the machine's introduction, with smaller farms closing because they did not have the land nor resources to make a profit given the steep price of a mechanical harvester (Carlisle-Cummins, 2015; de la Peña, 2013). This consolidated wealth into the hands of a few powerful owners and led to disproportionate effects on small, rural farming communities.

The tomato harvester is a prominent case study in Winner's (1980) foundational article on the politics of artifacts. Winner does not claim that the mechanical tomato harvester was a plot to benefit large private corporations and hurt small farmers and agricultural workers. Yet, he argues that the mechanical harvester had inherent politics that favored specific social interests (private corporations) while harming others (farm laborers and small farmers). Winner also points to mechanization projects that *were* "almost conspiratorial" in their intentions (Winner, 1980, p. 125). In the mid-1880s, the McCormick Harvesting Machine Company in Chicago spent much money on new pneumatic molding machines. These machines did not simply increase manufacturing efficiency; they also concentrated power with the factory owners by weeding out the skilled workers who were engaged in labor organizing and replacing them with unskilled workers to operate the molding machines. As Winner writes, "After three years of use the machines were, in fact, abandoned, but by that time they had served their purpose – the destruction of the union" (Winner, 1980, p. 125).

Labor automation will continue to displace jobs and political power with inequitable results, concentrating more wealth within a very small subset of people who form the capitalist elite. For example, autonomous vehicles will put more than three million commercial drivers out of work, many of whom have less education (Autor et al., 2020). One common counter-argument to these large-scale job losses is that automation also creates new jobs. However, as the Massachusetts Institute of Technology (MIT) *Future of Work* report points out, the benefits of automation are only achieved equitably in our existing capitalist framework if governments make large-scale investments to retrain displaced workers for new jobs and support communities built around disappearing industries (Autor et al., 2020). This support has rarely materialized as automation has increased.

Alternatively, another solution to the inequity of automation is an emancipatory approach in which communities work together with engineers to determine the automation that *they themselves* want and need. In this way, the focus is not only on the technical aspects of automation; instead, it is on the sociotechnical impacts and emancipatory power of automation to elevate the lives of everyone. *What might it look like to provide structural power to marginalized communities so they can access the benefits of automation without incurring the harms? And what role do engineers play in this process?*

Creating automation technology is a prominent aspect of many mechanical and aerospace engineers' careers. Therefore, students in these fields need to understand the full impact that these technologies have on people and systems. This requires reflection on one's own positionality and power. Once students have engaged in this self-reflection, they can work to understand the perspectives of others – particularly of those with less power who are significantly impacted by automation technology, such as the farm workers in the case of the tomato harvester. Students can learn how automation affects minoritized people and society through critical discussions and investigation of historical case studies and current events (the teaching methods chapters on

teaching ethics using case studies, Chapter 20, and reflective and dialogical approaches in engineering ethics education, Chapter 25, provide detailed explanations regarding how to use these approaches).

Sustainability efforts

The choices we have made about technological development, use, and disposal in our recent history have impacted planetary systems to a point where the environmental conditions that enable human survival have become a primary societal concern. It is particularly critical now that future mechanical and aerospace engineers be made aware of the potential impact of their work on society and the environment. Sustainability efforts focused on reducing emissions from automobiles and airplanes are a prominent topic in these disciplines, but this is a narrow focus. All phases of a technology life cycle can impact the environment; sustainable engineering design must consider them all while acknowledging that each phase has its own set of environmental impacts and requires specific sustainable approaches and decisions.

In mainstream engineering terms, sustainable engineering involves improved efficiency, optimization of resources, and disposal. However, it is equally important to encourage students to examine the purpose of the use of the technology and think critically about why the technology exists in the first place and whom it serves functionally, financially, and politically. A specific technology's production, use, and disposal can be considered 'sustainable' according to engineering professional standards by using recyclable materials, applying zero-emission manufacturing processes, and/or being decomposable. However, if that technology's very existence and purpose foster an unsustainable sociotechnical order, we argue that this type of engineering is unsustainable and unethical. For example, even if machines such as mining excavators, harvesting equipment, and oil drill rigs are developed following all current sustainable processes and standards, we argue that these technologies are unsustainable because their purpose is to enable unsustainable practices and processes. This furthers the degradation of our planetary conditions, which has been shown to disproportionately impact oppressed populations such as those in the Global South (Ngcamu, 2023; *Social Dimensions of Climate Change*, n.d.; Suri, 2023).

Take the commonly used case study of the 2015 Volkswagen emissions scandal (Atiyeh, 2019), in which engineers developed and installed software in diesel engines to cheat emissions tests run by regulatory agencies. Using this software was clearly unsustainable, unethical, and illegal, but this case presents an overly simplistic picture of unethical practices in mechanical and aerospace engineering. It gives the impression that unethical (and unsustainable) behavior is easily identifiable and that simply following codes, laws, and regulations makes engineering practice sustainable and ethical. In the case of vehicle emissions, reducing emissions can certainly decrease negative impacts on the environment and people's health. However, a sole focus on transitioning to low-emission and electric vehicles serves to mask our current unsustainable sociotechnical order. Low-emission and electric automobiles and aircraft tackle only the 'use' stage of technology. The make-to-stock manufacturing of vehicles still requires enormous extraction of natural resources – including the lithium needed for batteries for electric vehicles extracted from Indigenous communities in Africa and South America – and ever-increasing landfills for car disposal.

Additionally, these vehicles still foster a car-centric culture that furthers individualistic and consumerist behavior. A sustainable solution would be to provide a public infrastructure that is both environmentally sustainable and structurally accessible (Marx, 2022). Increasing efficiency and reducing emissions can create a 'less unsustainable' society, but it is far from contributing to a truly sustainable and just sociotechnical order.

Engineering sustainability should not only consider the land; it must also consider people's relationship to land. A prominent example is the effect of 'development' on the livelihood – and existence – of *quilombolas* in Brazil, who are descendants of runaway enslaved African and Indigenous peoples. The *quilombola* communities on Boipeba Island, which is under federal environmental protection, are currently threatened by a mega real-estate tourist development that encompasses approximately 20% of the island and includes a resort, residential properties, a golf course, a harbor, and a small airport (*MPF cobra revogação de autorização para megaempreendimento na Ilha de Boipeba, em Cairu (BA)*, 2023; Uzêda & Sabrina, 2023). Mechanical engineers are implicated in this project through their work on the machinery's design, construction, and use; the project's water distribution systems; its buildings' heating, ventilation, and air-conditioning systems; and the manufacturing of materials and parts necessary to make this endeavor a reality. Similarly, *quilombola* people living near the Alcântara Satellite Launch Center on Brazil's northern Atlantic Coast are threatened by a proposed expansion from a joint Brazilian-United States technology agreement in 2019 that would allow US companies to launch rockets from the facility (Fox, 2021; McCoy & Traiano, 2021). The location is attractive to spaceflight companies because of its proximity to the equator, which reduces the fuel a rocket needs to boost a satellite to orbit. However, this expansion would forcibly displace hundreds of people from their *quilombola* communities for the purpose of increased profits for the capitalist elite.

Employing an emancipatory lens, it becomes irrelevant how efficient the buildings are, how 'green' the concrete is, or how non-polluting the vehicles are, as unsustainable projects are defined by their negative impact on people and the environment.

It is essential that students critically analyze cases that do not have a clear-cut answer to whether they are considered ethical or sustainable. If we want to educate engineers capable of making truly ethical and sustainable decisions, students must start asking questions that go beyond the immediate use of technology and consider what types of sociotechnical orders they enable through their practice. *Are the engineers making a project come to life holding paramount the protection of environment – as professional societies (Engineers Canada, 2016; NSPE, 2019) have agreed upon as an ethical responsibility of engineers? Can it be in any way ethical to consider 'profit' concurrent with 'people' and 'planet' as the 'triple bottom line' of sustainability?* If we understand sustainability as the balance between human societies' 'needs' (mostly wants) and the planetary conditions that enable human survival (and advocate for justice and emancipation), then engineering practice must be revisited. Therefore, we suggest that ethical and sustainability issues presented to mechanical and aerospace engineering students go beyond life-cycle analysis and focus on *how* and *if* the technology promotes sustainable sociotechnical orders.

Humanitarian engineering

As Mitcham and Muñoz (2010) explain, humanitarian engineering is "the artful drawing on science to direct the resources of nature with active compassion to meet the basic needs of all – especially the powerless, poor, or otherwise marginalized" (p. 35). It is rooted in the broader humanitarian movement, focusing on relieving human suffering in situations of natural disasters (e.g., earthquakes, tornadoes/hurricanes/tsunamis, and extended droughts) or other calamities (e.g., wars, epidemics, and human-caused disasters).

Humanitarian engineering – sometimes referred to within educational spaces as 'service learning' – offers an alternative framework for applying mechanical and aerospace technologies for explicitly 'humanitarian' purposes. (How to teach using service-learning and humanitarian engineering education is the dedicated focus of Chapter 23.) Many mechanical and aerospace engineering students

participate in humanitarian engineering projects as a part of their undergraduate curricular or co-curricular experience. At Michigan State University, for example, 'humanitarian projects' within their mechanical engineering department "aim to make daily tasks such as processing food, retrieving water, caring and storing crops, and simplifying other manual tasks to make them more efficient" (Michigan State University Department of Mechanical Engineering, n.d., p. 3). As another example, the Michigan Sustainability Applications for Aerospace Vehicle Engineering program has worked with Air Serv International to develop an uncrewed aircraft explicitly for use in humanitarian missions (*Air Serv and University of Michigan Partner to Improve Aerial Support in Humanitarian Aid*, n.d.).

The most visible perspective within Western engineering education on humanitarian engineering efforts is one that supports these initiatives as attempts to redirect resources to support those in need. However, from an emancipatory lens, the poor and oppressed deserve both the agency to determine what sociotechnological solutions will best serve their communities and the power and resources to enact these solutions. Humanitarian engineering, on the other hand, typically relies on models of charity and philanthropy, in which powerful entities decide on behalf of marginalized groups 'what these people really need' and then decide how much they would like to provide from the position of a savior. This has race, class, and gender connotations, thus fitting into (rather than challenging) the existing oppressive model of engineering practice (Keshavarz, 2020).

The worldwide and heavily resourced organization of Engineers Without Borders (EWB) is one case study through which we can consider the implications of humanitarian engineering. EWB collaborates on "long-lasting and sustainable infrastructure solutions" through more than 370 "community-driven projects" in 40 countries (EWB-USA, n.d., ¶ 3). The most visible perspective on the organization and its efforts is one of sharing resources to make progress toward meeting basic human needs using engineering technology. From a more critical perspective, EWB provides elite students in highly resourced Western institutions with the valuable opportunity to do hands-on work that doesn't often provide meaningful support to the communities in which they work and which people in those communities do not have agency over. Complaints from target communities suggest that projects often inadvertently create additional work for the people and deplete additional resources rather than providing viable support (Thompson et al., 2022). Meanwhile, Western institutions benefit significantly from the positive publicity generated by EWB, especially within the neoliberal capitalist context in which publicity is monetized. The organization itself began to publicly grapple with its problematic positioning through a short report released in 2019 on the dangers of 'voluntourism' (EWB-SCU, n.d.), and its participants and leaders have also published their own critically reflective analyses (Partida, 2019; Thompson et al., 2022).

The complexity of the macro-ethical implications of humanitarian engineering is significant. Existing critiques describe the personal conflict between a desire to enact positive change in the world and awareness of problematic positioning as a savior. A more profound macro-ethical critique that engages directly with marginalized populations will be required to build structural solutions that serve emancipatory purposes. Emancipatory solutions must provide agency to marginalized populations that allows them to set the goals, determine the methods, and decide how to assess the success of engineering projects themselves. This will significantly affect how mechanical and aerospace engineering students participate in service-learning and community engagement projects, as described in the following section.

Mechanical and aerospace engineering education for social justice and community emancipation

We conclude this chapter by articulating how instructors can bring these conversations into mechanical and aerospace engineering classrooms. First, acknowledging the realities of oppres-

sion and power in engineering is a non-trivial step towards ethical and just behavior. Faculty members should make space for students – and themselves – to learn about and consider these issues. Case studies, which prominently feature in micro-ethics education, can also be used for this macro-ethics education in *any* mechanical or aerospace engineering course – especially those usually considered purely 'technical.' Instructors can present their students with articles and information about the examples discussed in this chapter – the sale of United States arms to Saudi Arabia, the mechanical tomato harvester, tourist and spaceport development in Brazil, and voluntourism – and others. Students can use these case studies to learn about minoritized perspectives, reflect on the role of engineering in oppression and emancipation, and engage in constructive dialogue with their peers about how to consider social justice and emancipation in their future engineering careers. For example, Aaron Johnson (an author of this chapter) has designed and implemented one-day dialogue-based macro-ethics lessons for aerospace engineering courses (Benham et al., 2021, 2022; Ennis et al., 2023). These interventions start with a brief lesson introducing students to concepts like positionality, rights-holders, and ethical lenses. Students then read a short issue brief or article about a macro-ethical issue in aerospace engineering, such as the military-industrial complex, orbital debris, or space settlement and resource utilization. Finally, students engage in small- and large-group discussions in which they consider the power of and impact on various stakeholders, also sharing their own perspectives. Importantly, students are encouraged to reflect on their own positionality in relation to the sociotechnical systems they are analyzing.

Beyond just acknowledging and engaging students in dialogues about macro-ethical topics in mechanical and aerospace engineering, it is also important to teach students about ways in which mechanical and aerospace engineering can *empower* people, groups, or communities marginalized by current sociotechnical systems. Empowerment can be defined as the "multi-dimensional social process that helps people gain control over their own lives" (Page & Czuba, 1999, p. 11) by fostering power in people and groups/communities to operate the changes they may want in their own lives, their territories, and in their society. Empowerment is emancipatory when it allows individuals and groups to improve their lives – that is, 'being more fully human' (Freire, 1970) – fight for their rights, or build other possible social realities and/or ways of living, without dwarfing other peoples' rights and search for self-determination.

Ways of using engineering for empowerment range from sociotechnical inclusion (i.e., giving people/groups access to some essential or desired service, such as clean water) to political emancipation (i.e., allowing/encouraging the community to advance its political potentialities and build alliances with other actors that enlarge its potential to enact desirable and agentic changes) (Kleba & Cruz, 2021). Other ways of cultivating a community's empowerment through engineering interventions and designs include acknowledging, cultivating, or sharing cultural differences, qualitative relationships, technical competencies, investigative competencies, social and economic emancipation, and environmental awareness (Kleba & Cruz, 2021). Notably, an emancipatory outcome is usually as dependent on the intervention process as on the sociotechnical product created.

Within the United States, programs and organizations dedicated to using mechanical and aerospace engineering for social justice can serve as examples for students. The research conducted by the Space Enabled Research Group at MIT seeks to advance justice on Earth through applications of space technology. Examples of such research include using satellites to monitor invasive species in Benin (Onuoha, 2019) and developing decision support tools to help local leaders in Indonesia manage coastal flooding (Lombardo et al., 2022). From an advocacy perspective, The JustSpace Alliance (2022) works for more inclusion and ethical consideration in space explora-

tion; in 2022, they campaigned for renaming the James Webb Space Telescope, as James Webb was an administrator in the US Department of State in part responsible for the implementation of government policy to remove employees who were not heterosexual (Prescod-Weinstein et al., 2021; Witze, 2022).

Lastly, there exist frameworks for helping students to intentionally consider social justice and community emancipation in their engineering practice. In Brazil, a movement known as 'popular engineering' (PE) – meant as grassroots engineering – aims to contribute to emancipating marginalized groups and communities through interventions performed by teams of students and teachers linked to a university extension center (Cruz, 2021a). The approach of these extension centers draws on Paulo Freire's emancipatory approach to popular education (Freire, 1970) and on action-research designing/intervention methodology (Greenwood & Levin, 1998) in order to construct – or (re)appropriate – existent technologies alongside (i.e. *with*, rather than *for*) the local community. The devised sociotechnical solutions embody the group's knowledge and values, make sense to the group's members, and can be operated, fixed, and improved upon by the community. Further, these projects take the ideal of solidarity economy and self-management as the intended broader sociotechnical horizon (Cruz, 2021a, 2022).

The construction of a flour mill in a rural community in the northeast of Rio de Janeiro state can serve as a valuable case study of PE for community emancipation. This project was pursued alongside a community of 63 *quilombola* families. The community has lived since 2014 in a settlement for landless rural workers. The community was already accustomed to autonomously organizing its collective work and internal functioning, and it was the community that initially sought support from the Federal University of Rio de Janeiro (UFRJ), Macaé for the mill's construction (Laricchia et al., 2021). In the design and construction of the flour mill, engineering support was provided by undergraduate students and teachers from mechanical, civil, and production engineering programs at UFRJ (Laricchia et al., 2021). Given the specific conditions of the settlement – such as lack of access to electricity – the community and the engineering team decided to use a bicycle to power the cassava shredder. To determine technical aspects of the shredder, the operator was assumed to be a middle-aged woman, since this assumption was realistic for the community's intended use of the mill. To increase safety and promote social interaction, the shredder was designed to allow the two people (the person who feeds it with cassava and the one pedaling it) to work face-to-face (Laricchia et al., 2021). The intentional design of this system enforces conviviality and allows for the inclusion of aged people and people with some disabilities into the productive process. The community determined all these characteristics of the built sociotechnical solution in an agentic manner and, once they were embodied in the designed artifact, these values emulated a sociotechnical reality that supports and strengthens their community (Laricchia et al., 2021).

Engineering practices employing a PE framework seem to align with the framework defined in Western academic spaces of Engineering for Social Justice (J. Leydens et al., 2014). This orientation for engineering practice is committed to fostering social justice and community empowerment through a co-constructed, bottom-up approach to technical design and technology to support realizing marginalized peoples' and communities' imagined alternative social, political, and economic orders. In so doing, it takes engineering itself as something to be widened, demanding expanded definitions of engineering knowledge, leading to alternative ways of conceiving reality and the field of engineering itself. This reframing of mechanical and aerospace engineering fields, then, would impact every step of the educational process, from content delivery and assessment to student projects and career pathways. PE, or Engineering for Social Justice, is an enactment of an alternative conceptualization of engineering.

Concluding remarks

In this chapter, we have employed a critical perspective of engineering ethics to push back against the hegemonic conceptualization of engineering as apolitical and neutral. We showed how mechanical and aerospace engineering have facilitated oppression and inequity and argued that developing students' critical consciousness should be a primary objective of engineering ethics education. The four topics central to mechanical and aerospace engineering presented here – the military-industrial complex, automation, sustainability efforts, and humanitarian engineering – provide examples of how critical perspectives can be applied to analyze the oppressive effects of the fields of mechanical and aerospace engineering and therefore are essential for the education of students in these fields. Educators can engage students in dialogues about these case studies using the summaries and references presented in this chapter. However, these are certainly not the only macro-ethical issues essential to present to mechanical and aerospace engineering students, and we encourage educators to apply a similar critical analysis to other applications that their students will likely contribute to in their engineering careers. Lastly, students need to understand how to use their future positions as engineers to work toward social justice. To help do this, we encourage educators to build on our discussion of mechanical and aerospace engineering education for social justice and community emancipation. Nurturing students' understanding of oppression caused and perpetuated by mainstream mechanical and aerospace engineering – and ways in which these same fields can be reconfigured to help oppressed peoples emancipate themselves – can develop the critical consciousness necessary to empower students to develop and use their skills as decolonial engineers. Such activity can foster a widened and decolonized engineering community capable of supporting the (co-)creation of more just and equal sociotechnical orders.

Notes

1 This chapter uses 'Western' and 'the Global North' interchangeably to signify those countries that engaged in, and benefitted from, colonization at the expense of the Global South (Braff & Nelson, 2022).

2 'Sociotechnical order' is a concept that highlights how 'technology' and 'society' are intertwined, not existing apart from one another (Feenberg, 2010; Karwat et al., 2015). That means that there is no technological reality totally separated from a social order (or, conversely, a social order separated from a technological one) in a way that any change in one of them does require or produce a change in the other. An *emancipatory* sociotechnical order is one in which overcoming oppressions of any kind (economic, racial, religious, sexual, etc.) is actively sought to allow people, individually and collectively, to flourish, or, as Paulo Freire says it, to *be more fully human* (Freire, 1970).

3 Coloniality and the liberation from it, *decolonization*, are meant here as they are defined by the Decolonial Theory. The hegemonic oppression faced worldwide has as its main bases capitalism (and the mainstream politico-military tools to enforce it) – *coloniality of power* – Western technoscience – *coloniality of knowledge* – and Western worldviews, ideals, and values (which articulate Christianity, racism, masculinity, and so on) – *coloniality of being*. Thus, coloniality as a whole is an offspring and continuation of the European colonial domination of the sixteenth to the twentieth centuries, but usually without military occupation. Each coloniality demands and supports the other two, so overcoming any of them requires overcoming all the three. Since engineering is a central player in the coloniality of knowledge, building another possible world (with alternative social arrangements and possibilities of living and making sense of life) demands enlarging the construct of engineering itself. For more on Decolonial Theory, see Chapter 9. For more on decolonizing engineering, see Cruz (2021a, 2021b, 2022).

4 In October 2022, Genocide Watch, a Washington D.C.-based non-governmental organization, referred to the conflict in Yemen as "the world's most severe humanitarian crisis" and declared the conflict an instance of genocide (Drysdale, 2022, p. 1). The fact that many of the deaths are caused by food scarcity, disease, extreme poverty, and the internal displacement crisis (Bachman, 2019; Drysdale, 2022; Wilken & Kane, 2020) – consequences of the violence rather than the violence itself – allows many governments and organizations to avoid characterization of the conflict as a genocide (Bachman, 2019). However, Bachman's

socio-political analysis explains that the Saudi-led coalition (supported by and including Western nations) "is conducting an ongoing campaign of genocide by a 'synchronised attack' on all aspects of life in Yemen" by targeting the medical and economic infrastructure of Yemen's civilian population (Bachman, 2019, p. 1).

There are significant political implications – as well as even more dire possible consequences – of refraining from characterizing the mass death in Yemen as genocide. The same is true in the case of the genocide occurring in Gaza at the time of writing (*Nearly 25,000 Palestinians killed*, 2023). See Bachman (2019) for discussion of the conceptualization of genocide and the application of the term in the case of the Saudi-led Coalition's genocide of the people of Yemen.

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16

ETHICAL ISSUES IN ELECTRONIC AND ELECTRICAL ENGINEERING

Susan M. Lord and John E. Mitchell

Introduction

As is evident throughout this handbook, ethics education in engineering takes many forms. In this chapter, we focus on the discipline of electronic and electrical engineering (E&EE) with the aim to provide examples of various discipline-specific approaches for the classroom. We hope E&EE educators may be inspired to try one of the approaches or develop their own. This, and the other discipline-specific chapters (Chapters 14, 15, 17, and 18 of this handbook) showcase the importance of including options for students that engage their disciplinary interests rather than only generic 'engineering ethics' where some of the examples might not resonate with students. For instance, the examples of failures of bridges or hotels may not seem as relevant to electrical engineering students as examples drawn from consumer electronics. It is important for E&EE students to conceptualize ethics as not merely an academic exercise but something relevant to their lives and professional practices. In addition, engineering ethics is an inherently interdisciplinary field, spanning multiple engineering as well as social sciences disciplines, and including ethics in engineering curricula helps students see the value of knowledge in other disciplines.

In this chapter, after describing our positionality as authors, we then compare ethics education in electronic and electrical engineering to other engineering disciplines to highlight why we believe that ethics has not had as much prominence in E&EE as it has had in some other disciplines. Following this, we discuss the importance of ethics within E&EE education. We consider the literature on ethics specific to electronic and electrical engineering education, including discussions of professional responsibility connected to codes of ethics and E&EE courses. We provide examples, from the literature and our own work, of integrating ethical considerations into specific technical E&EE courses. We demonstrate that although there are ethics-related case studies available, relatively few relate directly to electronic and electrical engineering; those relevant for E&EE mostly cover general engineering considerations or codes of conduct rather than E&EE scenarios. We propose that instructors draw from the teaching models and ethical issues and concerns presented below; various educators and researchers have identified these concerns and used these models to integrate ethical issues within the delivery of technical content. The presented set of E&EE-related concepts and models is not exhaustive; rather, we aimed to identify broad topics and identify some specific problematic examples from E&EE culture that instructors can use to

DOI: 10.4324/9781003464259-20

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facilitate in-class discussions. We believe these can be integrated within the context of technical content delivery.

Positionality

We, the two authors of this chapter, are electrical engineering educators with decades of experience in academia. We have held leadership positions in the IEEE Education Society. Susan Lord is a white cisgender woman with undergraduate and graduate degrees in electrical engineering who is a full professor at a US university focusing on teaching. Her research is in engineering education. Her experiences of marginalization as a woman in E&EE have contributed to her desire to change the culture of E&EE to be more welcoming and inclusive. John Mitchell is a white cisgender man with undergraduate and graduate degrees in electronic and electrical engineering and a full professor at a research-intensive UK university. His career started with research focused on communications systems but has developed to focus mainly on engineering education, particularly curriculum design. Both have been involved in developing integrated programs where technical and transferable skills, such as ethics, are combined within the curriculum. We acknowledge that our positions of privilege have informed our approach to our work, including the writing of this chapter. Thus, our examples are drawn from the published Western literature that we know best.

Ethics in E&EE compared to other disciplines

Fleddermann (2000) stated in the opening of his paper on ethics case studies that "Rarely is electrical technology at the focus of the classic case studies used in engineering ethics courses and textbooks" (p. 284). Our research for this chapter has demonstrated that this continues to be the case two decades later. Although ethics is undoubtedly taught in electrical engineering courses, the case studies tend to be situated outside the discipline or to address professional issues where the setting is electrical and electronic engineering but the E&EE context is not central to the ethical issue at hand (e.g., whistle-blowing in semiconductor manufacturing that could be in any manufacturing process). Although research (e.g., Barry & Whitener, 2011) has suggested that electronic and electrical engineers are reasonably well prepared to handle ethical issues, this may be because ethics classes have typically considered generic, professional ethics rather than issues directly linked to the discipline (Bielefeldt et al., 2018).

We argue that it is important for students to grapple with ethical considerations (a) explicitly relevant to their discipline, (b) integrated into their technical studies, and (c) taught by engineers. To accomplish this, it is crucial to understand why electrical engineering case studies are rare. We hypothesize that while other branches of engineering produce infrastructure that is directly public-facing – civil engineering (buildings), chemical engineers (chemical plants), mechanical engineers (cars and planes) – electronic engineers especially (but also electrical engineers) build components that are within all these engineered systems or products. This degree of separation means that direct ethical considerations related to electrical and electronics engineering are less obvious than those within other engineering disciplines. Of course, this is just a perception – and one that in many areas has never been entirely true and becomes less so with computer systems and electronic control being at the heart of the modern world. This view is encompassed in the growing refrain that modern engineering and modern engineering graduates must be equipped to grapple with the ethical and social aspects of engineering and the technical aspects. To do this effectively, they must perceive ethical considerations as central to the problem-solving process of their discipline, recog-

nizing that a 'good' solution must be ethical, sustainable, and inclusive just as much as technically feasible, manufacturable, and financially viable.

Although many of these ethical considerations will be social or professional in nature, some will be technical and can be quantified with calculations and technical arguments. The next section describes enacted and proposed approaches for bringing ethics into the electrical engineering classroom.

Importance of ethics in E&EE education

For many programs, the teaching of ethics within electronic and electrical engineering is based on codes of conduct or codes of ethics (see Chapter 5). This is unsurprising, as many of these codes are linked to professional bodies – and often to the accrediting bodies that will evaluate the content of courses (see Chapters 32–36 for more on accreditation). As such, these codes provide an interesting starting point for discussing ethics within the electrical engineering curriculum.

In our realm, the best known of these is the IEEE Code of Ethics (IEEE, 2020), which provides a normative ethical framework of both consequentialist (e.g., those relating to health and safety) and deontological (bribery and corruption) positions. These are spelled out in even more detail in examples from the National Society of Professional Engineers (NSPE) Code of Ethics for Engineers (NSPE, n.d.). Chapter 2 identifies and defines a wide range of applicable theoretical frameworks.

Many professional engineering institutions have produced similar statements of ethics and professional values worldwide. However, they typically apply to the engineering profession in general rather than to electronic and electrical engineering specifically. The Association of Computing Machinery (ACM) (2018) emphasizes several issues of particular importance to those in computing systems – a topic that has considerable overlap with E&EE (e.g., issues related to privacy, issues when modifying or retiring systems, and systems that become integrated into the infrastructure of society) – and identifies relevant ethical and professional codes. In the UK, the Institute of Engineering and Technology (IET) (2019), formerly the Institute of Electrical Engineers) also produces a rules-of-conduct document and in it (in keeping with rules of conduct), ethics and professional codes are combined. Codes in the E&EE realm frequently include considerations for upholding the image and reputation of the engineering profession; for example, the IET (2019, p. 3) specifies that "Members shall neither advertise nor write articles (in any medium) for publication in any manner that is derogatory to the Institution or to the dignity of their profession."

Although professional bodies are often considered the bastions of professional standards, they face their own ethical challenges. For example, a lively debate ensued when the IEEE announced that they would ban Huawei scientists as reviewers in their publications in response to legislation within the United States (Mervis, 2019). This highlights that while we usually consider that codes encompass moral binarism, stating clear and delineated positions on professionalism and professional ethics, applying and upholding them is far more complex. As we will explore later in this chapter, some of the most interesting applied moral discussions invoke an element of moral relativism in forming personal ethics. (For more on moral development theories, see Chapter 10. For more on relativism, see Chapter 28 on epistemological development.)

Although the professional bodies oversee frameworks for the engineering profession, it falls to accrediting bodies to distill these codes into required learning of professionally accredited programs. These typically follow the specifications set out by the Washington Accord (International Engineering Alliance, 1989) and are all relatively similar in scope and coverage, as seen in Chapter 32.

Approaches to ethics in E&EE education

Historically, electrical and electronic engineers have been involved in ethics education. Many engineering ethics textbooks have been written by electrical engineers, including those by Martin and Schinzinger (1996; Martin is in philosophy and Schinzinger in E&E engineering), Unger (1994), Fledderman (2008), and Baura (2006). Joseph Herkert, another E&E engineer, collaborated on a review of engineering ethics that included a section on education (Barry & Herkert, 2015). Herkert also edited a volume on *Social, Ethical, and Policy Implications of Engineering* (2000) that drew from work conducted in the IEEE Society for the Social Implications of Technology. The Society publishes the *IEEE Technology and Society* magazine, which often includes work related to ethics (IEEE Technology and Society, n.d.).

Colby and Sullivan (2008) published a review of approaches to engineering ethics education that focused on standalone courses in ethics (taught by engineering or other faculty), discussion of professional responsibility (often tied to codes of ethics), and modules (typically delivered within two or three class periods). Colby and Sullivan based their work on in-person visits to undergraduate programs in mechanical and electrical engineering at universities in the United States. We use Colby and Sullivan's categorization of discussion of professional responsibility tied to codes of ethics, standalone courses, and modules to frame our review of the literature in this section.

Discussion of professional responsibility tied to codes of ethics

In their teaching, faculty members in E&EE have incorporated discussions of professional responsibility tied to codes of ethics (often the IEEE Code of Ethics). They often incorporate these codes when discussing case studies that fit into one class period. This type of discussion, or content presentation, could be facilitated at any level of study; therefore, we provide examples below from Master's courses and (first-, third-, and fourth-year) undergraduate courses.

The literature shows that discussions of professional responsibility may be specifically focused on the IEEE Code of Ethics. In a project-based course titled 'Electric Power Engineering' for Master's students at Chalmers University in Sweden, Ehnberg et al. (2022) helped students tie the IEEE Code of Ethics to their specific projects. Students used the code as a tool to identify ethical dilemmas or risks for their project and then explored ways to avoid these dilemmas in their final reports. To illustrate, in a project that was designing an electrical brake for a wind turbine, students considered the ethical canon of "To hold paramount the safety, health, and welfare of the public, to strive to comply with ethical design and sustainable development practices, to protect the privacy of others, and to disclose promptly factors that might endanger the public or the environment" (Ehnberg et al., 2002, p. 2) from the IEEE Code of Ethics. Students identified a risk/dilemma of the brake system failing in strong winds and described preventive action of making sure to be well aware of the safety issues related to installation; students highlighted these in their report. After this experience, the students reported that they believed the IEEE Code of Ethics was relevant to their projects and future careers. Ehnberg et al. (2022) stress the importance of focusing on discussion with real-world examples. An interesting aspect of this research is that the interventions were done in-person, online, and pre-recorded; all three approaches had similar outcomes in terms of student engagement and responses to questions about the relevance of the codes to their projects, suggesting that instructors have flexibility in how they choose to implement such work.

Some discussions of professional responsibility focus on case studies and use the codes of ethics to suggest ways to analyze or move forward. Clancy et al. (2005) developed a module for one 3-hour laboratory period for a first-year module on circuits at Worcester Polytechnic Institute in the United States. Students examined case studies on engineering ethics drawn from internet

resources (National Institute for Engineering Ethics, n.d.). Case studies were chosen to be relevant and easily understood by first-year students but not to be specifically relevant to E&EE topics. Students learned about and used the IEEE Code of Ethics to suggest action plans. Data collected from focus groups indicated enhanced student awareness of ethical issues. The authors provided useful advice for others interested in incorporating ethics into their classes. They believe that the adoption of the program by other faculty members was made easier due to the availability of an unused laboratory period in the first week of the term. Materials were also provided, and one of the authors assisted the instructor during the first offering of the program.

Ekong (2015) developed a module for teaching ethics that could be incorporated into any E&EE undergraduate course using case studies from electrical and computer engineering practice and exposure to codes of ethics from the IEEE, IEEE Computer Society, Association for Computing Machinery (ACM), and National Society for Professional Engineers (NSPE). Ekong implemented this intervention at Mercer University in the United States in two 50-minute class periods in a third-year microcontroller class with introductions, case studies, assignments, and quizzes. Ekong stressed the importance of having an electrical engineering faculty member lead the instruction because "a student is more likely to relate to the topic, e.g. Engineering Ethics, if that topic is taught by a professor in the student's discipline" (p. 1). Ekong chose E&EE-related case studies because the case studies "in ethics courses are usually taken from mechanical and civil engineering disciplines. Electrical and Computer Engineering (ECE) students may have difficulties relating to these cases" (p. 1). The case studies from NSPE focus on E&EE topics: software security, quality of products (defective chips), copyright (using unlicensed proprietary software to create a new software product), and compliance with ADA guidelines. Students must connect the specific NSPE code of ethics to the case study (NSPE, n.d.). (See Chapter 18 on ethical issues in software engineering for more on this realm.)

Another example of case studies tied to the IEEE Code of Ethics has been implemented in several capstone design courses within electrical engineering. Motivated by the desire to incorporate achievement of ABET (2000) Student Outcomes related to ethical responsibility and contemporary issues, Jiménez et al. (2006) developed a 2-hour module on 'Social and Ethical Implications of Engineering Design,' which aims to help students "reflect on issues and challenges associated with (i) professional integrity, (ii) engineering, industry and social responsibility, and (iii) technological impact, societal and global awareness" (p. 1). This module was incorporated in capstone courses in various areas of electrical and computer engineering taken in the last year of undergraduate study at the University of Puerto Rico-Mayagüez. The 2-hour module included an introduction to ethics, a discussion of professional integrity in engineering, and a consideration of ethical frameworks, including the IEEE Code of Ethics. Each course featured a case study relevant to the topics of that capstone course in the discussion of integrity in engineering. For example, in the 'Communication System Design: Signal Processing' course, the ethics case study centers on digital rights management (DRM) and the tension between consumer rights and the right of companies to protect their digital content. In the 'Communication Systems Design: Circuits and Antennas' course, the ethics case study considers the health hazards of electromagnetic waves – focusing on the high-power radio transmission towers in Cesano, Italy, and their impact on the local community (Hellemans, 2005). Jiménez et al. (2006) reported that students who responded to surveys at the end of the courses were positive about the impact of these modules on their own development. All students said it was very important for 'ethics to be integrated into engineering courses.' Over 80% indicated that the module had a high impact (5/5) on their 'willingness to be guided by ethical principles in professional work' and 'to be alert to and sensitive to ethical problems.' However, only 60% strongly agreed that others would be motivated to act ethically due to this module. In their

comments, students expressed a desire for more opportunities to practice using ethical principles in their electrical engineering designs and curricula.

Standalone courses on ethics taught by E&EE faculty members

Passino (1998) provided an example of using Martin and Schinzinger's (1998) textbook to support the teaching of professional and ethical aspects of E&EE to a class of 120 students at Ohio State University in the United States. This example is primarily an ethics class taught by an E&EE faculty member but not focusing specifically on E&EE topics. The course was intentionally placed in the final year of the curriculum in hopes that most students had already worked on an engineering job and could bring this experience to the discussions. The main topics were "(1) safety and risk with case studies; (2) engineering as social experimentation and its link with design with case studies; and (3) professionalism and organizational issues with case studies" (Passino, 1998, p. 274). The author recommended avoiding:

> spending too much time on ethical theories at the expense of getting the students to debate case studies (engineering students tend to identify much more closely with case studies and become convinced of the importance of the material easier than via ethical theories). It is important to make some connections between technical design issues in engineering and safety, risk, ethical, and professional issues. Certainly, some time should be spent on codes of ethics.
>
> *(Passino, 1998, p. 274)*

At the University of Illinois at Urbana-Champaign in the United States, Michael Loui created an elective course, 'Engineering Ethics,' for third- and fourth-year students studying electrical and computer engineering (Loui, 2005). The course emphasizes "ethical issues in engineering including professionalism, responsibility, confidentiality, conflict of interest, risk and safety, relationships between engineers and managers, loyalty, whistle-blowing, codes of ethics, licensing, and choosing a vocation" (Loui, 2005, p. 384). Loui specifically chose case studies related to E&EE, such as the Bay Area Rapid Transit (BART) case, where technical problems arose from the electronic sensors, electrical signaling, and software controls. The BART case is historically important because it was the first and only time that IEEE filed an amicus brief in support of whistle-blowing engineers (Unger, 1973). Loui found that students benefited from cases of actual incidents and activities that included discussions with diverse perspectives.

Modules integrating ethics in electrical engineering curricula

Colby and Sullivan's recommendations for engineering programs to better prepare students for "the ethical-professional dimensions of their work" (2008, p. 335) include (1) defining ethics and professional responsibility broadly specifically going beyond codes of ethics; (2) integrating with other learning goals; and (3) using active pedagogies, since professional responsibility includes skills and habits in addition to knowledge. Current efforts to incorporate ethics into electrical engineering curricula follow these recommendations; they integrate ethical issues related to electrical engineering into required courses where the ethics are tied to technical learning outcomes using active pedagogies and not directly tied to the IEEE Codes of Ethics. This is consistent with the pioneering work of Donna Riley, who developed a handbook on thermodynamics that provided specific examples of tying social and ethical content to technical content (Riley, 2011).

Below, we summarize such efforts in E&EE for the standard electrical engineering courses of 'Introduction to Circuits' and 'Controls Systems.' We also provide some guidelines for doing this type of work and examples of student responses.

Circuits class

The 'Introduction to Circuits' class is typically the first course that students in electrical engineering that students majoring in EE&E encounter. It is a required course for students in other engineering disciplines as well. It is often taken in the second year of the curriculum. Finding ways to incorporate ethics into this course has powerful implications for students seeing the relevance of ethics to the field of E&EE.

Conflict minerals: Lord et al. (2018) incorporated a module on conflict minerals into the 'Introduction to Circuits' course at the University of San Diego in the United States. Conflict minerals include tantalum, tin, tungsten, and gold mined in areas such as the Democratic Republic of the Congo (DRC) where the money from their production supports armed conflict. The module was designed to connect conflict minerals' ethical implications to capacitors, a typical topic in this course. Learning objectives included:

- Analyzing capacitors as electrical devices
- Defining conflict minerals and describing at least two social issues surrounding them
- Describing where conflict minerals are used
- Describing potential options for engineers concerned with societal implications of conflict minerals

Before the module, students completed calculations about tantalum (Ta) in capacitors and cell phones and identified where Ta is mined. During the in-class module, the instructor defined and introduced some history about conflict minerals. Students discussed conflict minerals and their societal and ethical implications and brainstormed ways to reduce reliance on conflict minerals as engineers. For homework after the module, students were each assigned a well-known company. They researched the company's conflict-minerals policies and presented their findings to classmates in a subsequent class. Students were asked to highlight social implications and concerns about these strategies. Additional modules explored electronics recycling and sustainable innovation (Lord et al., 2018).

*EV batteries and circular economy:*In another module for 'Introduction to Circuits,'students explore electric vehicle (EV) batteries, tying them to the technical topic of voltage dividers which is typically covered in this class and the concept of the circular economy, where products are reused or recycled for as long as possible (Judge et al., 2022; Lord & Finelli, 2023). Learning objectives include:

- Designing a voltage divider for a DC source to illustrate repurposing EV battery packs
- Estimating energy available in end-of-life EV batteries
- Describing societal risks introduced by recycling EV batteries that could be alleviated by applying circular economy principles

Instructional activities include listening to a podcast about the circular economy and answering questions, estimating the energy demand existing end-of-life EV batteries could meet, discussing how the circular economy relates to circuits' concepts and EV batteries, and discussing ways to

use the circular economy to repurpose batteries. For example, batteries no longer suitable for EVs could be used in residential solar energy systems. Exercises for homework include designing a voltage divider to provide a specific output voltage from a repurposed EV battery and exploring the effect of energy degradation on EV battery repurposing. The module helps students explore the ethical issues of the circular economy as an alternative to the traditional economy, how engineering design is connected to the end of product life, and electric vehicles and sustainability.

Lord and Finelli (2023) are working on a US National Science Foundation (NSF) grant to incorporate sociotechnical modules, including ethical considerations, into the 'Introduction to Circuits' course. They will implement the conflict minerals and EV batteries modules in larger classes and develop more modules (Finelli & Lord, 2023). They are recruiting partners who teach 'Introduction to Circuits' in other universities to implement these modules elsewhere.

Controls class

The concept of social justice can be an interesting way to incorporate ethical considerations into electrical engineering courses. Researchers at the Colorado School of Mines in the United States incorporated social justice concepts in an 'Introduction to Feedback Controls' (IFCS) course/module (Johnson et al., 2015; Leydens et al., 2021). The course is for electrical and mechanical engineering students and is taken as a technical elective in students' third or fourth year of undergraduate study. The interventions related to social justice included a guest lecture by a faculty member from social sciences focused on the 'Engineering for Social Justice' (E4SJ) criteria (Leydens & Lucena, 2018) and a reading from Riley's (2008) *Engineering and Social Justice* book on mindsets in engineering. As Johnson et al. (2015) state, "Social justice defies a universal definition, but is related to the vision that people and communities have the right to equality (in various senses), to health, to dignity, and to opportunities" (p. 1). Thus, social justice considerations for engineering often explore ethical questions of impact. The authors provide an example of modern agricultural machinery where:

> Advanced control systems have made crops more affordable. These same systems have reduced the sustainability of the family farm, significantly changing the agricultural lifestyle, which has had far-reaching implications on rural communities and their economies. Thus, for engineering practice, a social justice framework encourages exploration of the following questions: In the short and long-term, from engineering designs, models, and other interventions, who benefits? Who does not benefit? Who suffers?
>
> *(Johnson et al., 2015, p. 2)*

In the controls class, instructors presented what they called 'social justice' examples that connected the controls' topic of resonance to the safety implications of an unbalanced washing machine and issues related to wind energy systems and active prosthetics. Homework problems were rewritten to include social considerations. For example, a problem about a water tank with no context was rewritten to explicitly ask students to balance protecting the pump, ensuring enough water for a village's needs, and not wasting water by overfilling. Using focus groups, the researchers investigated students' responses to incorporating this content in courses where the social justice content was made visible (Group A) compared to the traditional approach where the social justice content is not visible (Group B). Their results indicated that some students valued incorporating ethical topics into this technical class while others did not. Many students focused on social justice as related to "individual ethics or responsibilities but did not recognize the obligations of the broader

group of professional engineers to society" (Leydens et al., 2021, p. 740). The findings highlight the importance of integrating ethical considerations into multiple classes since one class is insufficient to help students learn to deal with the complexity of ethical issues and develop robust ethical knowledge and ideas.

Pedagogical guidelines and student responses

Incorporating ethical issues effectively in electrical engineering classes is challenging for many reasons. The instructor's identity can come into play because those educated with a deep technical focus may not feel prepared to engage in discussions about ethics or venture into challenging topics related to race, gender, and so on. Instructors need some degree of comfort in dealing with ambiguity, which differs from how most technical work is taught (for more on this, see Chapter 28). For example, there is a 'right' answer to calculating the voltage at the middle of a voltage divider but no 'right' answer to the question: *Should I get a new cell phone every year now that I know the impact on people's lives?* Learning these ethical concepts by doing only mathematical problems is impossible. On the other hand, ethical issues can be incorporated using collaborative methods such as discussions and active learning (Colby & Sullivan, 2008; Loui, 2000). (Please see Chapter 25 on reflective and dialogical approaches in engineering ethics education.) It is also important for instructors to follow good pedagogical practices, including having clear learning objectives and assessments (see Chapter 29). Gelles and Lord (2021) proposed a framework for integrating sociotechnical content into engineering classes that might also help teachers consider integrating ethical content, which is sociotechnical. The (Gelles & Lord, 2021) framework involves the following steps:

0. Identify possible sociotechnical collaborators.
1. Identify a salient course topic that has broader social and environmental implications.
2. Identify, add, or update existing course learning objectives and/or the ABET student outcome(s) that this sociotechnical course topic aligns with.
3. Create learning objectives for specific sociotechnical modules.
4. Create modules by designing activities for homework before and/or after class session(s) and class session(s) that integrate technical content and calculations students are familiar with and social and environmental context.
5. Include low stakes assessment for the module (e.g., homework) and consider including sociotechnical questions on exams.
6. Conduct formative assessment and/or engineering education research on sociotechnical modules to get student input and improve module offerings in the future.
7. Refine modules and identify possible sociotechnical collaborators for the next course offering.

In contrast to some efforts to include ethics, which are met by resistance among engineering students, this integrated approach with modules tied to technical content has been shown to be appreciated by students. Students see ethics as 'real world' engineering and beneficial for their learning. For example, after the conflict minerals module, students participated in surveys and interviews (Lord et al., 2018, 2019). In the survey, all students who responded said the topic mattered to them as engineers. In the interviews, several students pointed out that these topics were not typical for engineering classes and that they found them engaging and would like to see more of this.

> *I thought it was a really interesting topic that has larger social consequences. It was cool to get away from the stigma of engineers only worrying about math and showing that engineering is able to have effect in other disciplines.*
>
> *(Lord et al., 2019, p. 5)*

The experience helped some students see connections between their personal and professional lives as ethical engineers.

> *Prior to this class, I did not even know what conflict minerals were or where they were used. The in-class group presentations on this topic were especially relevant because I researched on Samsung, while having a Samsung phone. I learned that non-conflict tantalum and tin are used in the circuitry of my cellphone. In the future, I can apply this knowledge as an engineer (in design) and as a consumer (in purchasing from companies that have specific procedures for dealing with conflict minerals).*
>
> *(Lord et al., 2019, p. 11)*

In the interviews, students indicated that they found modules to be well-integrated into the class. When asked about engineering as a field, every student brought up the modules, emphasizing that they saw the ethical context as important for developing their sense of engineering in the real world and its potential. It did, however, challenge some students' definition of 'engineering' as indicated below:

> *Obviously, we looked at a lot of stuff that wasn't engineering including the conflict minerals, and the Sunshine Box which I thought was really cool. And that was very clearly … I mean it was engineering but at the same time it was very clearly like looking at it from different angles.*
>
> *(Lord et al., 2019, p. 12)*

One measure of the success of these modules is that a student specifically recommended keeping them and even expanding them because they were beneficial for learning as "*real examples of how the things worked"* (Lord et al., 2019, p. 13).

In interviews that Lord and Finelli conducted after the EV battery module, students said that they found the module interesting, impactful, and relevant.

> *I came in hating electrical engineering, like it was just not for me. So I think like actually doing the voltage divider and using that for like sustainability purposes and the circular economy was really cool to like actually be like, okay, the stuff we're learning is like being used for something … I liked that part of it.*
> *… we are a part of the issue if we don't decide to fix it.*
>
> *(Lord & Finelli, 2024, p. 4)*

Opportunities for exploring ethical issues in electronic and electrical engineering

In considering ethics in E&EE, there are many opportunities to explore issues that have not been explored in the literature. In this section, we begin with some issues that could easily be included in a technical class to align ethical discussions with the delivery of electronic and electrical engineering content. As highlighted above, although there are some publications in these areas, they

have not all focused on the pedagogical implementation of discussion of such issues in the electrical engineering classroom. In the subsequent section, we identify topics specific to the culture of electronic and electrical engineering that may provide useful talking points for instructors who are comfortable leading discussions of sensitive ethical issues. Both sections provide instructors with ideas regarding ethical considerations they could use with students. We encourage electrical engineering educators to take up the challenge of doing this important work and explore how to directly connect consideration of ethics with the technical electronic and electrical engineering content they are teaching and the culture in which they are embedded.

Broad issues for future E&EE curricula

This section provides some examples of ethical issues that could be covered in electrical or electronic engineering classes in Table 16.1. For each, we include an example of a potential reading that might be introduced in the class. These are examples for future development, not citations of work reporting curricular interventions.

Ethical issues in the culture of electronic and electrical engineering

In addition to tying to technical issues, we encourage readers to look locally and consider the culture of E&EE and how implicit and explicit aspects of the academic and professional culture can impact students in the E&EE classroom. *What messages do engineering educators send by the language that we use?* Here, we discuss some terms prevalent in E&EE that can adversely impact the sense of belonging for many people, including women and students of color: the language of master/slave, male/female connectors, resistor codes, and the 'Lena' photograph used in digital image processing (an area within electronic engineering). These are opportunities for E&EE instructors to identify their own potential blind spots and explore relevant ethical issues with their students in classes. This is probably more challenging than the sociotechnical integration we call for in the previous section. We suggest a pedagogical approach for having these discussions at the end of this section as well as some important considerations for educators who want to take on the challenge of doing this work to change culture.

Technical jargon

Taheri (2020) explains the importance of language as

> one of the most powerful tools we have as humans that incorporates personal assumptions, social norms, and cultural ideologies. It is therefore important to consider language critically and to watch for biases in usage. Language reflects the world it is used in, but it is also active in maintaining or redesigning that world. It can be a tool of discrimination or of empowerment.
>
> *(Taheri, 2020, p. 151)*

Taheri goes on to review some discriminatory and non-inclusive language in science, technology, engineering, and mathematics (STEM) fields to raise awareness and suggest alternatives. Two of these examples are particularly relevant to electrical engineering and are discussed further here.

Master/slave: Historically in E&E engineering, the phrase 'master/slave' is used for digital designs where one section of software, hardware, or firmware, the 'master,' controls another, the

Table 16.1 Ethical topics that could be integrated into E&EE classes

E&EE Topic/ Class	*Summary of ethical issue*	*Supporting material/ case material*
Electro-magnetics	There have been numerous news reports expressing concerns about the abundance of electromagnetic energy sources, primarily for communication but also power lines, and their effect on human health. Although the risk is low, there is evidence that it is not entirely non-existent. Engineering class discussions might consider what level of risk is acceptable, who should decide, and potentially even link to technical evaluations such as the power from a mobile phone mast at a distance over the power from the phone itself.	Examples – Ghorbani et al. (2018); Reading – Hardell and Sage (2008)
Integrated circuit design	Instructors could structure a case study (Chapter 20) and facilitate reflection and discussion (Chapter 25) around the responsibility to disclose to customers information on flaws in the design of an integrated circuit (IC), the Intel Pentium Processor. Students can be prompted to consider the company's responsibilities and ethical position on releasing potentially damaging details to the users of its products.	Example Fleddermann (2000); Reading Crothers (1994)
Electronic design	Consumer electronics have developed to a position where typically, repair is not possible. If a significant fault occurs, replacement and disposal are the only options. Recently, a global movement backed by legislation on the 'right to repair' has sought to reverse this. The aim is to require manufacturers to design in the possibility for repair where possible. However, doing this is not always easy and may have unintended consequences. Cunningham and Hobbs (2023) may be used for a debate on how far 'right to repair' should go and where it is appropriate.	Reading – Cunningham and Hobbs (2023); O'Reilly (2020)
Off-grid, smart-grids, and renewable energy	The topic of electricity supply is a rich area for discussions that links ethical and sustainability considerations with technical evaluation. Exercises may ask groups to consider energy supply in a remote area, where considerations about the users are as central to the engineering design solution as the technologies available. The design could be supported by energy usage monitoring and discussions of available materials to minimize environmental impact.	Example – Louie (2018)
Image technology and privacy	With the growth of imaging and security technologies come potential privacy issues. Harrington (2014) highlights the level of intimate detail that can be observed. Engineering class discussions could consider how privacy and security could be balanced, and if alternative technologies might be possible to increase security while preserving privacy.	Reading – Harrington (2014)
Weapons systems	Many electronic engineering employment opportunities can be found in the military and defense sector and its supply chains. This is an ideal opportunity for a discussion on personal ethics and how each person must use their own moral compass to determine where they feel comfortable. Some may feel perfectly fine working on autonomous weapon systems, whereas others would refuse this type of work. Chapter 15, on ethical issues in mechanical and aerospace engineering, discusses such dilemmas in detail.	Reading – Hersh (2022)
Wireless systems class	Apple Airtags have a wide range of useful and innocuous applications, yet they have also raised security and privacy concerns. *What responsibility do technology companies have for the unintended uses of their products? Is it on firms to consider the implications and produce 'fixes' that reduce potential harm?*	Reading – Roth et al. (2022)

'slave.' The use of this terminology began in 1904. Eglash (2007) explored the term's history, its relationship to racialized social connotations, and why it is so popular in engineering, and then posed recommendations for alternatives. In 2004, this term was listed by the Global Language Monitor (Reuters, 2004) as one of the most politically incorrect terms of the year. In the wake of the 2020 murder of George Floyd in the United States and the heightened awareness of systemic racism, stronger calls have emerged for the elimination of this terminology including by students (Steele, 2020), EE professionals (Ellis, 2020), and the media (Canales, 2020).

> Santiago Gomez, a graduate student in computer engineering, was so perturbed when he encountered the terminology in a textbook – *Digital Design, 6th Edition* – for his Logic Design course … that he wrote to the textbook's publisher, Pearson, calling for the language to be changed.
>
> *(Steele, 2020, p. 2)*

Alternatives have been proposed and adopted. For example, in 2021, Microchip issued a product change notification that listed "1) Replaced terminology "Master" and "Slave" with "Host" and "Client" respectively" (Microchip, 2021, p. 1). It is interesting that the reason for change stated was "To Improve Manufacturability" (p. 1). Other proposed alternatives include 'leader–follower,' 'primary–secondary,' 'writer–reader,' 'primary–replica,' and 'coordinator–worker' (Taheri, 2020, p. 153).

Danowitz and colleagues (2021) investigated the importance of this terminology for student inclusion and belonging at an undergraduate predominantly white university in the United States. They found that "42% of students surveyed either agree or strongly agree that use of master–slave terminology is problematic, including 100% of female and 100% of African American students, and that the use of the terminology may create conditions to evoke Stereotype Threat" (Danowitz et al., 2021, p. 1). The authors also critiqued the usefulness and accuracy of the terminology for learning.

Male/Female connectors: Traditionally, connectors used throughout E&EE are referred to as 'male' and 'female,' referring to whether they have a plug or a socket. As described by Wikipedia, "the female connector is generally a receptacle that receives and holds the male connector" (Wikipedia Gender of Connectors, n.d., ¶ 1). Wikipedia has a long entry on this topic, which conflates gender and sex, does not critique or suggest problematic aspects of this terminology, and provides detailed descriptions of hermaphroditic connectors and gender changers. This terminology can be seen as reductionist, reducing gender to physical attributes as well as being uncomfortable in terms of its focus on sexual intercourse with the joining of these connectors called 'mating.' This terminology also can be seen as unnecessarily sexualizing and enforcing a heteronormative narrative. Wikipedia lists plug, pin, and prong as options for 'male' connectors, and receptacle, socket, and slot for 'female' connectors (Wikipedia Gender of Connectors, n.d.). Various creative alternatives to 'male' and 'female' have been proposed such as 'worm–apple,' 'pen–cap,' 'bottle–cork,' and 'sword–sheath' (Eveleth, 2015) or 'outie–innie' (Pearlstein, personal communication, 2019).

Resistor codes

Another problematic aspect of the culture of E&EE is the mnemonic code or phrase used for remembering the resistor color code. In 1961, Alan Dundes wrote a paper exploring the richness of mnemonic devices as examples of folklore in various fields where he flatly states the following for 'resistor code reminder' (1961, p. 41). Although the language is horrifying, it is sometimes

used by educators today, and to show what students over the past half century have been exposed to, we quote Dundes' explanation:

> This device gives the resistance in ohms of a coded resistor. The code is: Black (0); Brown (1); Red (2) ; Orange (3); Yellow (4) Green (5); Blue (6); Violet (7); Gray (8); and White (9). 0 1 2 3 4 5 6 7 8 9 (ohms) B B R O Y G B V G W (colors). Thus if a resistor had a red, a green and a black band, the resistance would be 2, 5, and 0, that is, 250 ohms.

The mnemonic device:

> Bad Boys Rape Only (Our) Young Girls But Violet Gives Willingly.
> Black Boys Rape Our Young Girls But Violet Gives Willingly.
> Bad Boys Run Our Young Girls Behind Victory Garden Walls.

There is no commentary in the Dundes (1961) article on the sexist or racist nature of these devices. One example of the impact of such codes is depicted in *Violet Gives Willingly*, a documentary film about a woman "confronting troubling memories of her short-lived career as an electrical engineer" (International Documentary Association, n.d., ¶ 3) in 1974 (Sanford, 2022). As a woman studying E&EE in the 1980s, one of us (Susan) was horrified by the codes she heard as an undergraduate student and did not want to commit them to memory due to the references to misogyny and rape.

Yet Dundes' (1961) mnemonic persists. A recent search we authors conducted for 'resistor color codes' using Quora[1] turned up a range of possibilities including some of the ones above. A more recent website suggests using a less offensive but still gendered pneumonic device "Bright Boys Rave Over Young Girls But Veto Getting Wed" (WikiHow, 2023). A response to the query we ran using ChatGPT in February 2024 on "What are several mnemonics for the resistor color code," provided "Bad Boys Ravage Our Young Girls But Violet Generally Wins" as the first option and the one with "Violent Gives Willingly" fifth.

Recently, the first author of this chapter, Susan, briefly referenced the resistor mnemonics in class, and several students (white women and men of color) stopped her and asked for more explanation. Developing a discussion around this topic could be a rich experience for E&EE students and faculty. However, we recognize that not all educators will have the skill to facilitate such an explicit discussion with undergraduates, but they can work to develop such skills to help E&EE education and culture evolve.

Image processing

Sometimes, what might seem innocuous and commonplace in a technical community may have a more problematic backstory. One such example is the Lena (or Lenna) image. If you have ever looked into work on image processing or image compression standards such as JPEG, you will likely have seen this image – a 512 x 512-pixel image of the face and upper arm of a young woman in a straw hat with a blue feather turning to look over her shoulder towards the camera. The image is of Swedish model Lena Forsén (Söderberg), but what is perhaps not so commonly known is that the complete image (of which a cropped area from the top portion of the origin photograph is the widely used test image) first appeared as a nude centerfold in the November 1972 issue of 'Playboy' magazine. The image was used by the Signal and Image Processing Institute (SIPI) at the University of Southern California for comparative tests between compression algorithms but has now become ubiquitous within the image processing community. It is argued that the widespread use of an image from what many consider to be a pornographic source only serves to rein-

force the underrepresentation of women in computer sciences and engineering (Culnane & Leins, 2019), including by Lena herself, as she stated in the 2019 short film *Losing Lena* (Bartley, 2019).

It is of particular interest how the male-dominated culture of the technology and computing industry has served to not only normalize the use of such an image over many years but continues to do so even to this day. As a journal editor, the second author, John, recently found himself in an altercation with an eminent professor of computer science at a top US institution over his suggestion that alternative images would be far more appropriate. This exposes a potentially interesting debate with students on several fronts, but mainly around why the provenance of such an image matters and how it impacts the drive for diversity and inclusion with engineering and tech. It is now common for journals to advocate for the use of alternative (and arguably technically better images) such as 'Cameraman,' 'Mandril,' or 'Peppers,' and some, such as 'Nature Nanotechnology,' have stated that they will automatically reject submissions using the 'Lena' image (Nature Nanotech, 2019).

In November 2023, decades after the image first appeared, the IEEE officially adopted this policy citing ethical reasons.

> IEEE's diversity statement and supporting policies such as the IEEE Code of Ethics speak to IEEE's commitment to promoting an inclusive and equitable culture that welcomes all. In alignment with this culture and with respect to the wishes of the subject of the image, Lena Forsén, IEEE will no longer accept submitted papers which include the "Lena image".
>
> *(IEEE Author Center Magazines, 2024)*

Moving forward

Hopefully, this offensive language is not taught to students in today's E&EE classrooms. We might prefer not to state language such as the resistor codes at all in hopes that they will die out. Yet, today, the tradition is still alive online, and addressing it as an ethical issue is essential to changing the culture. Discussing such issues with a critical eye as to how they have contributed to and been reflective of the culture of E&EE is difficult but important work. It is worth noting that E&EE continues to have a very low representation of women compared to other disciplines in both the United States (Lord et al., 2015, 2019) and the United Kingdom (Bellingham et al., 2023). Educators need to do more to identify and address cultural factors pushing women and people of color away from E&EE studies and careers. To confront ethical issues in E&EE related to language, belonging, and discrimination, we encourage E&EE educators to reflect on their own language choices. Educators can develop skills in discussing these sensitive topics by educating themselves; talking with colleagues outside of engineering in fields such as sociology or ethnic studies or education; discussing these issues with colleagues, friends, and graduate students; and then introducing them to undergraduate students. We recognize that some dangers exist for raising sensitive topics in class, particularly for students and instructors from vulnerable minoritized groups. Instructors must carefully navigate these discussions, ensuring a safe and inclusive learning environment while addressing the historical and contemporary issues in the field. Chapter 25 on reflective and dialogical teaching approaches may be a helpful resource.

The University of Washington Information Technology group has a website with an Inclusive Language Guide, which contains information regarding problematic terminology in information technology, including 'master–slave,' 'male–female' connectors, and others such as 'blacklist-whitelist' and 'blackhat/whitehat.' This is a good resource for exploring technical terminology and why it can be considered problematic (UW-IT, 2023).

A pedagogical strategy that might be helpful for educators who want to lead discussions of these ethical issues is the confront-address-replace (CAR) framework. According to Asfaw and colleagues,

> The CAR Strategy is meant to be a proactive and modern pedagogy which encourages discussion and thought on whether or not we should replace questionable aspects within engineering. The CAR Strategy does not force students to replace 'master–slave' or any terminology from their vernacular – it simply welcomes it.
>
> *(Asfaw et al., 2021, p. 25)*

Specifically, this CAR strategy involves *confronting* the historical significance of a term such as 'master–slave.' The strategy then moves to *addressing* the inaccuracies of this problematic terminology. Finally, the discussion turns to recommendations for alternatives to *replace* the problematic language, although these are suggestions rather than prescriptions. Researchers have studied the impact of a CAR strategy as a teaching framework in computer engineering at a predominantly white undergraduate university focusing on 'master–slave' terminology (Asfaw et al., 2021). Over two-thirds of the students surveyed in that study agreed that this was an effective framework for eliminating offensive terminology and that they would like to see it used in other engineering classes where it is applicable. The researchers suggest that this strategy could be used for other examples of problematic E&EE terminology.

Conclusions

In this chapter, we have identified the importance and historic scarcity of case studies that are directly and technically relevant to students of electronic and electrical engineering. We have highlighted example topics where instructors can go beyond the typical professional subject of ethics and codes of conduct to introduce ethical issues. We have suggested discussion points and case studies that can be integrated in the core curricula and classes that form electronic and electrical engineering. We have also demonstrated that within electronic and electrical engineering there exists a number of contemporaneous case studies and issues that would make excellent discussion points within the technical classes of an electronic and electrical engineering program. These include sociotechnical cases where ethics trade-offs are a central consideration and issues of the culture and language within electronic and electrical engineering which can be directly linked to engineering content and bring the discussion of ethics into technical classes. We have presented cases that can provide active learning activities to breakup lecture classes by introducing participatory debates on topics directly related to the theoretical concepts under discussion. We consider that this explicit connection of ethical concerns with technical content is vital to engage students and allow them to see that ethics is not an afterthought or an irrelevant and abstract concern but central to good electronic and electrical engineering practice and best practice in electronic and electrical engineering design. We hope that instructors will find these cases useful and will be able to integrate them into their classes.

Acknowledgments

We thank Tom Børsen for inviting us to write this chapter. We acknowledge the work of Erin Henslee and Pinar Denis Tozun, who worked on another draft of this chapter that they were not able to complete. We thank Michael Loui for his careful reading of a draft.

Note

1 https://www.quora.com/Whats-the-best-way-to-remember-resistor-colour-codes

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17

ETHICS IN CHEMICAL ENGINEERING

Jan Mehlich, Tom Børsen, and Dayoung Kim

Introduction

In this chapter, the ethical dimensions of chemical engineering (ChE) activities, including their effects on the environment and human lifeworld, will be addressed. This description aims to identify possible content for ethics education geared toward chemical engineering students. After pointing out the specific and characteristic features of chemical engineering in contrast to other fields of engineering and to academic chemistry, various cases and scenarios of chemical engineering efforts with ethical implications are presented. These are contextualized along the lines of different domains of responsibility. The underpinning idea is that engineering ethics education should prepare chemical engineers to exercise different forms of responsibility in their forthcoming careers and that familiarity with matching content will enable this pedagogical objective (Barry & Herkert, 2015; Bielefeldt et al., 2019; Børsen et al., 2020; Shallcross, 2010; Shallcross & Parkinson, 2006). Presenting historical and contemporary cases will allow the ChE student to understand their quintessence and apply this orientational knowledge to assessing other cases. Since the practical application of engineering ethics competence plays out in discourses in interdisciplinary team assemblages, the chapter presents a short practical guide on the role of chemical engineers in such professional settings. Hopefully, this chapter will inspire educators who design ethics courses or teach ethics in ChE to choose content that ChE students and their future employers find relevant in exercising professional and ethical responsibilities.

The author team includes a female Korean engineering educator with a chemical engineering background now living and working in Virginia (United States), a Caucasian male Danish chemist-turned-techno-anthropologist, and a Caucasian male German chemist-turned-ethicist. All three work as academic scholars/researchers at universities and are primarily trained in Western intellectual traditions in their disciplines, yet with different cultural (Asian and European) upbringings. We acknowledge that these personal backgrounds have undoubtedly shaped the choice of content, perspectives, comments, and conclusions presented in this chapter. For example, all authors are formed by Western ethical frameworks and share a belief that chemical engineers have both epistemic and social responsibilities.

DOI: 10.4324/9781003464259-21
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Chemical engineering and its specific features

The demarcation of ChE against other engineering fields and against the academic scientific discipline of *chemistry* is important for understanding the specific ethical dimensions, and it explains why ChE, with its specificities, deserves its own chapter in this book. ChE graduates employ scientific knowledge of chemical substances, their characteristics and reactions, and their effect on organisms and ecosystems, for the elaboration of large-scale synthesis and production processes, transportation and storage methods and technologies, and various applications for industrial and consumer products. The chemical industry, as the main employer of chemical engineers and as the primary locus of research and development (R&D) with a strong innovative driving force, is an economically significant key industry in most developed countries around the globe. This translation of knowledge about new and synthetic substances and reactions sustains the creative character of the chemical sciences and, more often than not, enters uncharted territory with new developments and designs. Thus, the ethical relevance of *chemical* engineering springs not only from its sheer quantity of impactful output but also from its inherent newness and multi-leveled risk potential.

Chemical engineering and chemistry as an academic science

Chemistry investigates the background of the science encompassing aspects of the organic, inorganic, analytical, physical, and bio-chemistry. ChE is more multidisciplinary and practical, applying engineering science to problems relating to heat transfer, fluid dynamics, equipment design, and so on. Most chemists work in laboratories as research scientists, analytic chemists, pharmacologists, biochemists, toxicologists, or forensics, whereas typical industries for chemical engineers are energy, mining, food and drink, pharmaceuticals, wastewater, pulp, paper, and so on (Edwards & Shelley, 2018).

The ethical dimensions of these two domains' work are different. Since most of the work that chemists do has only little direct impact on people and the environment, codes of professional conduct and good scientific practice are at the center of science ethics. While there are certain codes of good engineering practice that apply to chemical engineers, there are external responsibility dimensions to engineering ethics that concern the impact of engineering products on individuals and societies as well as the ecosphere. The urgency of the societal and environmental impact is, on the one hand, caused by the mere scale of the output. While chemists work in labs with relatively small amounts of chemicals in closed experimental set-ups, most engineering work is done at the industrial scale with processes that involve a large throughput of substances and energy. On the other hand, it arises from the nature of ChE that a large group of stakeholders – people for whom something is *at stake* because of ChE work output – is entangled in ChE compared to academic chemistry. ChE stakeholders include entrepreneurs, managers, workers, regulators, clients and customers, consumers, and civil society. Therefore, the work of chemical engineers can more directly involve macro-ethical issues, which distinguishes it from academic chemistry (where work involves more micro-ethical issues like research ethics; see below for further details). Overall, chemical engineers have more touch points with interdisciplinary perspectives given their work's natural engagement of multiple stakeholders.

Chemical engineering and other engineering domains

Simply put, ChE deals with chemicals while other branches of engineering do not. This fact needs to be reflected in ethics teaching for ChE students. Chemical engineers design and manufacture

materials and products using scientific principles from chemistry, biology, mathematics, and physics. They may come up with innovative processes to use and transform energy. They can also work with microorganisms, food, pharmaceuticals, and fuels. ChE encompasses:

- Biochemical, biomolecular, cellular, and microbial engineering
- Biomedical, pharmaceutical, and tissue engineering
- Biotechnological and genetic engineering
- Food, textile, and paper engineering
- Materials and molecular engineering
- Metallurgical, welding, and corrosion engineering
- Petroleum, plastics, and polymer engineering
- Process engineering

In all these fields, working with substances and materials is central to daily activities (Denn, 2011; Green & Southard, 2018; Shallcross, 2017). Chemical engineers are experts in exploiting knowledge about material properties and characteristics for applications in devices and processes. These materials, their conversion or transformation products, or their waste products impact people, society, and the environment. Given the large scale of industrial processes in which ChE expertise is employed, these impacts can potentially be disastrous. Thus, an increased responsibility for overseeing and controlling the impacts of the material output of professional work may be formulated for ChE.

Given the scope of this chapter, we will not introduce details of each subdiscipline within ChE listed above. Instead, we propose contents and teaching strategies that can be jointly taught across different chemical engineering subdisciplines, focusing on common issues and themes applicable to various subdisciplines.

Cases for teaching ethics in ChE

This section presents prominent and insightful historical and contemporary ethical case studies relevant to chemical engineers. The compiled cases are extracted from different sources: a collection of ethical case studies in a special issue of the *International Journal for Philosophy of Chemistry HYLE* (re-issued in Schummer & Børsen, 2021), engineering ethics textbooks (Bowen, 2014; Harris et al., 2018; Johnson, 2020; Whitbeck, 2011), and searches in the academic journals *Science and Engineering Ethics*, *Science, Technology, & Human Values*, and *Journal for Business Ethics*. The selection of ethical case studies was then analyzed, and those considered most ethically relevant for ChE were selected. These are presented below to illustrate the normative implications of chemical engineering activity and possible solution pathways for conflicts encountered in professional contexts. As each case is linked to different forms of responsibility chemical engineers need to consider, educators can select which cases to use based on their class topic.

Ethical issues related to the unpredictable nature of chemicals

As outlined above, a main characteristic feature of ChE is the production, processing, or application of substances and materials for harvesting their specific properties for certain purposes. Inherently, as ChE innovation brings about new substances and materials, or new conditions under which they are processed, transformed, transported, stored, or recycled, one of the main sources of accidents, problems, and conflicts is the unpredictable nature of chemicals. Chemical accidents

and other adverse effects of ChE products in the environment and society can be attributed to chemical engineers' inability to foresee each and every factor of the complex network of chemicals, facilities, people, and system conditions that affect the safety and operability of larger engineered systems. Adverse effects are not necessarily caused by irresponsible conduct or someone's intentional wrong behavior – they may be regarded as the result of system failure or the unmanageable material complexity of ChE work.

One example is the unexpected side effects of chemical drugs. In 1957, a drug, Thalidomide, was launched in Germany by the medicinal company Grünenthal (Ruthenberg, 2016). It was considered a harmless drug to mitigate pregnant women's morning sickness but was ultimately shown to have serious implications for babies conceived. Thalidomide, when taken during certain stages of pregnancy, could lead to infants being born with limb deformities. Testing for unintended effects over longer terms was not conducted early on, and it was therefore impossible to foresee those outcomes. Adverse effects were not linked back to Thalidomide until the drug had been in use for several years. A lesson of the Thalidomide case is that we cannot expect fully effective but entirely harmless drugs. While modern medical practices now report long-term effects of drugs, establishing direct links between conditions and specific medications remains challenging. Thus, in designing and processing chemicals, potential adverse outcomes of the products need to be seriously considered. In other words, chemical engineers hold a so-called 'external' or 'social responsibility' to deliberately think of the potential outcomes of the products they design and produce. In this case, the implications regarded unintended harmful health effects.

Beyond direct impacts on patients' health, chemical drugs can also have unintended effects on human culture. An example involves psychotropic drugs such as anti-depressive medicines. These chemicals affect emotions, mood, motivation, and behavior. These drugs can help many people with diagnoses such as depression, obsessive-compulsive disorder (OCD), anxiety, post-traumatic stress disorder (PTSD), and other mental illnesses. According to Klavs Birkholm (2016), they also influence our culture if used intensively – by changing our perception of illness and health. Francis Fukuyama and others argue that humans strive for recognition and that psychotropics' effects on the state of mind of healthy humans might challenge this quest (Fukuyama, 2003), lead to passivity and apathy, and create false expectations of constant happiness through the misleading vision of perfect control over one's mental states. These sociocultural implications resemble ethical discussions on human enhancement. They are related to works of science fiction like Huxley's (2010) *Brave New World* and their warning against drugging the population to maintain an inhuman society.

The following case concerns the false assumption that chemical waste can be easily decomposed. From 1942 to 1952, Hooker Electrochemical Company dumped chemical waste at a place called Love Canal, which later developed into a settlement (Fjalland, 2016; Levine, 1982). A school was built close to the waste dump. Inhabitants complained about the odor. Public authorities initiated an investigation to establish whether it was unhealthy to live in Love Canal. They concluded that living at some (specified) distance from the waste dump was safe. This did not align with the inhabitants' registered occurrences of illness. Eventually, it was found that the model used by the authorities was wrongly chosen. Love Canal was declared unhabitable and was abandoned. Lessons learned here include that assuming that chemical waste can be stored safely is not straightforward and public authorities must listen to early warnings from concerned experts and local citizens. The case calls for participatory approaches to uncertainty, and related ethical issues deal with the science–society or expert–laypeople relationships and their respective epistemic authority.

Unexpected consequences of chemicals occur, at times, in proximity to the exposure, for example, to those taking drugs or their offspring or to those living next to a waste facility. However, this is not always the case. Sometimes, the implications are global.

The intensive use of dichlorodiphenyltrichloroethane (DDT) is an illustrative example of how the work of chemical engineers can turn into a global environmental issue (Børsen & Nielsen, 2017; Bouwman, 2013). DDT is part of a cluster of chemicals called persistent organic pollutants (POPs) that are hard to break down, are attracted to and soluble in lipids or fats (lipophilic), and can disperse over long distances. DDT disrupts the hormonal system and is recognized for its detrimental effects on reproductive development. It has toxic effects on internal organs as well as on the neural system. It is suspected to be carcinogenic. It builds up in the food chain because of its long breakdown halftimes. When DDT is used extensively, resistance becomes apparent, leading to the eventual loss of its desired effects. During and after World War II, DDT was used as an agricultural insecticide to control crop pests and as a tool to combat insect-borne diseases such as malaria. Its use rose exponentially until its residue was registered all around the biosphere, even at the poles. Its use in agriculture is now banned and its use to combat malaria is restricted. The ethical debate surrounding DDT revolves around balancing its effectiveness in disease control with its detrimental environmental and health effects.

Other examples where chemicals have caused global environmental pollution are those of Bisphenol-A (Martin et al., 2021; Resnik & Elliott, 2015), other POPs (Godduhn & Duffy, 2003), and PFAS (Kwiatkowski et al., 2020). If chemicals are used on a global scale, global environmental challenges can merge, where everybody on Earth is affected by chemical substances and their unforeseen and undesired implications. When chemical engineering has such powers, responsibility means stewardship for the whole Earth and is driven by a view of humanity as part of the world and the environment (for more on this, see Chapter 6).

Quintessentially, the ethical issues arising due to the unpredictable nature of chemicals as introduced in this section illustrate that chemical engineers' good professional practice does not only mean absence of misconduct but also paying as much attention as possible to what could go wrong and to the validity and legitimacy of factual claims.

Ethical issues related to misuse of chemical engineering

This section describes a different kind of ChE-related problems. Rather than the unpredictability of material effects, the cases in this section have in common that they are the result of irresponsible decision-making in research and innovation contexts that involve ChE and in which chemical engineers are part of a bigger collective of agents including managers, businesspeople, regulators, workers, civil society, and others. In these settings, interests might diverge and corrupt each other. In the following cases, ChE expertise involved activities beyond purely technical or scientific professional conduct.

Military application of chemistry is one area where ChE or ChE-enabled technologies are purportedly misused. There are several illustrative examples of military uses of ChE, including the use of poison gas in World War I, for example, as well as military applications of Napalm and Agent Orange during the Vietnam War. In this context, 'misuse' denotes the deliberate exploitation of chemical substances originally intended for specific applications, such as the textile industry or agriculture, with the intent to cause harm or produce destructive outcomes. This underscores the dual nature inherent in the use of numerous chemical and ChE innovations, as discussed by Tucker (2012). Many questions can be asked concerning the internal and external forces that drive engineers and other actors into unethical behavior:

- *Is it always wrong to develop or use chemical weapons?*
- *What ethical values are in play when assessing chemical weapons?*
- *What are the responsibilities of chemical engineers?*
- *To what extent were the implications of chemical weapons known?*
- *How do the responsibilities of chemical weapons engineers relate to the responsibilities of governmental representatives, politicians, military people, merchants, and the management of the weapons industry?*
- *What forces drive chemical engineers to misuse their knowledge, skills, and competencies?*

An illustrative example of misuse of ChE appears in the Fritz Haber case, which is thoroughly introduced in journal articles (e.g., Schummer, 2018) and books (Charles, 2011; Stoltenberg, 2004). The case literature is extensive. It regards the implementation of ChE in the development of chemical weapons such as chlorine gas and mustard gas that were applied during World War I. The case also illuminates the backdrop of weapons development, shedding light on the interconnected academic-military-industrial-governmental complex. Haber personified all spheres as he was a university professor, a captain in the army, involved in the chemical industry, and appointed to serve in the German war department. The case shows that Haber himself explicitly chose not to follow established ethical values. As described by Schummer (2018):

> First, he argued that he had never cared about the Hague Convention [an international agreement banning gas weapons]. Second, he was convinced that in times of war ethical standards are to be replaced by patriotism, such that warfare engagement becomes a moral duty for scientists … Third, he was fully aware that his weapons program initiated an arms race among the enemies.
>
> *(p.13)*

The details and discussions of the case in the literature show that the development and use of chemical gas weapons cannot be justified by any major ethical framework (see Chapter 2). Discussing the case with students gives them ample opportunity to reflect on the limits of what may be ethically justified and on current practices in honoring influential persons in their field of study.

A related case is that of Agent Orange. Arthur Galston discovered a chemical compound, 2,3,5-triiodobenzoic acid (TIBA), that inhibits the growth of leaves and, instead, increases the number of floral buds on soybean plants. In higher concentrations, though, it leads to defoliation (abscission of leaves) and the death of the plant. Galston learned later that the US army, interested in this chemical's defoliation effects, produced derivates of this compound and used the most powerful ones (known as Agents Orange, White, and Blue) in the Vietnam War, with disastrous effects on the ecosystem, food chains, and aboriginal culture and lifestyle. Galston remained extremely concerned about the implications of his work, not shying away from confrontation with the US government. He became a public voice of science that reminded fellow scientists of the inherent dual-use potential of every scientific discovery. According to Galston, "The only recourse for a scientist concerned about the social consequences of his work is to remain involved with it to the end!" (Galston, 1972, p. 223). Certainly, the same applies to engineers. It is important to note that the issue is not Galston's responsibility as causal contributor to the United States' use of Agent Orange in the Vietnam War. He was not able to predict that a fertilizer that has defoliation effects in high concentrations would be exploited deliberately as a warfare product. The responsibility attributions of him to himself and, as he hopes, all scientists and engineers to themselves, refers

to being aware of what happens with research and innovation output and, if necessary, to actively seeking participation in its discourse (Jacobs & Walters, 2005).

These cases open discussions of how to set up institutional regimes to prevent the development and application of chemical weapons by referring to the Chemical Weapons Convention – an international treaty that prohibits chemical weapons (Frank et al., 2018).

Perhaps counterintuitively, most so-called *chemical incidents* fall into this category, too. According to the World Health Organization (WHO, 2023), chemical incidents refer to:

> An explosion at a factory that stores or uses chemicals, contamination of the food or water supply with a chemical, an oil spill, a leak from a storage unit during transportation, deliberate release of chemicals in conflict or terrorism, or an outbreak of disease that is associated with a chemical exposure.
>
> *(¶ 2)*

Only in a few cases do chemical incidents occur due to the unpredictable nature of chemicals. Most, indeed, can be reconstructed as the result of a chain of questionable decisions. They are viewed as misuse if commercial interests overshadow the safety concerns of workers and civil society. They are viewed as the result of (unavoidable) uncertainty and risk amidst the complexity of chemical knowledge and expertise. One cannot generally judge whether a chemical incident reflects misuse or the unpredictable nature of chemistry. Chemical accidents that happen due to neglect of ChE expertise (and, perhaps, ChE warnings) in the pursuit of commercial or profit interests reflect a form of misuse, especially when public health is put at risk.

In December 1984, an accident happened at a factory producing the insecticide 'Sevin' in Bhopal in central India. Between 3,000 and 10,000 people were killed and more than 100,000 were injured. The company that owned the plant claimed that vandalism committed by a dissatisfied employee was the course of the accident. This accusation was never proven, and Eckerman and Børsen (2018) present an alternative explanation. Different safety standards were installed and operated in India than in a similar facility in Virginia (United States): The manufacturing pathway – including the toxic compound methyl isocyanate (MIC) that explodes when in contact with water – was risky, but in Bhopal it was chosen over more expensive, but safer, alternatives. The MIC was stored in huge tanks and not in several smaller tanks. The safety system was under-dimensioned and manually managed. Ingrid Eckerman and Tom Børsen portray the Bhopal accident as misuse because of slack safety precautions, insufficient training, poor management, and so on. This case illustrates relationships among the chemical industry, public regulation of the industry, independent non-governmental organizations (NGOs), and critical journalism.

Other examples of chemical incidents include the Seveso accident (Moser & Dondi, 2016), the Deepwater Horizon oil spill (Beever & Hess, 2016), the Beirut explosion in 2020 (Al Daia & Yaacoub, 2021; Al-Hajj & Kazzi, 2021), and many others. Given the severity and frequency of such incidents, central issues in teaching ethics to chemical engineers are safety maintenance and accident prevention through chemical engineers' proactive engagement in multi-stakeholder discourses (see next section).

A third type of misuse of ChE has to do with the right to patent chemical substances and processes. In short, the discussion revolves around materials that could potentially solve important problems but are limited in their widespread application because of patenting for commercial interests. Patenting is important in business models and corporate innovation practices (Grubb, 2016). Chemical startups, especially, secure their intellectual assets by patenting substances or chemical transformation processes. Without patents, monetizing their innovation activities is jeop-

ardized. Patents can prevent more widespread use of the products or processes, which has many effects. When the use potential involves tackling an ethical (e.g., societal or environmental) challenge, deliberate patenting may be interpreted as a misuse of ChE competence and expertise for profit – when considering the missed chance of doing something good for people, society, or the planet. Ethical issues like equality, global justice, and transparency are at stake. The conflict is amplified when the patenting leads to hype (and, perhaps, 'patent racing') around a technological branch such as synthetic biology or genetics (Schummer, 2016).

Another ethical issue in the patenting context arises when the patenting of a synthesis of a natural chemical substance disregards the interests of those who – perhaps unknowingly – have lived or worked for centuries with this substance (McGonigle, 2016). Gerber and co-authors (2021) describe the case of psilocin and psilocybin, compounds from the psilocybin mushroom, successfully patented by Swiss pharmaceutical company Sandoz after their R&D team found a way to synthesize these molecules. While the company benefitted from a huge market for psychedelics, the Mazatec tribe, indigenous to Mexico, who discovered the psychedelic and other health effects of the mushroom, were left without any mention of their innovation or knowledge around its use. Despite the UN Convention for Biological Diversity (1992) that urges to "respect, preserve and maintain knowledge, innovations and practices of indigenous and local communities" (p. 1), Gerber et al. (2021) assert:

> There are still a series of important issues pending and ongoing debates on how to best achieve just and fair consultations and agreements, respecting ethical, epistemological, and ecological concerns, and how to properly share the benefits in the case of private properties of pharmaceutical corporations based on cumulative collective knowledge of indigenous peoples.
>
> *(p. 574)*

Lessons to be learned from cases where ChE is misused are that chemical engineers have a responsibility to reflect on the intentions and ethical legitimacy of their endeavors. Often it is difficult for engineers to formulate and make explicit a coherent ethical orientation system. Knowledge of different normative frameworks can assist in the formation of a feeling of what is right and wrong. Such formulation also requires toxicological and ecological knowledge to know whether a substance is harmful to humans or the environment. Hence, misuse of ChE requires a focus on the actual effects of ChE that can be compared to the intended ones. The harmful consequences for fellow humans, the environment, or even humanity that often accompany the – obviously morally wrong – misuse of ChE may also lead to a loss of public credibility of ChE in its entirety. This is more pertinent when misuse is not admitted and spun in secrecy and half-truths, making it ever more difficult to decide whom or what to trust.

Efforts to prevent misuse and manage uncertainty through interdisciplinarity

While the previous two subsections address ethical dimensions of ChE practices from the 'problem' perspective, this subsection intends to highlight the role of ChE in finding solutions. It often does so as an element of a network of many actors rather than regarding its own core competencies. The following examples illustrate how ChE practice can be supported by other disciplines and expertise realms, and how ChE can support multi-stakeholder efforts to tackle ethical challenges.

Concepts of sustainable and green chemistry as strategies to address environmental issues in chemistry and ChE are sometimes co-shaped by chemical and non-chemical experts. Green chem-

istry (GC) is an attempt to operationalize value-sensitive design in a chemical engineering context. GC was proposed by Paul Anastas and John Warner (1998) to translate ethical concerns into chemical engineering. It is defined as "the utilization of a set of principles that reduces or eliminates the use or generation of hazardous substances in the design, manufacture and application of chemical products" (Anastas & Warner, 1998, p. 2). It builds on 12 principles related to human and environmental health, safety, and security. Other ethical concerns can be added (Friedman & Hendry, 2019).

Illes and co-authors (2021) provide an example of how GC can be enacted in polyvinyl chloride (PVC) production that, according to them, poses three ethical challenges: *How can industry introduce safer technology? How can they be motivated? How can the complexity and uncertainty of new pathways be addressed?* GC deals with several stages in addressing these issues: extraction and conversion of raw materials, manufacturing, end-uses, end-of-life disposal, recycling, and reuse. Ethical values can be reflected in all steps. The idea is that chemical engineers can design their way out of ethical challenges – such as pollution and human health threats. Ethical dilemmas can be mitigated through altered synthesis design, the development of new reaction pathways, or the creation of novel chemicals that pose no threat to human or environmental health. GC is an interdisciplinary endeavor combining chemical engineering, ethics, toxicology, and ecology. It also involves different branches, including industry (management and employees), academia, legislative levels, and civil society. Hence, a potential obstacle arises from conflicts with other interests, including commercial ones and those associated with novel practices. Additionally, the challenge of uncertainty persists.

Carbon dioxide (CO_2) capture, as a form of climate engineering or geoengineering, aligns with the principles of green chemistry. Addressing climate change can involve either the chemical capture or removal of CO_2 or the chemical shading of sun radiation. Ethical concerns extend beyond the uncertainties surrounding climate engineering effects; the so-called 'moral hazard argument' is also pertinent. It states that attention given to unrealistic or uncertain technological solutions to climate change will make the world population less willing to change behavior and, in that way, worsen climate change: *Why bother when there is always an engineered technological solution at hand?* To prevent technocratic decision-making and overly optimistic reliance on technical solutions for addressing global challenges, chemical engineers must approach their interdisciplinary and public communication of possibilities and promises with care and caution.

A movement in the same direction as GC, but with significant dual-use potential, is the recent development of artificial intelligence (AI) technology applied to chemical contexts, such as identifying possibly harmful substances. Such application may support GC motifs by avoiding toxic or hazardous substances, for example, in drug discovery (Urbina et al., 2022). Yet, responsible use by chemists and chemical engineers is paramount given concerns regarding foreseeability, risk management, and accountability.

Ethical codes of conduct are often seen as a possible solution to the misuse of chemical technology. When a researcher or engineer pledges to serve the interests of clients, colleagues, humanity, and the environment and not intentionally misuse chemical technologies, the rationale is that the susceptibility to misconduct and malicious intentions is significantly reduced. Many companies and national or international ChE organizations have crafted codes of conduct for chemical engineers, for example, the American Institute of Chemical Engineers (AIChE), the Institution of Chemical Engineers (IChemE), the European Council of Engineers Chambers (ECEC), or the Philippine Board of Chemical Engineering (see links in the References section). The Organization for the Prevention of Chemical Weapons (OPCW) has compiled an overview of more than 120 codes of conduct for chemists and chemical engineers from around the world (link in the References section).

In the private-sector chemical industry, ChE activities are firmly entangled with corporate social responsibility (CSR) entrepreneurial strategies (see Chapter 6). Although some CSR efforts can be debunked as greenwashing or ethical whitewashing – the false claim of acting sustainably or philanthropically for marketing or publicity purposes – there are examples of positive effects of truthful CSR commitments on ChE practice in corporate settings. When German chemical company Bayer bought the infamous US-American company Monsanto, it saw itself exposed to criticism and loss of reputation. In response to this criticism that targeted disputable business and innovation practices at Monsanto, Bayer installed an internal ethics and integrity board tasked with overseeing R&D practices and giving competent advice on societal and environmental issues. Whether the board will have any visible effect remains to be seen. In any case, however, credible CSR approaches take discourses on the social impacts of ChE straight into R&D departments. This initiative is not unique. The chemical industry has institutionalized voluntary CSR activities in its 'Responsible Care' program (Belanger et al., 2014); the framework has been enacted in many countries around the world (Evangelinos et al., 2010; Gamper-Rabindran & Finger, 2013; Givel, 2007).

A case of successful collaboration of chemical experts with other experts involves the EU's development of regulations for the 'Registration, Evaluation, Authorization, and Restriction of Chemicals' (called REACH) and their enactment into law on 1 June 2007. REACH has been extended and developed iteratively over time. The European Chemical Agency (ECHA) oversees the enactment of regulations and supervises their execution and fulfillment. A core element of REACH is providing one coherent framework for new and existing chemicals. It seeks a balance between a reasonable workload of updating the database and the flood of new compounds and substances (considering, e.g., the countless nano-scaled compounds with different properties at different particle sizes). The idea is that the registry prioritizes the reporting of those substances that are relevant for industry and applications (sorted by various steps of production/consumption in tons per year). This form of organization only works with effective communication along the supply chain, from manufacturers, importers, and distributors to downstream users. Because of this, there is a shift away from public authorities towards industry. What at first may look like a burden for industrial stakeholders (and analytical chemistry) is, in fact, an increased efficiency in regulation. Given the vast number of new materials from chemistry and chemical engineering, no agency or other public authority would be able to manage the testing and regulation of all these chemicals in a feasible way. Through shared competencies and a clever combination of them, REACH has become one of the most sophisticated and helpful chemical registries worldwide. It is considered to be an institutionalization of the precautionary principle (Llored, 2017).

Responsibility domains

When discussing cases of ChE outputs that have ethical implications, the focus is usually on the professional competencies of the involved people. This means seeking to understand and clarify the specific responsibility that chemical engineers have regarding their professional competencies (in contrast to them being members of society). The attribution of professional responsibility hinges on the legitimacy of expectations on skill, knowledge, and the ability to comprehend and follow rules and guidelines (Mitcham, 2005; Williams, 2012). In the context of chemical engineering, it means that the decision-making and action of chemical engineers are the ethical focus only insofar as it concerns their chemical knowledge and their professional institutional or organizational context.

The principle of 'backward-oriented accountability' means that a chemical engineer should have known (or may be expected to know) about the specific effects of a design choice. Regarding the principle of 'forward-oriented responsibility,' a chemical engineer may be expected to anticipate or proactively assess a variety of effects and impacts. For a more nuanced understanding, it is helpful to distinguish among (a) internal and external domains of responsibility; (b) moral, legal, and institutional responsibility; and (c) individual and collective responsibility (Nihlén Fahlquist, 2017). Note that responsibility is an intrinsically ethical concept in such a way that taking responsibility always concerns ethical and other normative values.

Internal and external responsibility

While codes of professional conduct and guidelines of good engineering practice constitute the internal domain of responsibility, the external domain concerns the impact of engineering activity on the environment and human lifeworld (see also Mehlich et al., 2017). Analogously, Herkert (2005) distinguishes *micro-ethics* from *macro-ethics*. While many voices (especially from the engineering community) wish to see ethical discussion focusing on (or limiting itself to) the internal domain of good engineering practice, others point to the significant impact of engineering activity on the environment and human lifeworld.

In the internal (or micro-ethical) domain, responsibilities are often codified through professional guidelines and codes of conduct that are institutionally or organizationally enacted. Academic institutions have ethics boards, and companies encourage employees to read and sign their commitments to professional conduct. Some companies identify explicit corporate values. In such cases, the call for ethical behavior is often modeled as a virtue approach – the 'good chemical engineer' is committed to virtues such as objectivity, truthfulness, fairness, or skepticism. Merton's communalism, universalism, disinterestedness, and organized skepticism as the central virtues of science (CUDOS) approach, and related lists have been refined for specific chemical engineering contexts and purposes (Merton, 1973/1942). This domain is especially relevant in the context of cases related to the unpredictable nature of chemicals – chemical engineers are expected to use their professional skills and knowledge to identify and mitigate risks. The atrocities caused by Thalidomide might have been prevented had the chemical experts conducted and communicated their toxicological studies more thoroughly without technophile or commercial interests in mind. The Love Canal pollution might have been avoided by rigorously following life-cycle and risk-assessment protocols or hazard foresight strategies.

The external or macro-ethical domain gets less attention in the scientific literature on the topic. There are far fewer operational and methodological instructions and codified guidelines. However, technology development, design, and innovation tools such as value-sensitive engineering, ethical design thinking, or ethics-by-design (for more on these, see Chapter 22) aim at providing practicable frameworks for chemical engineers to systematically consider normative dimensions in their work. This domain mainly concerns the innovation entanglement of ChE in which chemical engineers defend and explain their choices in front of other actors and stakeholders such as corporate executive boards and management, regulatory bodies, or the public and civil society. An illustrative case is the development of chemical weapons. Responsibility, here, is not about the safe handling of the chemicals or the scientific rigidity of the study of their properties but about the known contribution to an unethical act (killing civilians with chemical weapons).

Disasters like the Bhopal and Seveso accidents illustrate how both domains can be interwoven. On one hand, scientifically sound study and communication regarding chemical properties is part of the internal responsibility domain of chemical engineers. Without reliable ChE knowledge, the

safety of industrial facilities cannot be maintained. On the other hand, it is precisely that knowledge that empowers chemical engineers to supervise R&D practices and to communicate concerns about safety and risks, if necessary, in firm opposition to economic, commercial, or political interests. This is ChE's 'external responsibility.' External responsibility is sometimes called 'social responsibility.'

The unpredictable nature of chemicals touches upon the interplay of (or tensions between) internal and external responsibilities. Chemical engineers and their disciplinary societies need to address the unpredictable nature of chemicals as part of good professional practice. Doing so is also an expression of caring for societal and environmental impacts. In that way, chemical engineers are responsible for upholding the reputation and good practice of chemical engineering (internal responsibility) and for protecting society and the environment (external responsibility). Chemical engineers hold a responsibility to engage with different perspectives and manage uncertainty when trying to assess the sometimes unpredictable implications of chemical products.

Ethical issues related to the misuse of chemical engineering also transgress the distinction between the internal and external responsibilities of chemical engineers. If chemical engineers consciously violate good engineering practice, it has implications for citizens and the environment, as we have seen in several cases above, regarding accidents, war crimes, and unjust distributions of risks and benefits arising from chemical engineering.

Thus, we suggest including reflections and deliberations on both the purposes and the broader implications of chemical engineering projects as part of good practice in chemical engineering (for more on how to do this, see Chapters 6 and 25). Good engineering practices are intertwined with the societal role of engineering expertise. Luckily, we need not start from scratch when we try to bridge chemical engineers' internal and external responsibilities. Existing mechanisms to prevent misuse and manage uncertainty are ways of bridging the internal and external responsibilities of ChE. They include science-based legislation, formulating engineering programs to explicitly address ethical challenges (including green chemistry projects), and the many initiatives to formulate ethical codes of conduct in academia and industry. Almost all cases presented in this chapter can be made sense of by considering these two dimensions of responsibility. In the authors' teaching activities, students are happy to learn about this distinction because they find it clarifying.

Moral, legal, and institutional responsibility

Responsibility is attributed with different intentions by different stakeholders in view of different aspects of decision-making (rules, knowledge, competence, authority, etc.). It is useful for our purpose to distinguish the following dimensions of professional responsibility:

- Moral responsibility: to act in accordance with moral codes and to respect ethical values and principles
- Legal responsibility: to know and obey relevant laws and codified sets of rules
- Institutional responsibility: to follow institutional, for example an employer's, rules and binding contracts

Ethical behavior, here understood as an activity driven by a commitment to norms of any kind (morals, laws, culture, etc.), is not limited to one of the dimensions defined above but usually requires a navigation of choices that avoids conflicts between the different domains. For example, chemical engineers are, by contract, embedded into an institutional (e.g., firm-internal) network of rules and codes that they are expected to follow loyally. Sometimes, it may be necessary to chal-

lenge these rules in the face of moral dilemmas that surface during the development of a technological item, a prototype, or during the real-world implementation of something new (a chemical compound, a process, etc.). Imagine a chemical firm interested in the market implementation of one of its innovative products. When chemical engineers involved in the environmental impact assessment process identify safety risks beyond required assessment standards, they may face a conflict between their moral responsibility to point out these risks and their institutional obligation to follow executive board directives concerning the public communication of internal R&D information. The Love Canal and Thalidomide cases and historical chemical accidents are illustrative examples of chemical engineers under such tension.

In the cases described in this chapter, we see examples of ethical values such as no-harm and human dignity, international legal regimes such as the Chemical Weapons Convention and the Stockholm convention regulating POPs, and institutional rules such as codes of conducts, green chemistry principles, and corporate social responsibility arrangements like Responsible Care, that define different normative arrangements that professional responsibility can be defined by.

In view of this distinction, chemical engineers cannot solely rely on the codified terms of accountability when doing their job; they need to be aware of the ethical dimensions of their work that require proactive engagement and have the readiness to face discourses with stakeholders who pull into other (perhaps unethical) directions. Competent professional agency, in this respect, exceeds the realms of compliance (with codes of conduct) and standard protocols but may enter supererogatory areas in which ethical behavior is not a matter of top-down duty but of a personal commitment to '*doing more than one's duty*' (as, e.g., whistle-blowing even in the face of negative personal consequences).

Individual and collective responsibility

To fully comprehend one's professional responsibility ascription and act accordingly, it is necessary to understand the difference between attribution of individual responsibility (for example, for professional agency) and collective responsibility (e.g., to firms or social spheres such as *science* or *innovation*). The latter can never mean delegating responsibility and duties away from an individual and towards an intangible, non-personalized entity. In many of the cases described above, it would be possible for chemical engineers to argue that they didn't do wrong because their contribution to creating a problem was only marginal or because the chain of decisions that led to the conflict didn't allow identification of them personally as the causal origin of the problem. For instance, Thalidomide is a marketed product, so some could argue that disseminators rather than developers are to blame. The waste-disposal site disaster was created due to a community decision. The Bhopal disaster resulted from a long series of questionable economic decisions meeting unfortunate local working conditions, and so on. Yet, as a counter-argument, many of the decisions were based on factual information that was contributed by the chemical engineers. Thus, rather than denying any responsibility, it would be prudent for engineers to clarify their own contribution to the collective and to engage in co-operative discourse that constitutes the collective responsibility of a multi-agent entity such as a firm or a public service organization.

Responsibility in action: chemical engineers in discourses

Chemical engineers usually work in highly diverse teams, so their specific role in research, development, design, and innovation processes is described in this section to highlight the normative dimensions of interdisciplinary collaborative agency. This issue predominantly concerns the external responsibility domain: the elicitation, comprehension, and anticipation *by design* of future

impacts of engineering, development, and innovation choices. In both academic and corporate realms, engineers make the normatively most significant decisions after connecting their factual contribution (research results, functionality and utility, risk and impact assessments, etc.) with values and norms that guide the outcome-oriented mission statements and purposes of the undertaken efforts.

For example, in their research about chemical engineers' experiences related to ethics in engineering in the health products industry, Kim and Kerr (2021) showed that when chemical engineers design products, such as drugs, they consider both benefits and risks to the patient, their customer, so that their products can ultimately improve patients' health outcomes. When engineers make a design decision, they ensure their arguments are based on data and scientific rigor so that their products can ultimately be safe and efficacious for patients. Kim and Kerr also showed that compliance with established technical standards and regulations is also an important consideration among engineers, and as engineers work in teams, acknowledging the contributions of different team members in the project is also very important. Likewise, chemical engineers consider multiple factors, including making data-driven and scientifically rigorous arguments, complying with technical standards and regulations, and cultivating a healthy teamwork environment to achieve their ultimate goal of serving the needs of their customers safely and efficaciously.

In these discourses, engineers are never purely knowledge exponents or providers of the necessary insights for the one right solution to a problem. Their input can potentially modify normative commitments and respective development paths. In this view, an ethical engineer is not an expert on ethics, but a responsible agent who stirs a constructive discourse among stakeholders to set factual knowledge and value claims into a plausible and scrutiny-withstanding relation. Pielke (2007) calls this the *honest broker*.

This practical discourse skill is not inborn but can and should be trained. Therefore, the education of chemical engineers in the ethical dimensions of ChE should be understood as practical coursework and not as a lecture on theories and principles. The case overview in the previous section indicates that the solution for problems arising from ChE undertakings, or the prevention of such problems, is not formal training in ethics and ethical principles but better development of responsibility, communication, critical thinking, teamwork, and interdisciplinary competencies. Chemical engineering education at most higher education institutions follows a highly practical curriculum. ChE students are used to approaching problems *at hand*. Ethical and other normative challenges are not different – they require pragmatic, practicable approaches that are best learned in a challenge- or problem-based fashion (Mehlich, 2022).

Conclusion and outlook

This chapter provided content for the ethics education of chemical engineers. Educators may consider broadly covering topics, including ethical issues arising from the unpredictable nature of chemicals, their potential misuse, and prevention efforts, employing specific examples we provided in this chapter. When teaching those issues, educators can consider the various responsibility domains we suggested. To support the design of teaching items in ChE, we suggest linking the presented cases with the different forms of responsibility discussed above. Since chemical engineering practice and its impact are interdisciplinary, educators may also need to consider how to help chemical engineers develop practical skills for navigating such environments.

Contemporary theoretical work on the philosophy and sociology of chemical engineering has laid out the foundations for understanding the practical agency of chemical engineers. This means the conceptual and methodological bridge towards practice and education has been built.

With appropriate and scientifically validated concepts of sustainability at hand, it is now possible to craft effective and efficient impact assessment procedures that not only study toxicity or life cycles but also address ethico-environmental issues such as biodiversity or environmental justice (Thompson, 2020), trust, or diversity. Thus, future chemical engineers will be empowered to anticipate these ethical aspects via their professional competence.

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18

ETHICAL ISSUES IN SOFTWARE ENGINEERING

Stephanie J. Lunn, Isis Hazewindus, Prajish Prasad, and Vivek Ramachandran

Introduction

From the first home computers in the 1970s to the use of algorithms in application engineering, technology has expansively progressed over the last several decades (O'Regan & O'Regan, 2008). The resulting hardware, software, and data generated have become ubiquitous in our lives, permeating numerous industries such as healthcare, finance, agriculture, manufacturing, and education. Creating and utilizing these technologies can wield tremendous influence, offering solutions across domains and society at large. Meanwhile, the conversation regarding ethical issues and implications surrounding these technologies expands.

Despite early twentieth-century critics of modern technology (Heidegger, 1977; Ellul, 1964; Mumford, 1934, 1967) warning about humans becoming slaves to technology, the prevailing view on technology has remained primarily utopian. Initially, technology and its artifacts were considered neutral, holding no intrinsic morality, with the user assessing its ethical aspects. Such approaches shift moral judgment from creators to users, subject to their end goals, thereby overlooking any underlying ethical concerns in the technologies themselves. Consequently, computers and software are perceived as objective entities, neglecting the influence of the software engineers behind the technologies, who bring their own viewpoints, beliefs, and biases into the products they and their teams create.

Nevertheless, the extensive integration of computing into society, the rise of 'Big Data,' and the impacts of created tools have necessitated questioning technology's purported neutral status. Although technology can enhance the quality of living, it can be harmful as well. Various scandals concerning algorithmic applications, data misuse, and security breaches make it increasingly evident that technology is dependent on the (conscious or subconscious) biases and decisions of the people and organizations building and maintaining them (Brown et al., 2021; Hysa et al., 2023). Computing solutions are applied to a range of topics and can enable access and dissemination of useful information, but can also exacerbate inequity, amplify stereotypes, and/or lead to issues around privacy and security – for example, through hacking into personal accounts (Xu & Tang, 2020). Thus, the influence of humans and the development and use of technology are inextricably linked, and decisions can have consequences not linked to the designers' intent.

DOI: 10.4324/9781003464259-22

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To help software engineers become aware of their responsibilities in the creation, dissemination, and maintenance of software, as well as their consideration of data use, educators must provide environments that promote not only learning how to code, but also how to recognize ethical issues and accurately respond to them. This is where ethics education can play a crucial role; scholars have emphasized the need to focus on *how* rather than *what* to think (Horton et al., 2022). Integrating ethics and encouraging social justice early on can promote students' agency and encourage them to consider the implications of decisions (Moore, 2020; Ferreira & Vardi, 2021). It is critical for ethics to be conceptualized as part of practice, and to transition disciplinary mindsets away from 'avoiding traps' toward anticipating 'sociotechnical risks' (Andrus et al., 2020, p. 77).

Incorporating ethics into education is not novel but constitutes an ongoing effort. Numerous organizations, governmental funding agencies, and institutions of higher learning have previously acknowledged the need for concentrated and comprehensive ethics education. A joint task force comprised of representatives from the Association for Computing Machinery (ACM) and Institute of Electrical and Electronics Engineers (IEEE) – two major international computing societies – worked together to define a guiding 'Software Engineering Code of Ethics' to encourage standards for teaching and practice (Gotterbarn et al., 1997, 1999). Although this code can provide a beneficial foundation for framing ethical issues, decision-making in the real world can include any number of variables and requires all individuals to take ownership and responsibility for choices. Towards this goal, 'IEEE 7000' is the first standard that describes principles surrounding unintended risks and encourages 'responsible innovation' with consideration of human and social values (Spiekermann, 2021). Recognizing the importance of personal responsibility in software design and use, IEEE 7000 centers around combining material value ethics and moral philosophy to drive value-based engineering (Spiekermann & Winkler, 2022).

Despite such endeavors, attempts to foster ethical mindsets are not always widespread (Connolly, 2020). Ethics are often taught either as standalone courses or are incorporated within existing courses by engineering instructors (Fiesler et al., 2020). Yet, integrating ethical awareness and decision-making cannot be undertaken in isolation; ethics instruction must intertwine with technical understanding (Martin et al., 1996). Presenting ethical concepts sporadically may be insufficient to alter students' thinking. Instead, a widespread approach is required, with lessons embedded throughout curricula and recurring periodically throughout a student's matriculation.

As societies evolve and along with them the definitions of technology, the field of computing likewise advances, and dynamic subfields – such as artificial intelligence (AI), computer vision (CV), human-–computer interaction (HCI), and machine learning (ML) – emerge and expand. Continually considering the ways in which technology can impact accountability and transparency is essential. Moreover, we must reflect on how ethics can be applied to advocate for social justice and encourage diverse voices (Cheong et al., 2021; Ferreira & Vardi, 2021). With this chapter, we hope to contribute to a long-term and sustained approach to molding ethically minded software engineers by providing didactic tools to establish educational and professional environments where inclusion and morality are at the forefront of the discussion. We encourage conversations on and integration of ethics throughout computing curricula, touching upon themes like responsibility and social impact.

In the section that follows, we will introduce ourselves as authors of this chapter and describe our positionality to frame the chapter. Then, we will then elaborate on ethics and its place in software engineering (SE), before diving into practical approaches to integrate ethics in SE pedagogy. Finally, we outline case studies that instructors can use to convey important ethical issues in SE. Our tools can be applied within different courses with distinct objectives and assignments, with reinforcement of ethical thinking as a final goal, while providing the opportunity for students to

practice ethics and empower them to "effectively actualize their (own) values in future professional decision making" (Cohen et al., 2021, p. 861).

Positionality

Given the role that personal values, experiences, and backgrounds can have in how topics are approached, interpreted, and defined (Martin et al., 2022), we want to be transparent as authors. We aim to disclose how we have influenced what is presented. In this section, we share information about who we are and the role that plays in the perspectives and approaches we use.

The first author, Stephanie, identifies as a white woman with a computer science (CS), computing education, and engineering education background. She is an assistant professor at a large, research-focused university in the southeastern United States. Her experiences with sexism and imposter phenomena in computing have driven her efforts to aid in students' technical and professional development and to cultivate more inclusive mindsets.

The second author, Isis, identifies as a white cis woman with a background in philosophy, data ethics, and the development of educational programs. Her role as a data ethics consultant and educator shaped her view on careful decision-making and ethics as a practice of continuous reflection and conscientiousness. Being the first in her family to receive training at university, she is conscious of the importance of introducing matters like ethics in ways explicitly tailored to the target group – to encourage enthusiasm and curiosity rather than alienation or indifference.

The third author, Prajish, identifies as an Indian man with a CS and educational technology background. He has experience working for an EdTech company as a software developer. He works as an assistant professor at a liberal arts university in India and teaches CS subjects like SE and introductory programming. His experiences in the software industry, as well as his research experience in educational technology, have made him realize that a more holistic instructional approach is required to help students engage with engineering problems equitably and effectively.

The fourth author, Vivek, is a non-binary Indian with a robotics, mechanical engineering, and engineering education background. His experiences in learning and teaching in Asia, Europe, and North America – where he was at times a member of the ethnic and/or racial minority – have shaped his worldview. His desire to see ethics and societal responsibility being prioritized in research and teaching in robotics and engineering domains drives his pedagogical practice.

Ethics and morality in software engineering

In philosophy, "if we are interested … in what our guiding ideals should be, in what sort of life is worth living, in how we should treat one another" (Shafer-Landau, 2018, p. 1), ethics is the domain that we turn to. This domain is a broad and can be categorized into three subdomains under which different types of looking at ethics can be grouped (Shafer-Landau, 2018):

1. *Normative ethics*: focuses on standards and what we ought to do. Irrespective of the circumstances, it considers *What are our moral duties?* and *Which actions are morally right, and which actions are morally wrong?* Normative ethics encompasses approaches like utilitarianism, consequentialism, deontology, feminist ethics, and virtue ethics.
2. *Meta-ethics*: focuses on the status of our moral claims and values. *Can we say moral theories or statements are 'true'?* We might even wonder about the existence of something like a universal moral viewpoint.

3. *Value theory*: focuses on questions concerning the nature of defining goodness and the good life: *What is a 'good life' and how do we pursue it?* It balances moral and natural goods.

More elaborate discussion of these ethical frameworks is provided in Chapter 2 of this handbook. Here, in this chapter, when we speak of ethics, we refer to normative ethics. We want to provide educators and students with the tools to help them grasp ethics and reflect on what is good in terms of actions and decision-making while keeping in mind that defining what is 'good' is complicated. Issues can be highly complex, contextual, and subjective, involving multiple actors with their own values, expectations, and worldviews. As such, we need a way to consider those issues cautiously and form as broad of a perspective as possible. With ethics, there may not be a simple solution offering clear-cut answers; rather, an ongoing process can guide and navigate conversations about these complex issues.

Cultivating ethical mindsets throughout the software development and data life cycles

We strive to instil moral values in computing students to empower them to make ethical decisions throughout the software development life cycle (SDLC) and the data life cycle (DLC). The term 'software development life cycle' describes the formal or informal process, or methodology, employed for the design, creation, and maintenance of software (Sommerville, 2016). It has been defined using multiple models (e.g., Waterfall, Spiral) (Ruparelia, 2010; Mall, 2018), and while the names for each of the phases and specifics vary, the core concepts remain more or less fixed. Linked to software design, the 'data life cycle' comprises distinct phases (Wing, 2019), and the term describes the process of generating, using, and managing data (e.g., datasets, databases, code). Once the data life cycle is complete, knowledge gained can result in feedback that can inform/improve additional data collection and creation. Throughout the SDLC and DLC, an 'Ethics by Design' approach can be employed to emphasize fairness, transparency, accountability, social and environmental well-being, privacy, data governance, and human agency (Dainow & Brey, 2021).

Teaching students to embrace 'Ethics by Design' involves tools, discussions, and assignments that promote ethical reflection and can aid in problem-solving. The approach can enhance choice consideration and inform planning, thinking, and actions in the life cycles. In Figure 18.1, we provide a conceptual model that unites the SDLC, DLC, and ethics (Ashok et al., 2022; Janeja, 2019; Karim et al., 2017; Wing, 2019).

Although each life cycle is presented separately, the arrows leading into each other are meant to emphasize that the two cycles are frequently interrelated. Data generation can follow software development, and data results can spark further software development. Ethical principles must be stressed across the process. We propose this new conceptualization, which we label the ETHOS model, as a way for educators to reconcile the components it encompasses:

Ethical (ethics)
Technological processes (software engineering life cycle),
Handling, and
Organizing data (data life cycle) for
Social impact (ethics)

These components appear in Figure 18.1 to make their connectivity explicit and highlight that consideration of ethics should encapsulate all decisions, choices, and actions taken – and all with

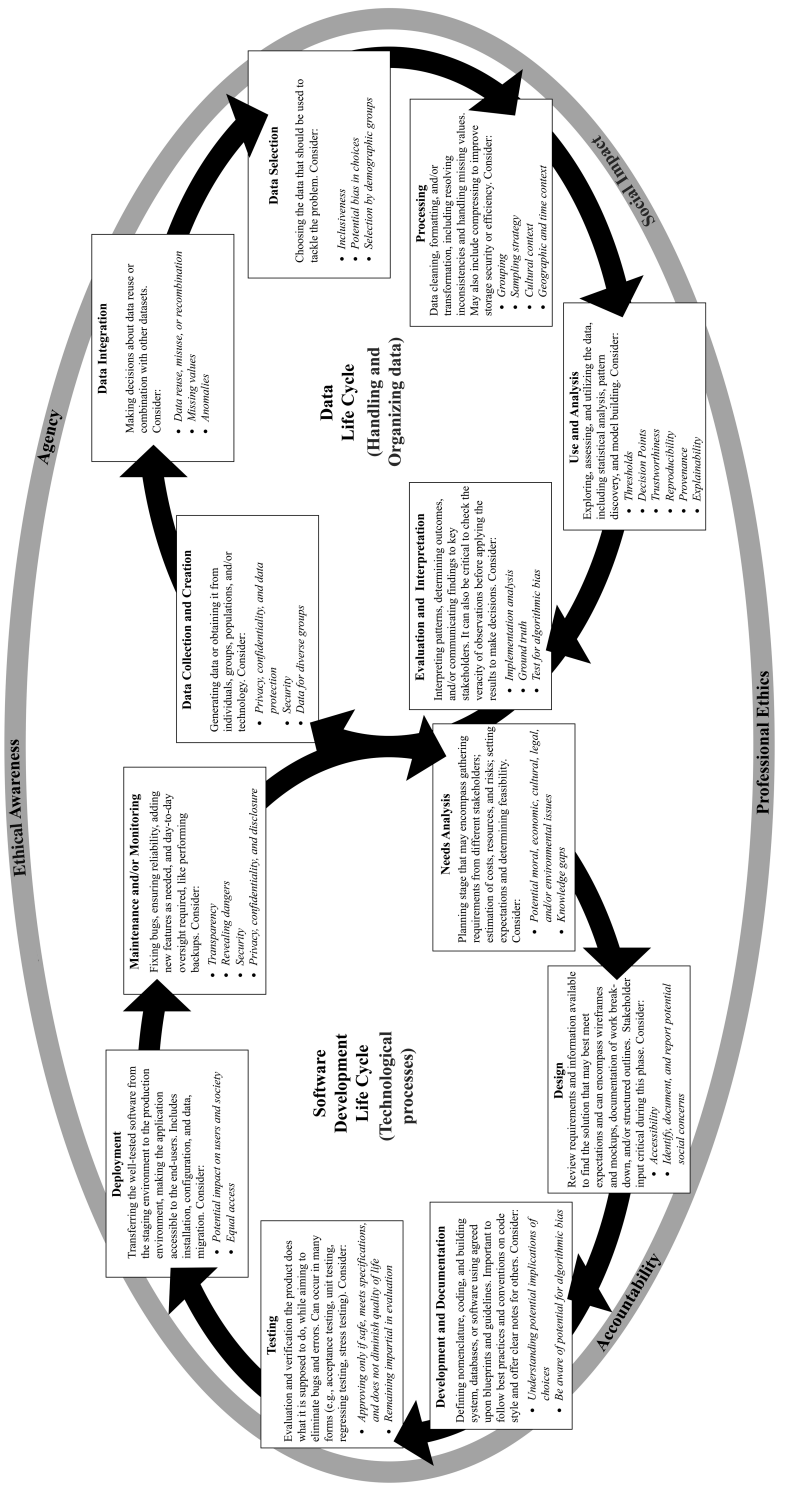


Figure 18.1 ETHOS conceptual model applying ethics throughout the SDLC and DLC (drawing from Ashok et al., 2022; Janeja, 2019; Karim et al., 2017; Wing, 2019).

regard for social impact. An 'Ethics by Design' perspective, as described by the ETHOS model, highlights the value of being mindful of and addressing ethical issues as early as possible. It also lends itself to following up on these issues and evaluating them throughout the process. As identified in the outer ring of this model, this necessitates consideration of additional concepts, including:

1. *Accountability and transparency*: To address the challenge posed by the 'many-hands problem' of attributing outcomes to a single individual in software projects that involve multiple people (van de Poel, 2015), the ETHOS model fosters a culture of individual ownership for actions, choices, and mistakes. In addition to complying with regulations, this involves collaboration, understanding expectations, ensuring accuracy, considering consequences, valuing quality, and being honest about decisions. Transparency involves communication, traceability (documenting or tracking goals, history, and assumptions), and intelligibility (comprehensibility and monitoring programs and systems to achieve outcomes).
2. *Agency*: Computing professionals make ethical decisions shaping the creation, maintenance, and application of products, programs, and data. They actively influence the ethical landscape through decision-making rather than passively following established norms. Thus, empowering individuals to address possible issues and propose solutions as they arise is valuable. Those involved ought to have autonomy over ethical deliberations, strategies, and actions that support the freedom and diversity of users. They should also reconcile differing interests and values while balancing project and product specifications.
3. *Ethical awareness*: Recognizing, analyzing, and navigating moral situations is crucial for individuals and organizations as it adds a human element of control. This goes beyond understanding the impact on various actors, including the environment. Hence, it is essential to prioritize the sustainability of products, programs, or data and anticipate potential repercussions at the conclusion of the item's life cycle. Proactively establishing safeguards in the SDLC and DLC to detect and mitigate problems is part of this, but ongoing appraisal and reflection are equally important for mindful decision-making.
4. *Professional ethics*: Conduct standards should be an integral part of the SDLC and the DLC. Individuals and organizations must uphold an ethical code – and values – centered on respect, transparency, honesty, fairness, privacy, and human-centric well-being. Individuals should understand diverse definitions and manifestations of these concepts, enabling them to make ethical decisions in various situations. Such ethical conduct may appear through fidelity, beneficence (i.e., promoting doing good or the well-being of others), non-maleficence (i.e., avoiding harming or injuring others), and justice (i.e., distributing benefits and burdens fairly).
5. *Social impact*: Ethical lapses in the SDLC and DLC can have serious consequences in real-world situations. Accordingly, it is important to extend beyond the technical aspects of software engineering to consider social, economic, cultural, and environmental impacts on humanity. Such thinking and behaviors demand global awareness, cultural competence, attention to accessibility, and the inclusion of diverse perspectives – encouraging proactive contributions to technology in alignment with varying ethical values. It necessitates addressing societal challenges, promoting social good, and identifying, seeking to prevent, and resolving potential negative impacts of programs, applications, and/or data.

We advise using the ETHOS model to emphasize human values and students' personal responsibility and encourage students to reflect on their own roles, biases, and decisions. Although we

present bullet points in Figure 18.1 to provide instructors with ideas for further discussion points, a simplified version may make the content more digestible in the classroom. Introducing individuals to existing codes like the ACM's Code of Ethics or having students create their own codes fosters awareness of their professional responsibilities. Everyone taking moral responsibility and striving for continuous improvement and just behaviors can contribute to a positive shift in our interdependent society.

The following sections describe several ways educators can approach these topics. We encourage the exploration of different (ethical) viewpoints. Instructors can initiate meaningful conversations in classrooms using several 'dialogical approaches.' Students can also engage with various 'reflection tools' to help them ruminate on ethical issues and concerns in projects. These dialogue approaches and reflection tools can be considered to further students' engagement with ethical concepts, issues, and thought processes. The sections below focus on software engineering, and, again, for a more general discussion on using reflective and dialogical approaches in engineering ethics education, we refer readers to Chapter 25.

Dialogical approaches

Reflecting on choices and engaging in open discussions about intercultural responsibility, tolerance, and consequences aid ethical decision-making (Ferri, 2014). Integrating ethics into the classroom can be approached in various ways to help students develop ethical mindsets and address any issue they may encounter.

The purpose of ethical conversations in software engineering

Issues can arise in any project or system involving software or data, and three key aspects must be considered (van Veen & Visser-Knijff, 2022). The first involves the project's technological possibilities and feasibility, leading back to the SDLC. The second regards the legality of data storage and application (i.e., the DLC): *Is data's intended use and processing compliant with regulations?* These two aspects can easily be solved by personnel like privacy officers and can be assisted with a control check like a data protection impact assessment (DPIA).[1] However, even when all technical or legal questions are sufficiently answered in any project, a third aspect remains – concerning desirability. This aspect demands careful reflection regarding software and data projects at different points in time to ensure we consider future ethical implications a project might have in addition to complying with the rule.

One way to empower students is by incorporating ethical conversations into educational practice. Gaining experience in having ethical conversations about real-world case studies can help students gain what Aristotle calls *phrónēsis*, or practical wisdom, which to him means knowing how to act in a specific situation (Kraught, 2022). Gaining *phrónēsis* is a process of training to cultivate one's virtues or good traits. By applying principles of action to real-world situations, we will train ourselves to assess each situation carefully and then decide on our actions, which should align with a virtue. This way of looking at ethics is crucial because 'doing' ethics is not a matter of ticking boxes or choosing the right action from a predefined set, but rather requires us to assess each situation and its context on its own and decide our best course of action from there.

In addition to teaching students about ethical implications, conversations can aid decision-making. Facing ethical issues can potentially pressure students to make the right decision. At the same time, many students lack formal decision-making training and practice (Bond et al., 2008; Keeney, 2020) while also underestimating how their own biases affect decision quality. Discussing projects in a group may reveal diverse perspectives, offering invaluable input on project decisions.

Conversations may yield new insights, promoting ethically minded decision-making and risk mitigation. Below, we will summarize three conversation methods and how they can be used:

1. *Socratic dialogue*: The Socratic dialogue is a conversation method which, according to Knezic et al. (2010, p. 1105), is defined as:

   > a philosophical group dialogue in which the participants guided by a facilitator and a number of ground rules strive to reach a consensus in answering a fundamental question on the basis of a real-life example or incident with the purpose of achieving new insights.

For those in computing roles, the Socratic dialogue may appeal for cultivating several skills:

a) Asking the right questions – as the dialogue is centered around one core question, students are pushed to carefully consider what questions should be asked in different stages of the ETHOS model.
b) Inquiring into underlying assumptions or ideas – at its core, the Socratic dialogue is focused on inquiring into one's ideas, beliefs, assumptions, and dispositions. It is thus pre-eminently suited to encourage students to examine their own attitudes and identify any bias they might unconsciously hold that could impact projects.
c) Promoting mutual understanding using the Socratic dialogue is a collective effort, pushing participants to reach a unanimous conclusion. This encourages students to adapt their viewpoints based on new information and sound arguments, fostering better teamwork and openness to diverse perspectives that can alter project directions.

2. *Moral case deliberation*: Moral case deliberation is a conversation method commonly used in medical and business ethics. It involves discussing a real-life case to move towards a concrete approach or action.

   As moral case deliberation aims to obtain concrete project actions, it proves valuable with case studies or as group intervention for students dealing with ethical challenges in internships. Discussing problems and extracting actionable suggestions boosts students' confidence to address such issues in projects, enhancing their ethical agency. As per the description in a 2012 article by Karssing (2012), students will also be encouraged to explore the case and thus increase their ethical awareness and sense of personal responsibility.
3. *Thought experiments and scenario thinking*: The last method to initiate a conversation is to use thought experiments or scenario thinking. Thought experiments (e.g., Philippa Foot's 'Trolley Problem') are widely used in philosophy. Philosophers often use thought experiments to visualize scenarios that cannot be carried out in the real world due to physical, financial, technological, or ethical limitations. The *Stanford Encyclopedia of Philosophy* (Brown & Fehige, 2019) describes engaging in a thought experiment as follows: "we visualize some situation that we have set up in the imagination; we let it run or we carry out an operation; we see what happens; finally, we draw a conclusion."

Using thought experiments and scenario thinking lends itself especially well to software engineering, as it focuses on trying to gain insight by:

a) Identifying issues by describing possible outcomes of deployment (by sketching utopian and dystopian scenarios).
b) Reflecting on the nature of the problem that serves as the basis for a project (by asking: *Is this solution the means or the goal?*).

c) Reflecting on the necessity of using specific data-driven or technological solutions and exploring alternative options (by asking: *What problem(s) does this solution solve? What problem(s) doesn't it solve? What new problem(s) does it create?*).

Reflection tools

Ethical reflection tools can provide further guidelines and resources to help identify and mitigate potential ethical issues that may arise. Although the ETHOS model can function as one such reflection tool, others have been developed which may offer additional resources and guidance. They adopt a variety of formats, that we broadly classify as: (1) ethics-by-design tools, (2) process tools, and (3) ethical impact assessments. What they all have in common is that they offer a way to start ruminating on a project in a structured way. They can help individuals in computing make more informed decisions, consider how to address ethical issues in their projects, and weigh the potential benefits and harms of different options. Some tools that may be helpful to consider are:

- The *Data Ethics Canvas*[2] is a reflection tool created to help identify and manage ethical issues in the DLC. Designed for project-wide use, it poses predefined questions to provide an overview of problems encountered in data-based projects. It can help acquaint students with using ethical tools. Alongside the canvas (i.e., matrix), the tool offers a comprehensive user guide that promotes using the tool for enhanced project planning and encourages a human-centered perspective towards the design from the project's outset.
- *Consequence scanning*[3] involves a set of activities that can be incorporated into the SDLC, to examine the potential impact on people and/or society. It consists of a manual, printed headings, a proposed structure to follow, and two sets of prompts focused on consequence and context. It provides opportunities for software teams to have conversations about the potential implications of what they have built, what they will build, and how they can mitigate potential harms before they happen. It can also help students cultivate awareness of ethical and societal problems arising when one fails to reflect on potential consequences of their designs.
- The *Data Ethics Decision Aid* (DEDA)[4] is a toolkit designed to be used by teams involved in data-based projects. With a worksheet designed like a board game and with a questionnaire and handbook, the DEDA offers an interactive experience. Using the DEDA is a structured process in which each team member has their place, with the worksheet serving as the foundation of a conversational session. Applying the DEDA in educational settings can help students gain experience with identifying values *and* value conflicts, and approaching software products using an ethics-by-design approach, while also learning to work as a team.
- The *Ethical OS Toolkit*[5] serves as a guide for predicting the future consequences of current technology. It aims to encourage various stakeholders involved in its creation, dissemination, and use to foresee potential issues. The tool includes 14 scenarios to start conversations, a risk mitigation manual, and a checklist for considering ethical questions and issues. Educators can use the toolkit in computing or design courses in multiple ways, for example,having students read about the proposed 'Risk Zones' and tasking them with collecting examples in the real world.

Case studies

Using case studies to promote ethical thinking

As educators, facilitating critical thinking via case studies can help students understand moral issues in real-world situations and prompt them to connect ethics to their practice. Analyzing case

studies can stimulate ethical awareness and a sense of personal and professional responsibility. Such activity can help students explore the values and interests of the various actors (Lim et al., 2011). Working with case studies can serve as problem-based learning, which may aid decision-making (Chowdhury, 2018) and improve critical thinking. By pairing reflection tools with the cases, teachers can provide students with further opportunities to internalize concepts and meta-cognition (Begley & Stefkovich, 2007). For specific advice on facilitating this type of discussion/activity, please see Chapter 20.

We note that 'flipped classrooms' can provide a viable alternative for educators concerned about finding time to integrate case studies into existing lessons. 'Flipped classrooms' is an instructional approach where traditional teaching methods are reversed, with students learning new material independently (outside of class) and engaging in activities, discussions, or case studies during class time. This approach can ensure material is covered while enhancing engagement and academic outcomes (Fuchs, 2021). Moreover, providing students with guidance as they solve problems (such as those described through hypothetical scenarios) can be aided via interactions with peers and/or instructors (Herreid & Schiller, 2013).

We now showcase three case studies to provide educators with possible starting points to engage students on topics related to current ethical concerns in software engineering. These case studies are hypothetical but are based on real-world examples. Although there are many ways to incorporate these cases into lessons, we encourage instructors to consider having ethical conversations (using the conversational methods described in the previous section) and providing students with reflection tools to tackle ethical issues arising from these scenarios. Alternatively, each could be integrated into small-group role-play (for more on that, see Chapter 20).

Case study 1: Algorithmic bias – fairness

Scenario

You head the Responsible AI division at SynGenAI, a software company based in Silicon Valley. SynGenAI is steadily gaining attention because it recently released Talk8ive, a free-to-access general-purpose chatbot that tries to answer users when they pose questions. Talk8ive is built on a large language model (LLM), a machine learning algorithm that can recognize, predict, and/or generate text based on large datasets of content, primarily text that the algorithm is fed as input training data. Talk8ive was released with much fanfare but within hours of its release, a number of users started flagging the chatbot for the content it was generating. Since Talk8ive is designed to answer any kind of question, all types of queries are treated the same way. Therefore, while queries with more deterministic responses (coding-related) generate less disputable answers, queries about predicting social outcomes (recidivism) produce untrue and racially biased responses. The behaviors displayed by the chatbot are attracting a lot of negative attention to the product and the company, and there are calls to revoke free access to the bot.

Discussion prompts:

1. *Did SynGenAI make the right decision to release Talk8ive? Why were the issues not flagged in the development stage itself?*
2. *Should the chatbot have been created to address all kinds of queries in the same manner? How did the algorithmic bias enter into the generated context?*
3. *As head of the Responsible AI division, how do you prioritize your responsibilities to the company, to the public, and to your team? What course of action should you adopt?*

Context:

Generative AI media, such as OpenAI's DALL-E or DALL·E2[6] and Speechify,[7] have gained popularity due to their ability to create realistic digital images or clone voices using machine learning algorithms (Chen & Lin, 2023; Pavlik, 2023). These algorithms are trained on vast amounts of data, allowing them to produce content that can be difficult to distinguish from human-created content. ChatGPT, developed by OpenAI, is one such example of a chatbot derived from the generative pretrained transformer (GPT) (Rudolph et al., 2023), that utilizes LLMs to generate responses in a conversational manner. Generative AI tools can be classified into three based on the kind of tasks they perform (Narayanan, 2019):

1. *Perception* tasks involve interpreting sensory data, such as images, audio, or text, to extract meaningful information.
2. *Judgment automation* involves making decisions or providing recommendations based on data analysis and predefined criteria.
3. *Social outcome prediction* involves predicting future events or results based on historical data and patterns.

An over-reliance and belief in generative AI's ability to solve problems can lead to complacency and additional problems. Although generative AI may positively contribute to society, it has limitations, because the underlying algorithms may be trained on datasets that reflect societal biases. If ignored, AI systems can 'learn' in ways that amplify incorrect points or replicate discriminatory patterns. For example, in areas such as hiring or loan approval, AI algorithms can reinforce existing prejudices and disadvantage certain groups, perpetuating social inequities rather than addressing them (Su & Yang, 2023). Broadly speaking, algorithmic bias can result in the following types of problems:

1. *Misrepresentation of minoritized communities*: Minoritized individuals in the tech industry often lack the influence to shape software development and data usage. Consequently, content related to these groups tends to be biased and reflective of societal stereotypes or exclusionary practices.
2. *Misinformation*: The presence of pseudoscientific content in the training data may result in its regurgitation by the AI when presented to users. Misinformation is compounded by automation bias because humans tend to trust the confident tone of interactive AI like chatbots.
3. *Automation and intellectual property theft*: Generative AI platforms are affecting the occupations of artists and journalists, whose occupations are being automated by large corporations because it is cited as being cheaper. Often, these automated generations neglect factors such as copyright or licensing in their training data – and effectively commit digital theft.

Educators can discuss algorithmic accountability using the ETHOS model to raise awareness of the ethical implications of creating and using AI. By ensuring that data sources are properly curated, algorithms are tested rigorously, and AI-generated outcomes are accounted for by developers, students will learn to develop a practice of developing ethical AI systems. Educators can use this case study in the classroom by integrating the dialogue approaches and reflection tools mentioned earlier. For example, moral case deliberation could be used to discuss algorithmic bias. Issues with unwanted output are a common problem with chatbots but often only seem to become apparent following release. Accordingly, we suggest holding a moral case deliberation on the question: *Should SynGenAI take Talk8ive offline completely?* We also suggest using the

consequence scanning tool in combination with this case study to encourage reflecting on more careful implementation. The main ethical issues in this case revolve around the (unforeseen or unintended) consequences of generative AI.

Case study 2: User interface design – empowerment

Scenario

Your current phone is falling apart – it is running out of storage, it has fallen to the ground multiple times – and as a result, it runs very slowly. So, you decide to buy a new phone. Although you do not have a clear idea of what phone you want to purchase, you do have a set budget. By querying a search engine, you find a list of shopping websites selling phones within your target range. You browse some of these websites, looking up different phones and their specifications. After some time, you pick a phone offered at a 30% discount. You read the reviews and are satisfied with the performance and other aspects of the phone. You decide to wait a few days to buy it since there is no real sense of urgency to purchase it immediately. However, as you try to navigate away from this page, you encounter a pop-up message, with a limited time offer for an additional 10% discount that is valid only until the end of the day. A countdown timer is also shown, ticking down on the hours left to buy the phone at this price. Since this seems like a good deal – and you do not want to regret buying it later at a higher price – you add it to your shopping cart. As you check out, you look at the final price and realize that the total is much higher than what was shown earlier. Upon closer inspection, you find that the final breakdown includes a screen guard and a one-year phone protection plan that you did not intend to purchase, and did not add yourself.

Discussion prompts:

1. *How does this experience make you, as a user, feel? What specific design aspects of the shopping website contributed to this feeling?*
2. *How does using interface design like those mentioned above affect your ability (as a user) to make a well-informed, autonomous choice?*
3. *Do you note any ethical issues in how the shopping experience has been designed? If so, what are they and what would your feedback be for the user-interface designer?*
4. *What might the different motivations be that govern the design choices for the website user interface? How might web designers redesign the interface to improve the user experience while addressing potential ethical issues?*

Context:

The scenario describes a not-uncommon shopping experience. Many users encounter similar situations where choices and decisions are manipulated during their interactions with a given website. Moreover, these types of manipulation are not restricted to e-commerce. Companies have tried to force certain actions on users engaged in unrelated pursuits. For example, at one point, users were automatically opted into eBay's marketing emails when they signed in with their Google accounts.[8] Some companies design their user interfaces to make it difficult to perform certain actions, particularly those that can harm the company's revenue. For example, users from Europe have complained about Amazon's confusing cancellation process regarding Prime service, which tries to distract users by issuing several warnings to deter them from canceling their subscription.

Some designers use knowledge of human behavior to implement deceptive functionality that is not in the user's best interest (Gray et al., 2018) to manipulate them (Haselton et al., 2015). The design interfaces and choices described in the scenarios mentioned above are commonly known

as 'dark patterns' (also known as 'deceptive designs'). Dark patterns are user interface design choices that nudge, manipulate, or deceive users into making unintended and potentially harmful decisions while using an online service (Mathur et al., 2019). The key issue in the above scenarios is that these services may sacrifice providing users with a positive experience for achieving the company's business goals. Dark patterns are asymmetric (i.e., available choices are presented differently to users), covert (i.e., the effect of the design choice is hidden from users), deceptive (i.e., they portray a misleading/false belief); and restrictive (i.e., they hide, obscure, or delay presentation of information from users) (Mathur et al., 2019). Tactics of dark patterns include sneaking (i.e., concealing or delaying information users might object to if overtly displayed), creating false urgency, misdirecting (i.e., using words or visuals to steer users), providing false 'social proof' (i.e., displaying fictitious user experiences to coerce a purchase), implying scarcity (i.e., falsely indicating product unavailability), causing obstruction (i.e., making it hard for users to exit situations), and forcing action (i.e., requiring tasks unrelated to current activities) (Mathur et al., 2019). Concerning the ETHOS model, such dark patterns can be associated with both the SDLC and the DLC, especially those regarding the design and development of software systems. Moreover, data from user interactions can be applied to design dark patterns, violating the professional and ethical codes of conduct that necessitate considering the well-being of the software's end-users.

Considering the prevalence of dark patterns, students are likely to have encountered such scenarios online. Hence, drawing from students' experiences can be an excellent way to introduce the ethics involved in user interface design. Instructors can describe examples of deceptive designs and ask students to classify designs into different categories. Researchers have suggested that interventions such as 'spot the dark pattern' and 'design bright patterns' can help create awareness regarding design patterns (Bongard-Blanchy et al., 2021) and hence can be used in classrooms. Instructors can also leverage software engineering projects to help students think of and design ethical user interfaces. Using dialogical approaches, such as the Socratic dialogue technique with a prompt like *What is autonomy?* can bring out diverse issues surrounding ethical user interfaces. DEDA can also be used as a reflection tool. Students can identify conflicting values between (a) companies that use dark patterns and (b) users confronted with dark-pattern user interfaces. As the DEDA encourages identifying (conflicting) values and actors, it can prompt helpful reflection on the implementation of dark patterns.

Although creating awareness of deceptive patterns is needed, a more crucial goal for educators is to help students integrate ethical reasoning into all aspects of the user interface design process. Elements of the ETHOS model, such as ethical awareness and accountability can prepare students to engage in critical debate about where to draw the line, based on the company's values and their own sense of ethics. Students should realize the power that user interfaces have over people's behavior and online interactions, and the social responsibility they as software designers have towards developing ethical user interfaces.

Case study 3: Privacy and surveillance – security

Scenario

You are a machine learning engineer in the computer vision department at Protos. This tech company manufactures virtual reality (VR) headsets and a social media space called Protospace for VR users' avatars to meet for work and entertainment. To enhance the experience, Protos has developed a headset called VISOR with inward-oriented cameras to obtain real-time information about users' gaze and non-verbal facial expressions. In addition, external, outward-oriented cameras emulate users' physical body movements in the virtual world. Protos states these sensors allow for deploying

customized, photo-realistic avatars and adaptive content based on the users' motion and reactions. Users must accept Protos' terms to use VISOR. Critics of the company are skeptical about this justification – they believe that Protos will store and apply the data captured by the sensors as training for future projects. Moreover, the purported immersive nature of Protospace will grant Protos access to users' behavioral patterns, which can then be monetized by manipulating advertising and/or selling the data to third-party enterprises. Furthermore, there are growing concerns about the potential use of such a technology by autocratic governments that seek to surveil their citizens to quell civic unrest, repress all forms of opposition, and establish political dominance.

Discussion prompts:

1. *What are the potential risks of collecting this type of sensitive data from users? What measures can be taken to mitigate these risks?*
2. *What should be included as part of the informed consent terms? Are there other ways to help users recognize what they are agreeing to? How can users be made aware of future uses of the data collected?*
3. *As a machine learning engineer whose work involves processing the data gathered by the cameras, how could you improve the design's privacy and/or security?*

Context:

Ongoing attempts to improve facial recognition, non-verbal expressions, and gestures have resulted in data collection beyond what users have agreed upon or consented to. The example scenario mirrors real-life concerns around Meta's Quest Pro, a device collecting multi-dimensional data (Johnson, 2022). Although the technology was described as valuable for deploying customized, photo-realistic avatars, concerns arose around the detailed information collected about individuals and how it might subsequently be used for surveillance or otherwise infringe upon their privacy. Furthermore, Meta's privacy policy was vague regarding how the data would be used and shared with outside services.

Despite multiple calls to regulate the collection, storage, and application of user data, it appears that companies frequently collect more data than necessary. Personal information, preferences, and behaviors are often captured without explicit consent, leading to a loss of control over one's own data. This can result in unintended consequences for privacy, such as data breaches, identity theft, or the misuse of sensitive information. Scholars have highlighted the need for enhanced consideration of the interactions between technology and the law (Bernes, 2022). They have encouraged those involved to consider the purposes, implications, and uses of data collected; the transparency of tools developed; and ways to limit what is gathered, recorded, and disseminated for purposes more strictly aligned with goals. The European Union's General Data Protection Regulation (GDPR) has specific guidelines on this that students should be made aware of; readers are encouraged to see Chapter 13 on law in engineering ethics education.

There are several approaches educators can take to deliver lessons on privacy and security. For instance, they can introduce concepts, such as the benefits of sharing; privacy protection and perceptions; codes of conduct; cryptographic protocols; and international laws and regulations. The ETHOS model could help align discussion around design, deployment, development and documentation, or maintenance or monitoring when referencing components of the SDLC. The scenario above can also fit into discussions about the DLC throughout its phases. Concrete examples and discussion scenarios can raise awareness of possible issues and promote the need for greater accountability. For example, students could be asked to consider the trade-offs of retail loyalty cards, which can offer discounts but may also track shoppers' habits and sell data to third

parties. Another example could be related to concerns regarding healthcare and genetic mapping and/or ancestry websites. By spanning multiple fields and topics, for example, from electronic payment systems to educational records, instructors can help illustrate the necessity of awareness, protection, and action to rectify issues.

Instructors can use dialogical approaches such as Socratic dialogue and scenario thinking in conjunction with this case study. For the Socratic dialogue, we suggest evaluating the concept of 'privacy by design' as a realistic development approach. For scenario thinking, imagining idealistic and problematic scenarios for collecting sensitive user data may shed light on data ethics and values. The Data Ethics Canvas can be an appropriate reflection tool introduced along with this case study. One of the central issues here is the collection and use of personal user-data to further the business interests of the company doing the collection. By utilizing the Data Ethics Canvas, students can engage with questions that address multiple facets of this case, including the legal implications of collecting and utilizing data on this scale; they can reflect on reasons for using this data and the need for transparency.

Conclusions

As technology evolves, so do concepts and definitions; engineering educators must continually update the definitions, pedagogical content, and activities we use. Although today's society is increasingly recognizing the urgent need to address ethical issues and the need for responsible software development, this recognition brings its own problems – problems that have an impact beyond the classroom. We face the very real danger of 'ethics' becoming a buzzword and another trend to be monetized by companies (the way corporate 'greenwashing' undermines sustainability initiatives, see Chapter 6). Ready-made courses or tools that seem able to solve our ethical issues rather than focus on imparting awareness and ethical thinking could exacerbate that potential problem. A criticism that has been raised regarding the Sustainable Development Goals (SDGs), which some organizations tout without achieving results in specific target areas, as mentioned in Chapter 6. Another possible concern involves the increasing reliance on tools; while ethics assessment tools can be very helpful in guiding conversations and steering us toward relevant questions, they can be used in more nefarious ways. They may be used to help companies identify or define what is easiest or cheapest rather than what is morally desirable (and desirability can also vary depending on who is asked). Using ethics assessment tools could devolve into merely another box-ticking exercise, rather than being used to prompt software developers to reflect on the possible implications of their decisions and the products they help create. Ongoing efforts must thus be made to provide meaningful examples, to articulate where potential issues may arise, and to help students make relevant and meaningful connections – to help ensure future generations of engineering practitioners are committed to ethical principles and that software engineers develop products and manage issues as equitably and justly as possible.

Notes

1 https://gdpr.eu/data-protection-impact-assessment-template/
2 https://www.theodi.org/article/the-data-ethics-canvas-2021/
3 https://www.tech-transformed.com/product-development/
4 https://dataschool.nl/en/deda/
5 https://oecd-opsi.org/toolkits/ethical-os-toolkit/
6 https://openai.com/dall-e-2
7 https://speechify.com/
8 https://twitter.com/darkpatterns/status/1470399874147438594

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